

Transformer Architectures *for* *Longitudinal Data*

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*Postdoctoral Research Associate
Northeastern University (US)*

Pre-conference Workshop

*International Conference on Social Computing
University of Oxford (Sep 2, 2026)*



About Me

PhD in **Computational Social Science** at the Technical University of Denmark

Advisors: Sune Lehmann & Lars Kai Hansen

- LLMs for Social Science
- AI Fairness and Interpretability

Thesis:

- Life Trajectories as Symbolic Language: Exploring Human Behavior with Language Models (*life2vec*)

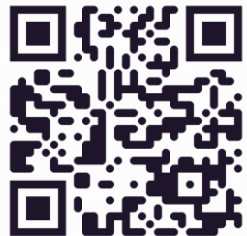
Postdoctoral Research Associate at Northeastern University

Mentor: Tina Eliassi-Rad

- Mechanistic Interpretability
- Formal Epistemology of LLMs
- Multi-agent LLM systems
- Belief Diffusion

Other Affiliations

- Pioneer Center for Artificial Intelligence (Copenhagen, Denmark)



Agenda

Time	Theme	Content
09:00 (Block A)	Natural Language Processing and Transformers	Topics: tokens, embeddings, attention, masked language modeling Pen & paper: one attention head by hand (15 min) Notebooks A1-A3: pretrained BERT
10:45	Break (15 min)	
11:00 (Block B)	A Person-Sequence Model, End to End	Topics: Tables, sequences, encoder, representations, AI Auditing Notebooks B1-B4: Synthea-BERT and 50,000 synthetic patients
12:45	Buffer and discussion (15 min)	

Goal

Block A:

1. Learn how tokenization works
2. Learn about transformers and masked language modeling
3. Read and audit embedding spaces

Block B:

1. Work with synthetic human data
2. Turn structured tabular records into an artificial language
3. Use a pretrained encoder's representations to predict severe-disease onset
4. Audit that prediction: baselines, ablations, and model interpretability

What we will not do: pretrain transformers live, Autoregressive LLMs, use qualitative sources (interviews), use real patient data



Workshop Page

Workshop Overview



Workshop Page

Assumed Background:

- Experience with Python and packages such as pandas or numpy
- Familiarity with Jupyter notebooks or Google Colab.
- Experience with data processing and basic stats.

Explicitly not assumed

- PyTorch, or any deep-learning framework.
- Hugging Face transformers.
- Any prior exposure to embeddings, representation learning or NLP.

What you need today?

- A laptop with a browser and a code editor (you can also run notebooks in [Google Colab](#)).
- A pen and paper: one full segment happens away from the keyboard.

Before we start: open notebooks/0-preparation and run it. It downloads the model and the data ahead of time.

Motivation

BEHRT: Transformer for Electronic Health Records

[Yikuan Li](#), [Shishir Rao](#) , [José Roberto Ayala Solares](#), [Abdelaali Hassaine](#), [Rema Ramakrishnan](#), [Dexter Canoy](#), [Yajie Zhu](#), [Kazem Rahimi](#) & [Gholamreza Salimi-Khorshidi](#)

[Scientific Reports](#) **10**, Article number: 7155 (2020) | [Cite this article](#)

ARTICLES · Volume 6, Issue 4, E281-E290, April 2024 · [Open Access](#)

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Foresight—a generative pretrained transformer for modelling of patient timelines using electronic health records: a retrospective modelling study

[Zeljko Kraljevic](#), MSc ^{a,e} · [Dan Bean](#), PhD ^{a,e} · [Anthony Shek](#), PhD ^{b,c} · [Rebecca Bendayan](#), PhD ^{a,e} · [Prof Harry Hemingway](#), PhD ^{d,f} · [Joshua Au Yeung](#), MBBS ^{b,c} · et al. [Show more](#)

Article | [Open access](#) | Published: 17 September 2025

Learning the natural history of human disease with generative transformers

[Artem Shmatko](#), [Alexander Wolfgang Jung](#), [Kumar Gaurav](#), [Søren Brunak](#), [Laust Hvas Mortensen](#), [Ewan Birney](#) , [Tom Fitzgerald](#)  & [Moritz Gerstung](#) 

[Nature](#) **647**, 248–256 (2025) | [Cite this article](#)

CAREER: A Foundation Model for Labor Sequence Data

Keyon Vafa, Emil Palikot, Tianyu Du, Ayush Kanodia, Susan Athey, David Blei

 Published: 22 Jan 2024, Last Modified: 17 Sept 2024  Accepted by TMLR  Everyone

Using sequences of life-events to predict human lives

[Germans Savcicens](#), [Tina Eliassi-Rad](#), [Lars Kai Hansen](#), [Laust Hvas Mortensen](#), [Lau Lilleholt](#), [Anna Rogers](#), [Ingo Zettler](#) & [Sune Lehmann](#) 

[Nature Computational Science](#) **4**, 43–56 (2024) | [Cite this article](#)

GETNext: Trajectory Flow Map Enhanced Transformer for Next POI Recommendation

Authors:  [Song Yang](#),  [Jiamou Liu](#),  [Kaiqi Zhao](#) | [Authors Info & Claims](#)

SIGIR '22: Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval

Pages 1144 - 1153 · <https://doi.org/10.1145/3477495.3531983>

MAPLE: Mobile App Prediction Leveraging Large Language Model Embeddings

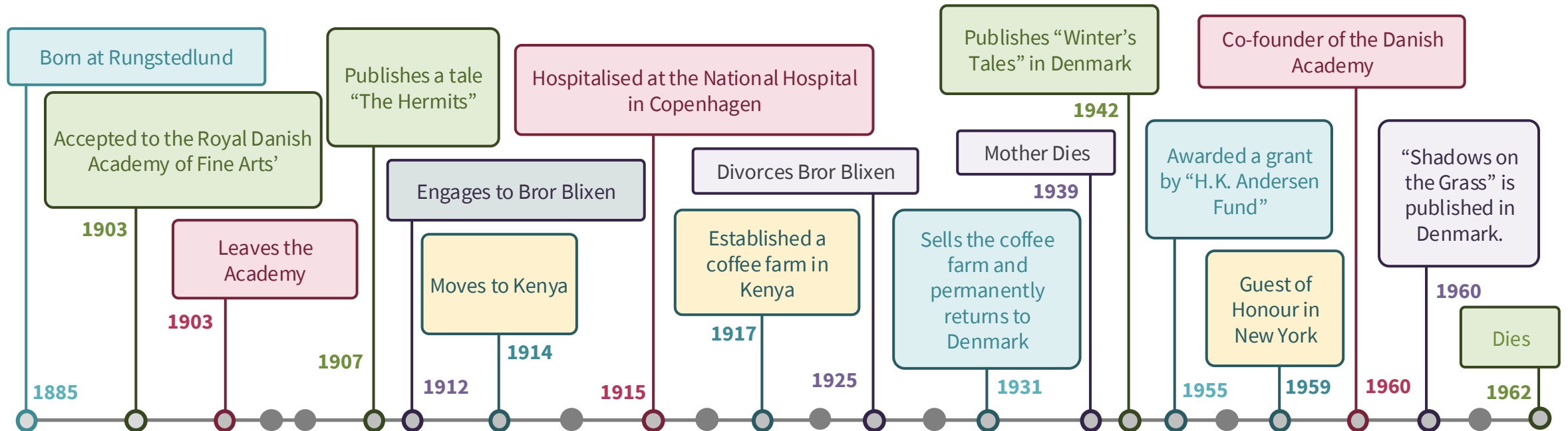
Authors:  [Yonchanok Khaokaew](#),  [Hao Xue](#),  [Flora D. Salim](#) | [Authors Info & Claims](#)

Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, Volume 8, Issue 1

Article No.: 10, Pages 1 - 25 · <https://doi.org/10.1145/3643514>

Life Trajectories

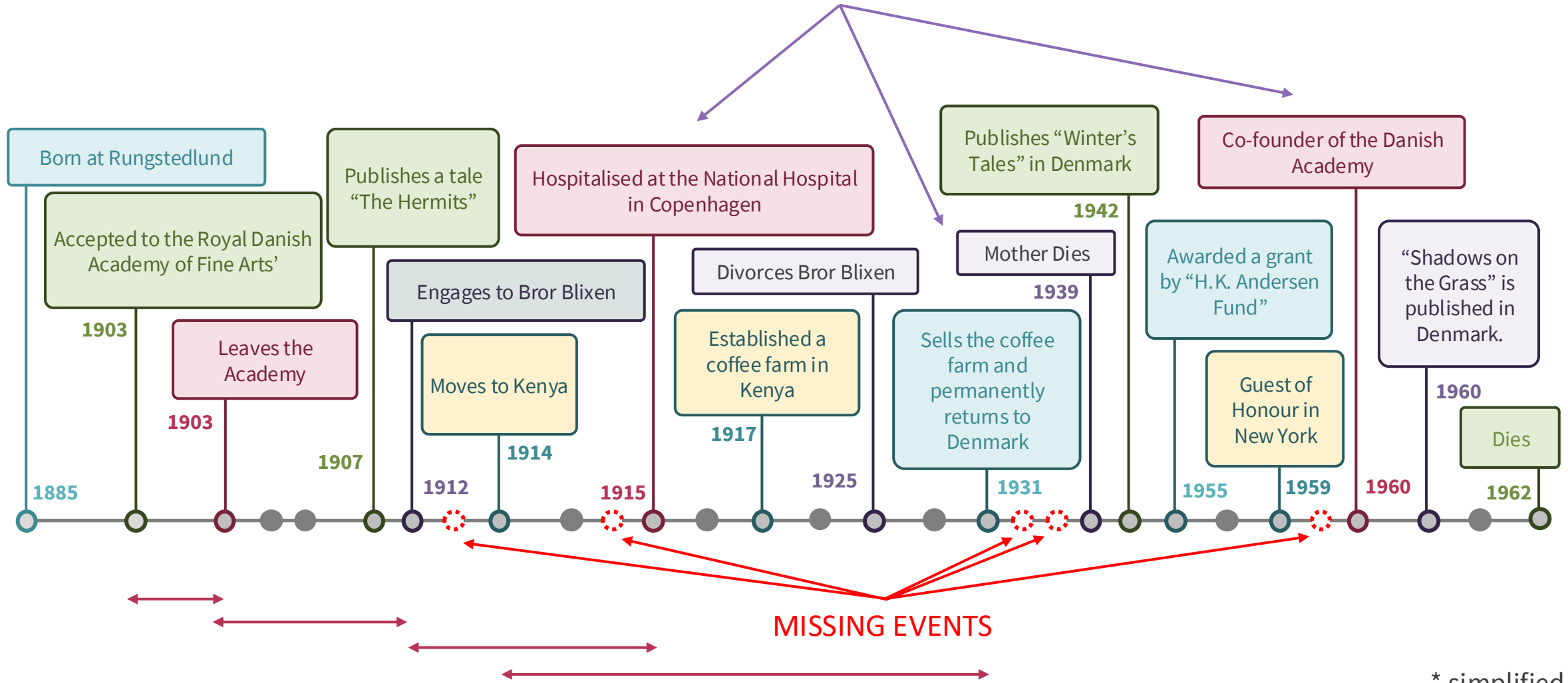
Life of Karen Blixen* (Danish author)



* simplified

Life Trajectories

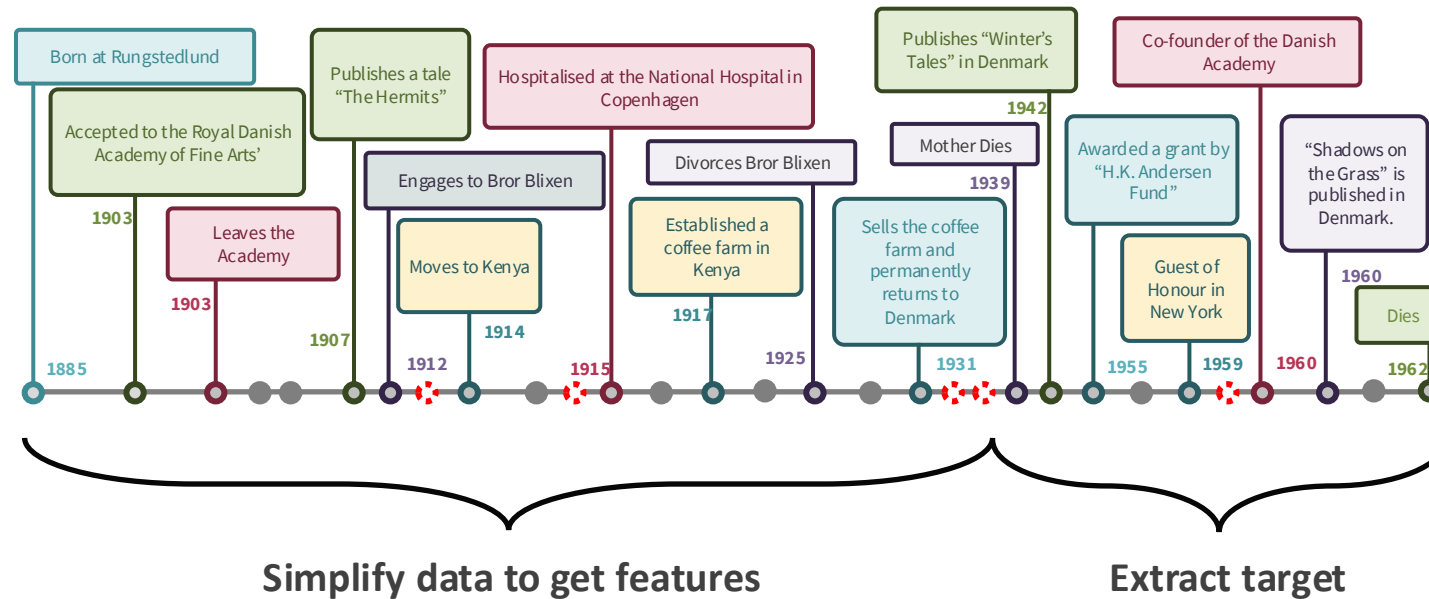
DIFFERENT SOURCES/MODALITIES



TIME VARIATION AND FREQUENCY OF EVENTS

How do we answer questions now?

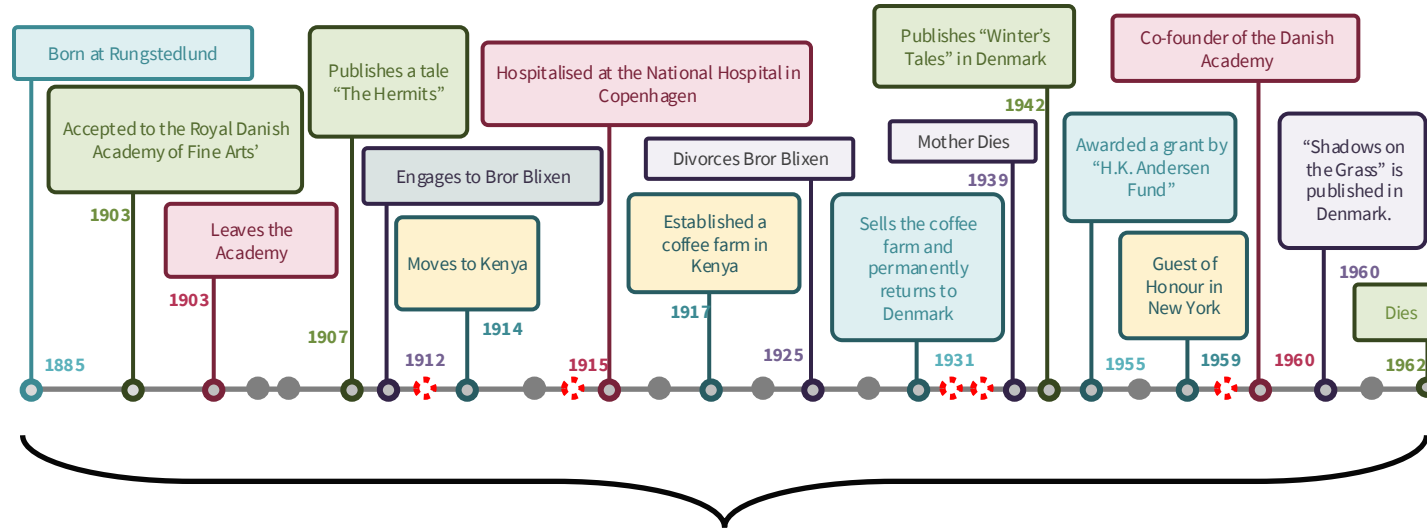
Research question: How many published works after reaching 55 years?



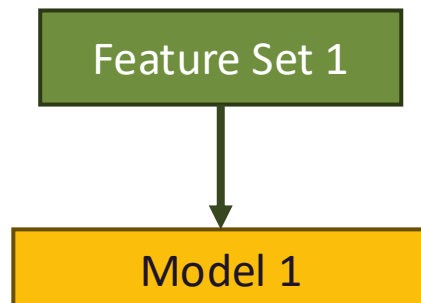
Average Income (past 10 years)	...	Ever Married	Hospital Admissions (past 5 yrs)	Publications (prior year)	Fraction of adult years abroad	Years since first publication	Current Residence
\$1,500	...	Yes (1)	1	0	0.29	34	Denmark

Generalized Linear Model, SVM, etc.

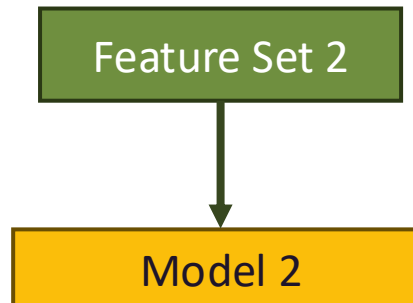
How do we answer questions now?



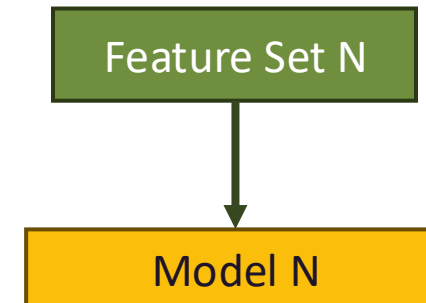
Research Question 1



Research Question 2



Research Question N



Motivation from Academia

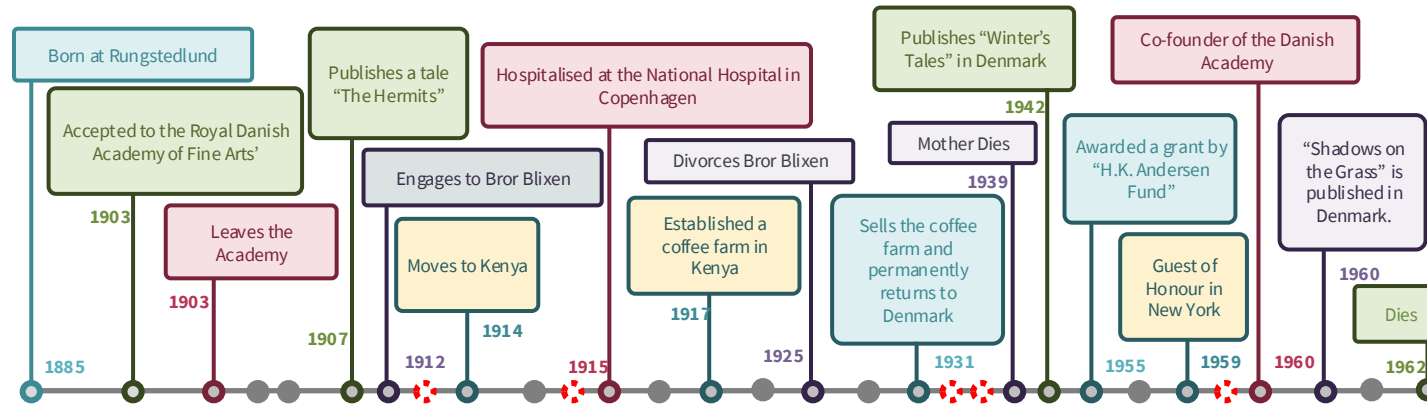
- The Fragile Families Challenge (Salganik et al., PNAS 2020)
 - A rich longitudinal study: **thousands of variables**, 15 years, one cohort,
 - **160 teams** applied a wide range of statistical and machine-learning approaches to predict six outcomes
 - Best models explained little of the variance ($R^2 \approx 0.20$ for GPA and material hardship, and around 0.05 for others).
- **Maybe...**The data was not the bottleneck, but how it was modeled

Salganik, M. J., Lundberg, I., Kindel, A. T., Ahearn, C. E., Al-Ghoneim, K., Almaatouq, A., ... & McLanahan, S. (2020). Measuring the predictability of life outcomes with a scientific mass collaboration. *Proceedings of the National Academy of Sciences*, 117(15), 8398-8403.

How do we capture and model this?

“From January 1, 2000 to June 12, 2019 James lived **with his parents** in the municipality of Amsterdam. From June 13, 2019 to December 1, 2019 James lived **alone** in the municipality of Groningen. From December 2, 2019 onwards James **again lived in the municipality of Amsterdam with Lisa**. Lisa had been living in Amsterdam her entire life.”

What transformers can give us?



We want a **single** model that takes **nuanced life trajectories**

General Purpose Model



Compressed **representation of life progression**

Predict the human behaviour
(on an individual level)

Study sociological phenomena
(on a population level)

Get **comprehensive insight** into the data

Everyone converged on the same five steps

events → tokens → sequences → transformer → representation

	events	Tokens	Architecture	Predictions
life2vec	Register Records	Diagnoses, labor (occupation, income) etc.,	Encoder, MLM	MLM, 4y mortality, personality
BEHRT	Hospital visits	Diagnoses (301 codes)	Encoder, MLM	diseases in next visit / 6m / 12m
CAREER	Job spells	Occupation codes + covariates	Transformer + causal modeling	Next job
GETNext	Check-ins	POI IDs + categories	Graph + Transformer	Next Point of Interest (POI)
MAPLE	Mobile apps opened	App IDs	LLM Embeddings	Next App
Delphi-2M	Diagnoses	Diagnoses + ages	Generative LLM	Whole disease trajectory

* Most of these predict a token from the same vocabulary as the input. life2vec predicts a label attached to the person.

Different fields (sort of). Different data. One recipe.

Natural Language Processing

Language and Machines

“Everything was beautiful and nothing hurt”¹



**AI-Generated Image

1. Slaughterhouse-Five, Kurt Vonnegut (1969)

Language Processing via Counts

“Everything was beautiful and nothing hurt”¹



Create a numerical representation of the text!



a	...	and	...	beautiful	...	everything	...	hurt	...	no	nothing	...	was	...	zyzz yva
0	...	1	...	1	...	1	...	1	...	0	1	...	1		0



Computers can work with numbers

1. Slaughterhouse-Five, Kurt Vonnegut (1969)

Language Processing via Counts

a	...	and	...	beautiful	...	everything	...	hurt	...	no	nothing	...	was	...	zyzz yva
0	...	1	...	1	...	1	...	1	...	0	1	...	1	...	0

If we reconstruct the sentence



“Beautiful was nothing and everything hurt”

“Everything beautiful hurt and was nothing”

“Everything hurt nothing and was beautiful”

Language Processing via Counts

Count-based representations



“Viktor prefers apples”

apples: 1
prefers: 1
likes: 0
spaceships: 0
kiwi: 0



“Maria likes spaceships”

apples: 0
prefers: 0
likes: 1
spaceships: 1
kiwi: 0

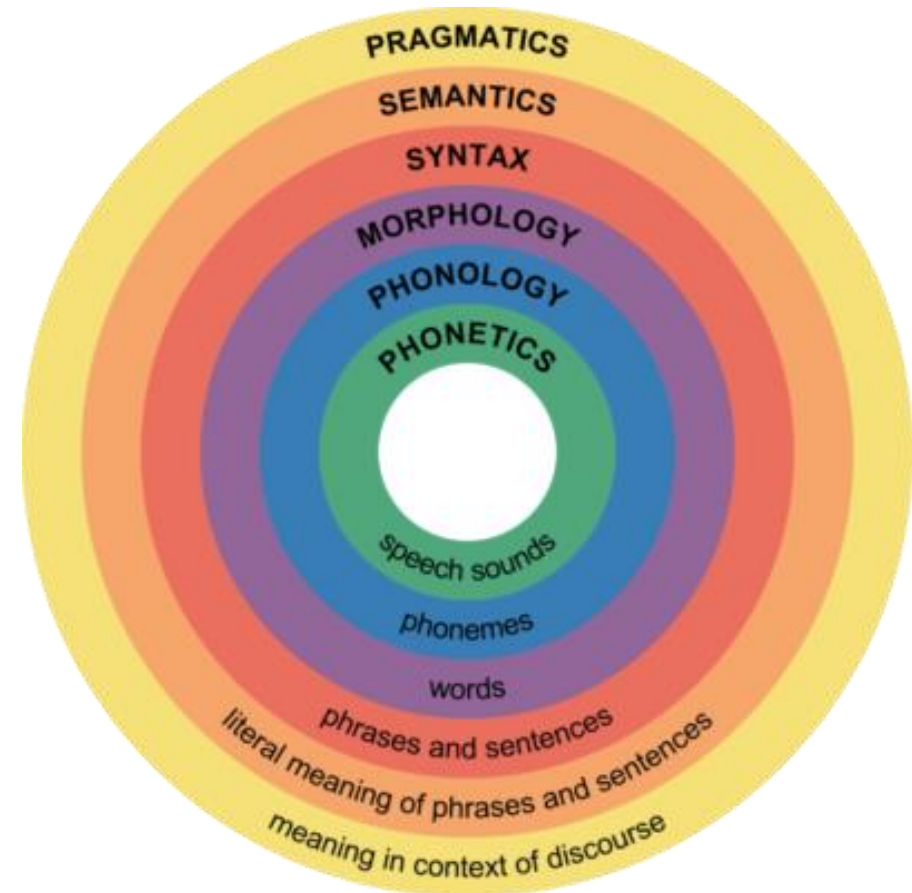


“Susanne likes kiwi”

apples: 0
prefers: 0
likes: 1
spaceships: 0
kiwi: 1

Layers of Language

Language is a **complex signal** with **many subtleties...**

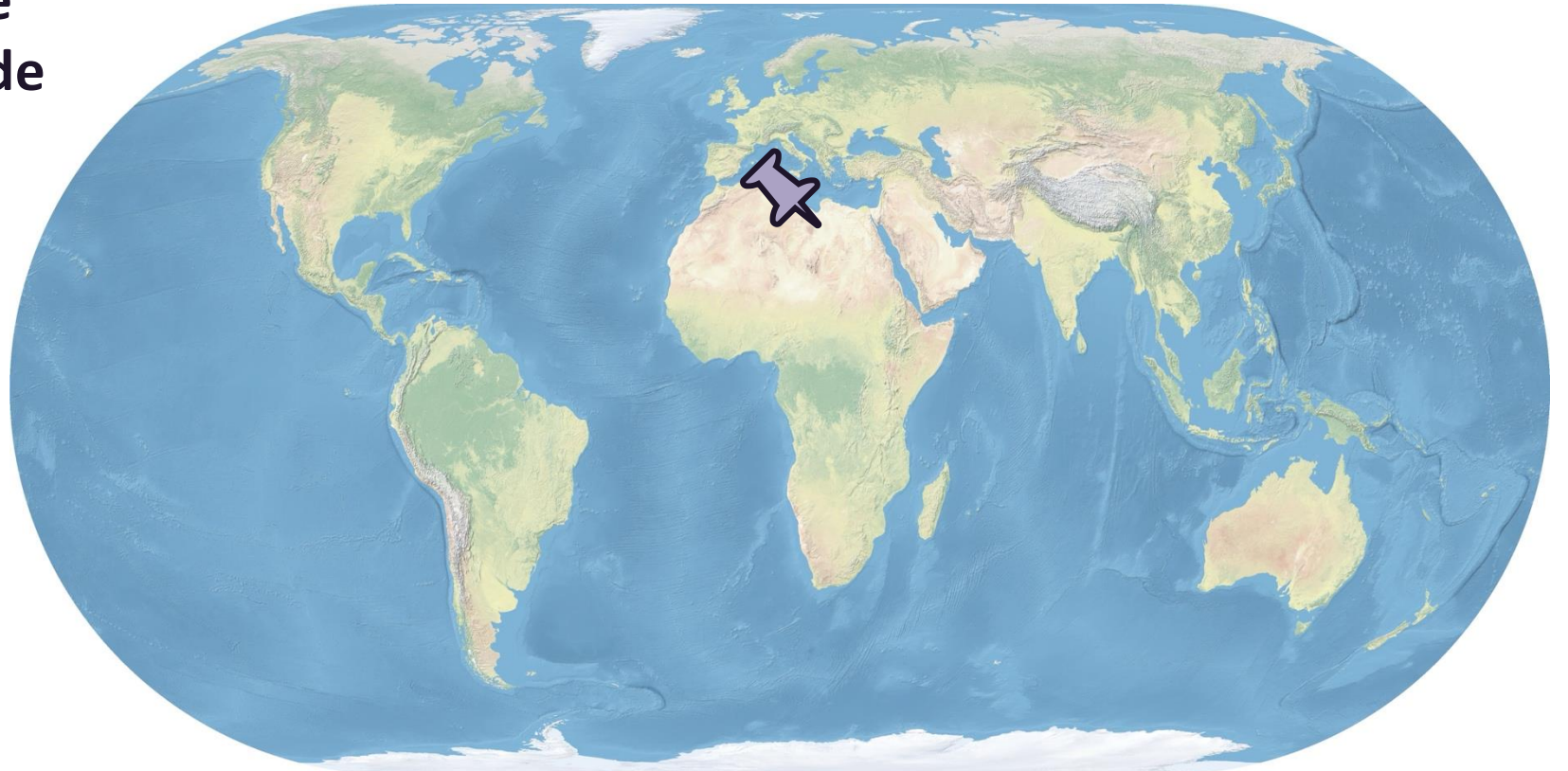


Representations and Embeddings

Let's consider these numbers:

29.9792° N latitude

31.1342° E longitude



Representations and Embeddings

	longitude*	latitude*
Great Pyramid	31.08	29.58
Petra	30.19	35.26
Machu Picchu	13.09	35.26
Colosseum	12.29	41.53



These values **capture spatial location**,
and allow us to **reason about the distances** (“similarity”).

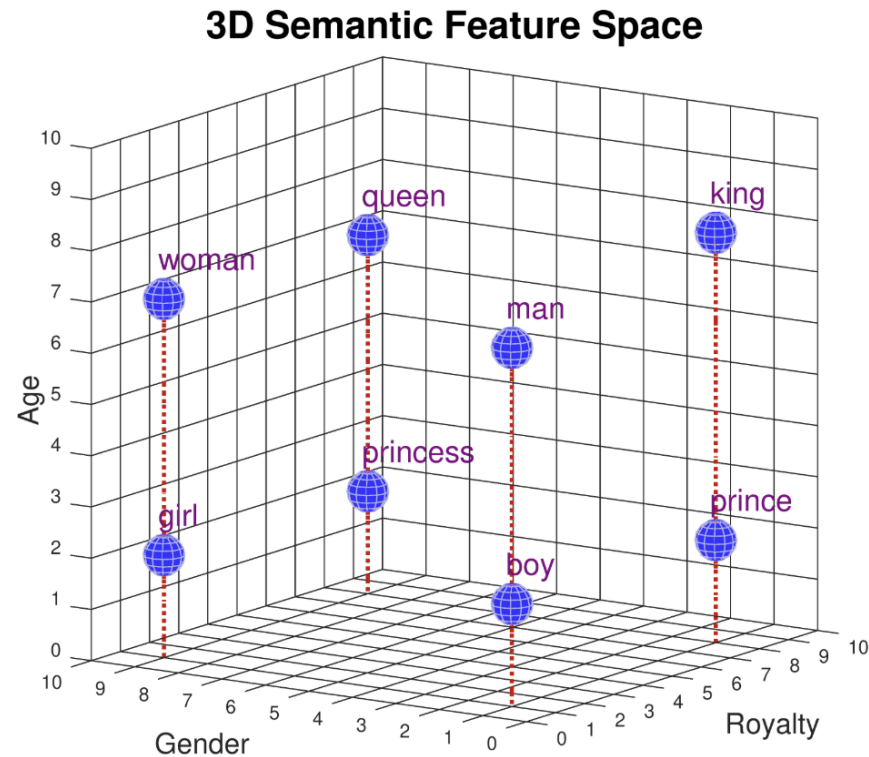
Language Processing via ~~Counts~~ Representations

Semantic Feature Space

Word Embeddings

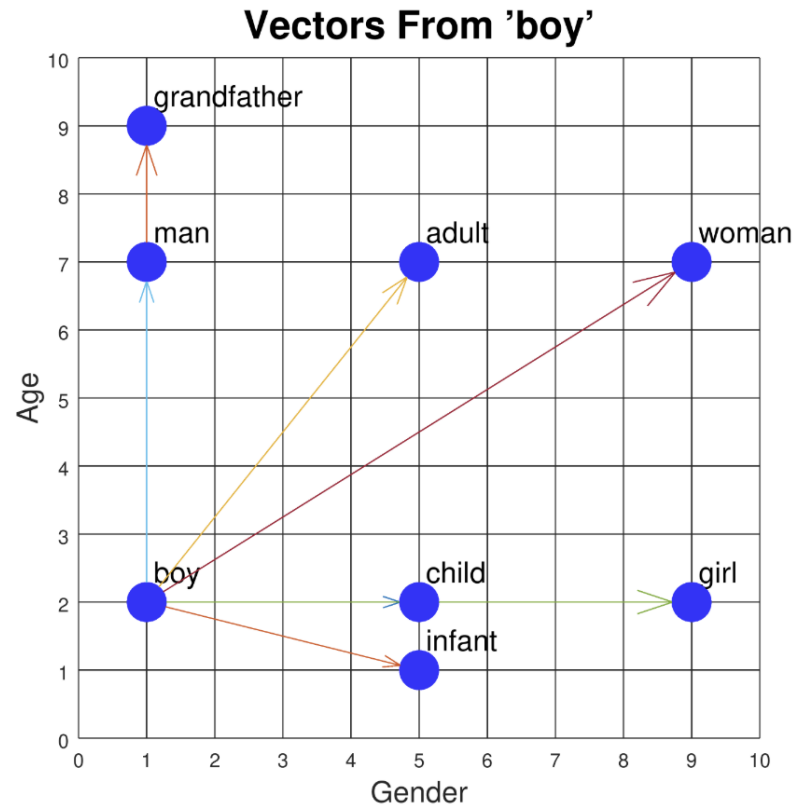
Word Coordinates

Word Vectors



Word Coordinates			
	Gender	Age	Royalty
man	[1,	7,	1]
woman	[9,	7,	1]
boy	[1,	2,	1]
girl	[9,	2,	1]
king	[1,	8,	8]
queen	[9,	7,	8]
prince	[1,	2,	8]
princess	[9,	2,	8]

Language Processing via ~~Counts~~ Representations



Distance from "boy"	
grandfather	7
man	5
adult	6.4031
woman	9.4340
boy	0
child	4
girl	8
infant	4.1231

Word Coordinates			
	Gender	Age	Royalty
man	[1,	7,	1]
woman	[9,	7,	1]
boy	[1,	2,	1]
girl	[9,	2,	1]
king	[1,	8,	8]
queen	[9,	7,	8]
prince	[1,	2,	8]
princess	[9,	2,	8]

Here, we specified each “dimension” ourselves.

Language Processing via ~~Counts~~ Representations

Word Coordinates				
	Gender	Age	Royalty	
man	[1,	7,	1]	
woman	[9,	7,	1]	
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king	[1,	8,	8]	
queen	[9,	7,	8]	
prince	[1,	2,	8]	
princess	[9,	2,	8]	



Here, we specified each “dimension” ourselves.

What are potential issues here?

Learning Representations

We can employ different methods to create the word embeddings:

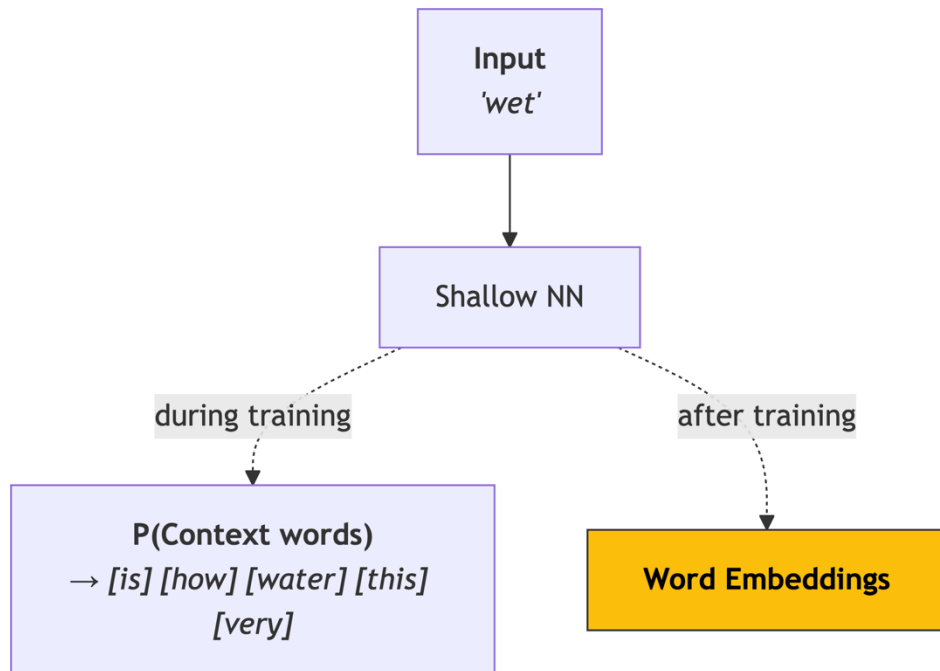
1. **Manually** assign values to each dimension (e.g., based on questionnaires)
2. **Count-based**: Count-Vectors, N-grams, Term Frequency-Inverse Document Frequency (TF-IDF)
3. **ML/Prediction-based**: SkipGram, GLoVE, word2vec + other deep learning methods (e.g., *Transformers*)

Read more: Jurafsky & Martin, *Speech and Language Processing* (3rd ed. draft), Ch. 5 "Embeddings."
Free online: web.stanford.edu/~jurafsky/slp3

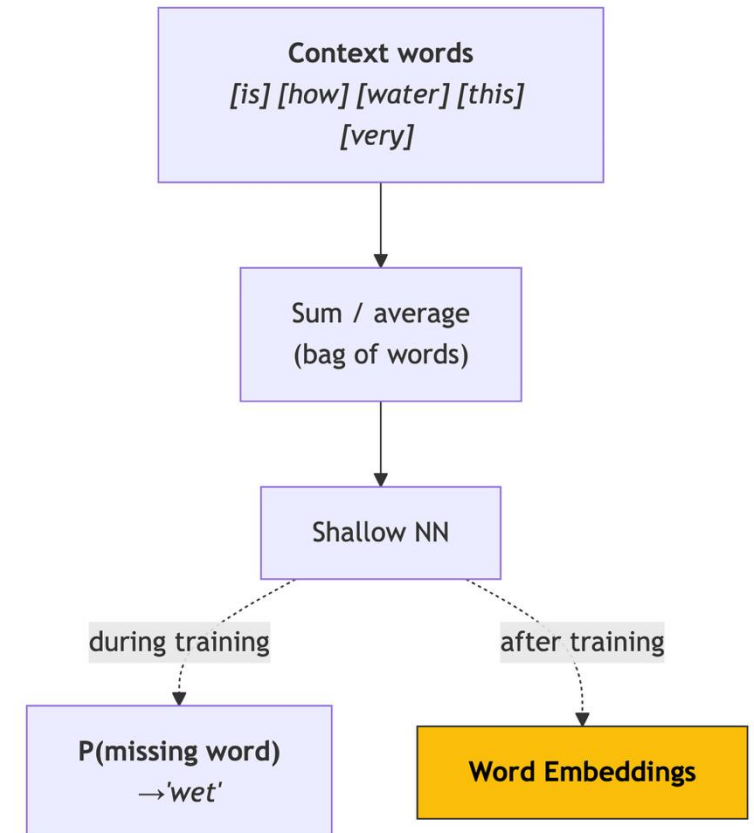
Learning Representation (word2vec)

Statement: How very wet this water is.*
Training sample: How very ____ this water is.
Target: wet

SkipGram



Continuous Bag-of-Words



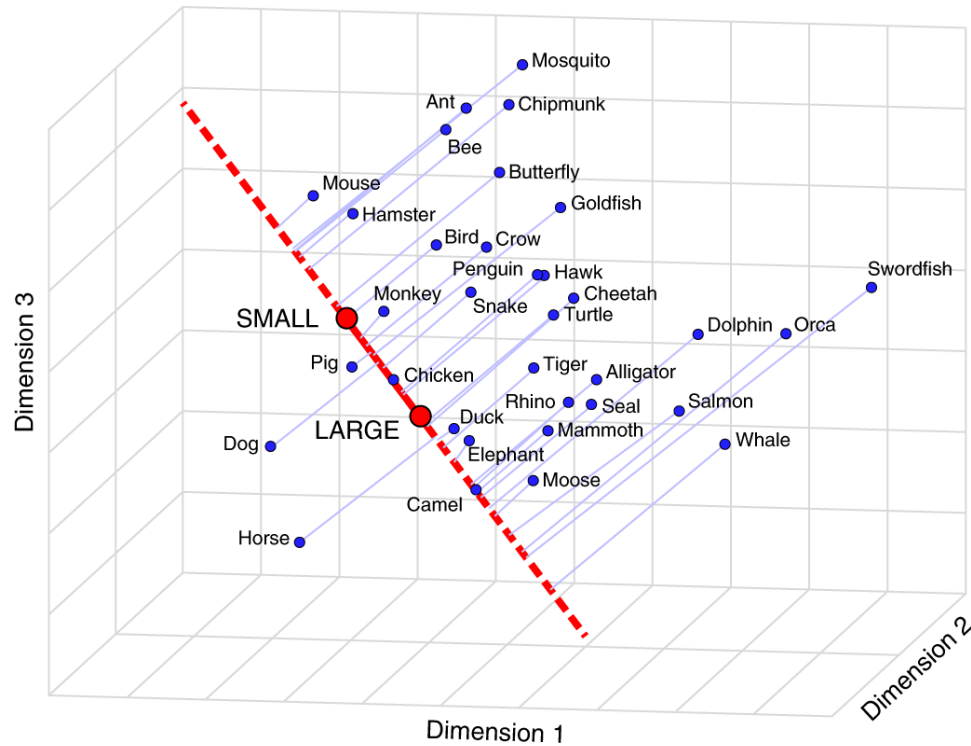


Fig.1: Example of semantic projection¹

In the embedding space (GloVe), “animal”-related words projected onto the “**small-large**” direction

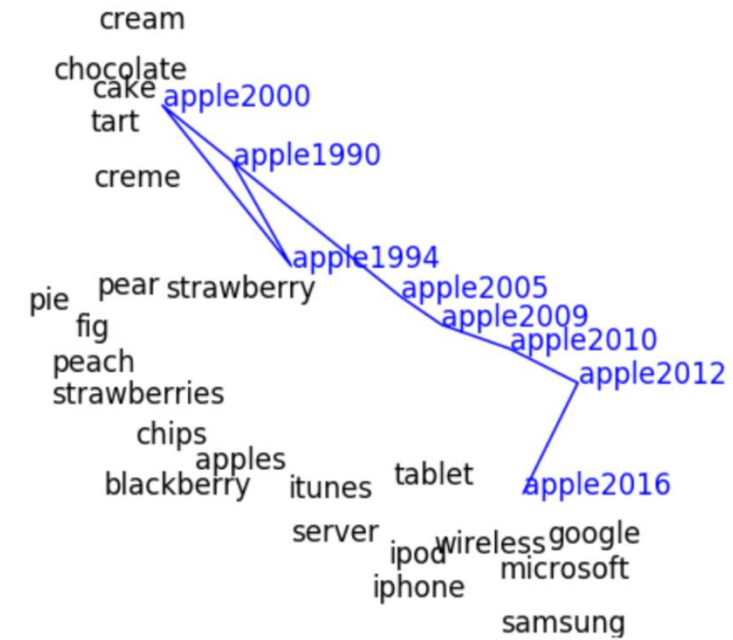


Fig.3: Trajectories of brand names²

Temporal evolution of “apple” with word2vec

1. Grand, G., Blank, I.A., Pereira, F. et al. [Semantic projection recovers rich human knowledge of multiple object features from word embeddings](#). *Nat Hum Behav* **6**, 975–987 (2022).
2. Yao, Z., Sun, Y., Ding, W., Rao, N., & Xiong, H. (2018, February). [Dynamic word embeddings for evolving semantic discovery](#). In *Proceedings of the eleventh acm international conference on web search and data mining* (pp. 673-681).

Large Language Models

What do you think this quote means?

*“Everything Was Beautiful and Nothing Hurt”**

Claude/ChatGPT/Kimi
Text to text

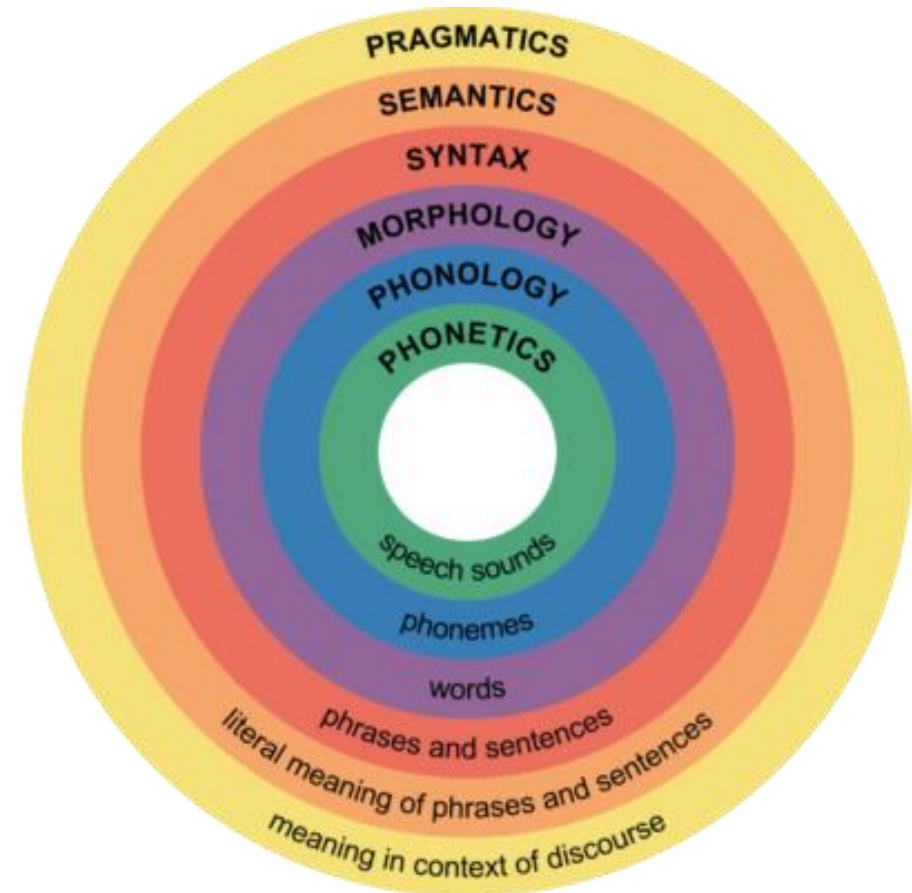


[...]The line captures a complex array of sentiments, many of which are rooted in the themes of the book itself, such as the trauma of war, the nature of human experience, and the fluidity of time [...]

Layers of Language

Language is a **complex signal** with **many subtleties...**

...just like a **life sequence**
(longitudinal data)



Crash course

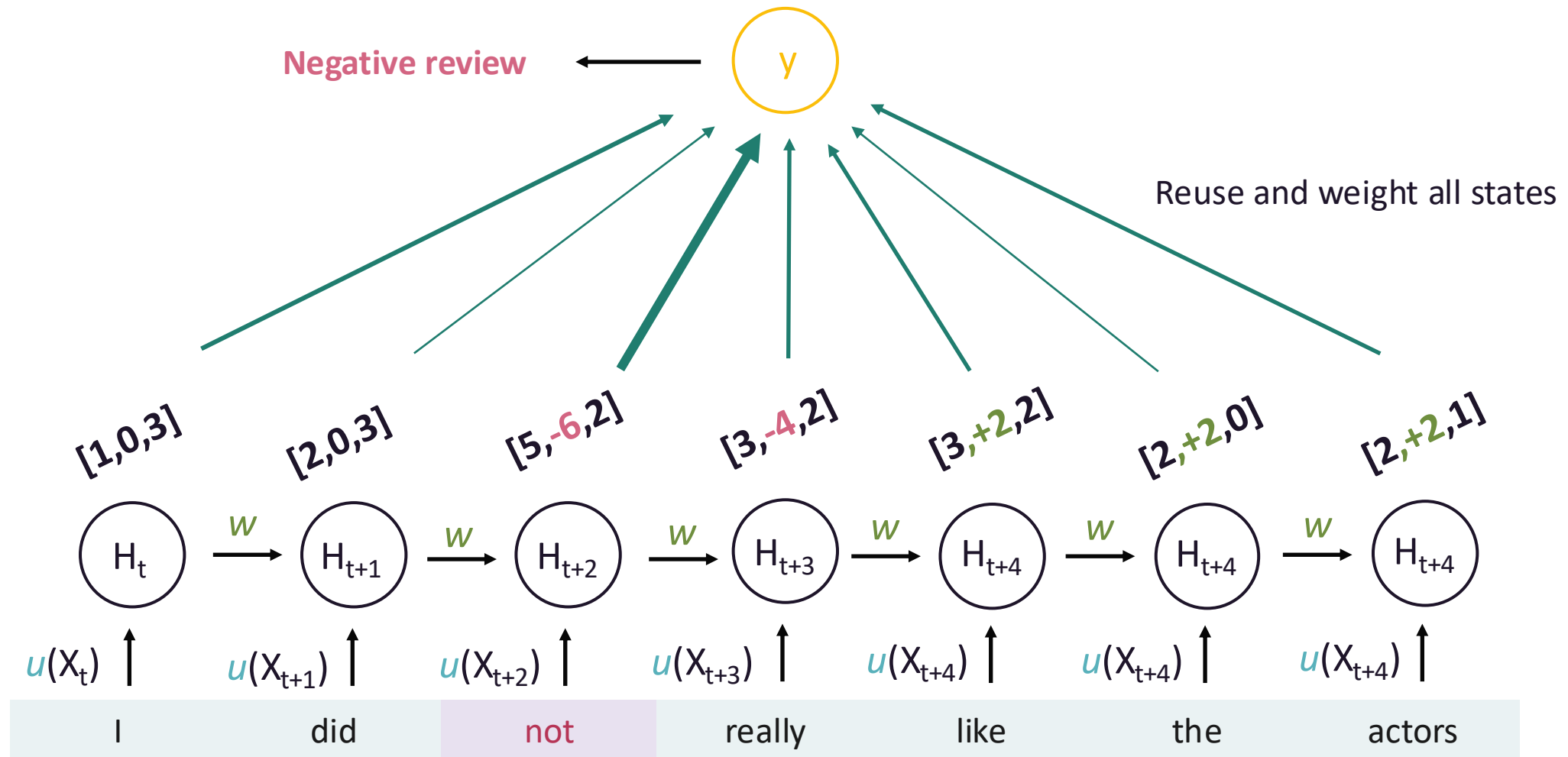
BERT: Bidirectional Transformers for Language Understanding

Recurrent Neural Networks



...but pure RNNs have issues (the information from previous steps gets lost)

RNNs and Attention

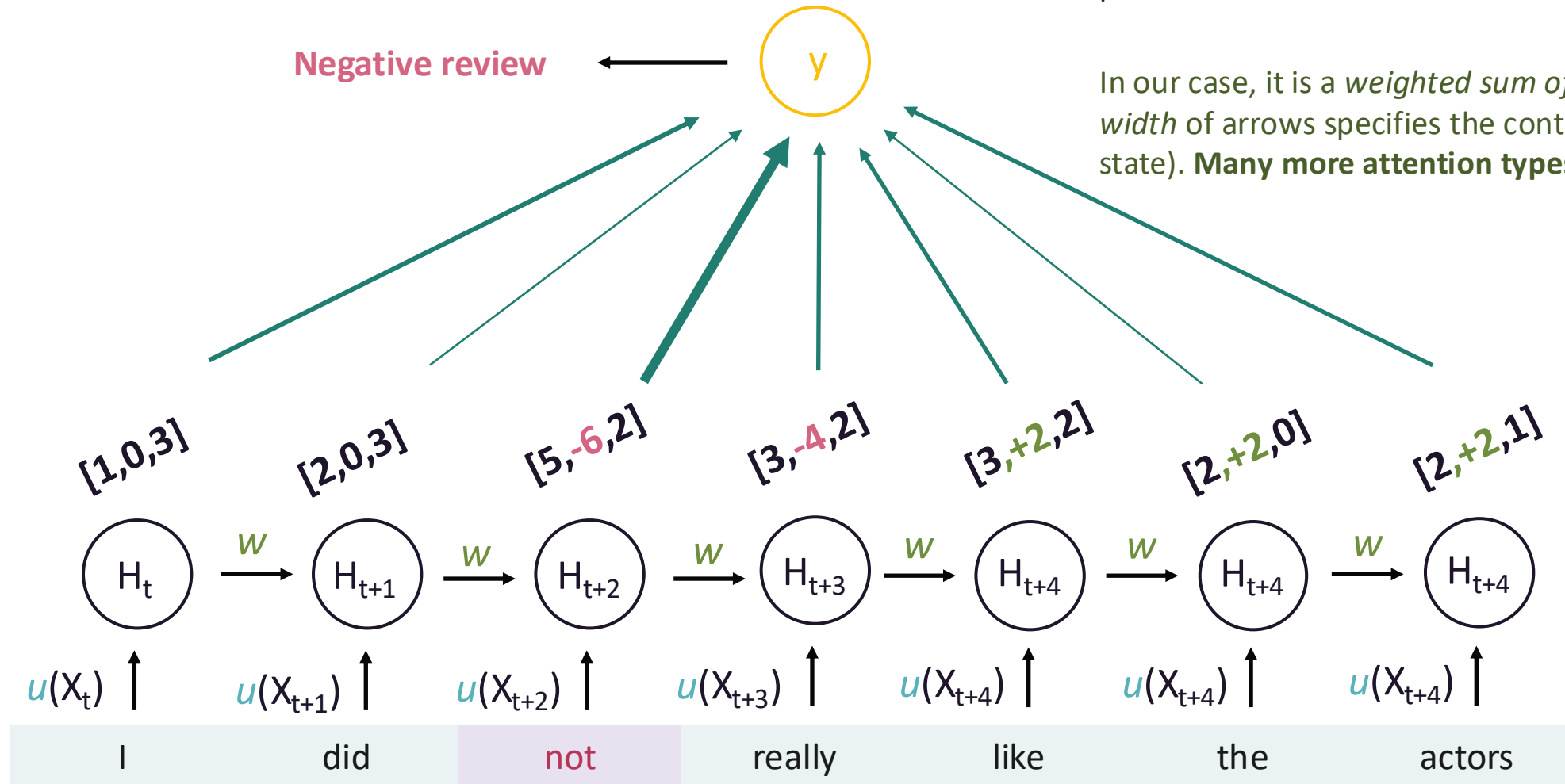


u, w, v (weights matrices)

RNNs and Attention

Attention Mechanism. The model calculates the contribution each hidden state should make to the final prediction.

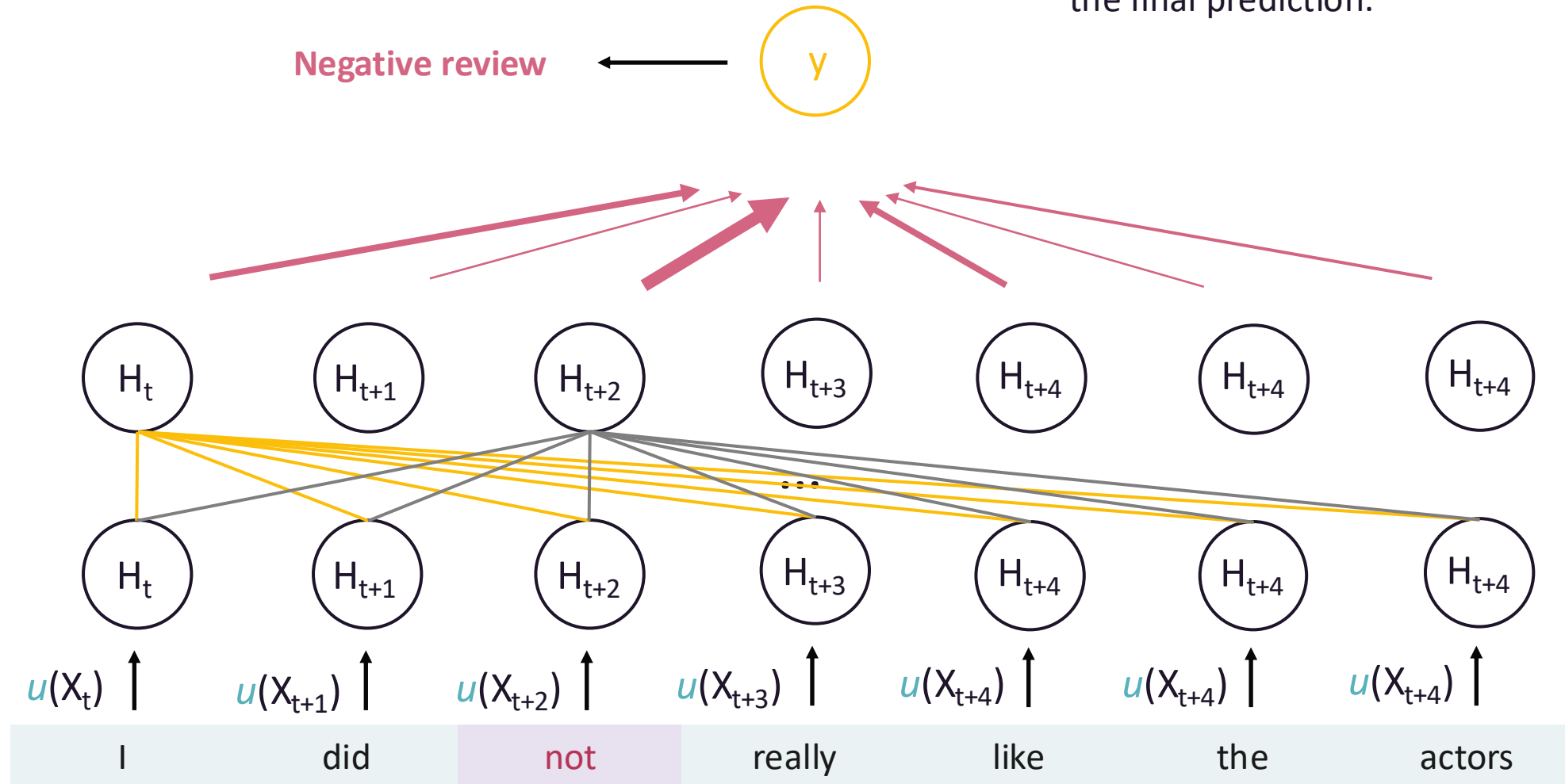
In our case, it is a *weighted sum of hidden states* (the *width of arrows* specifies the contribution of a hidden state). **Many more attention types exist.**



u, w, v (weights matrices)

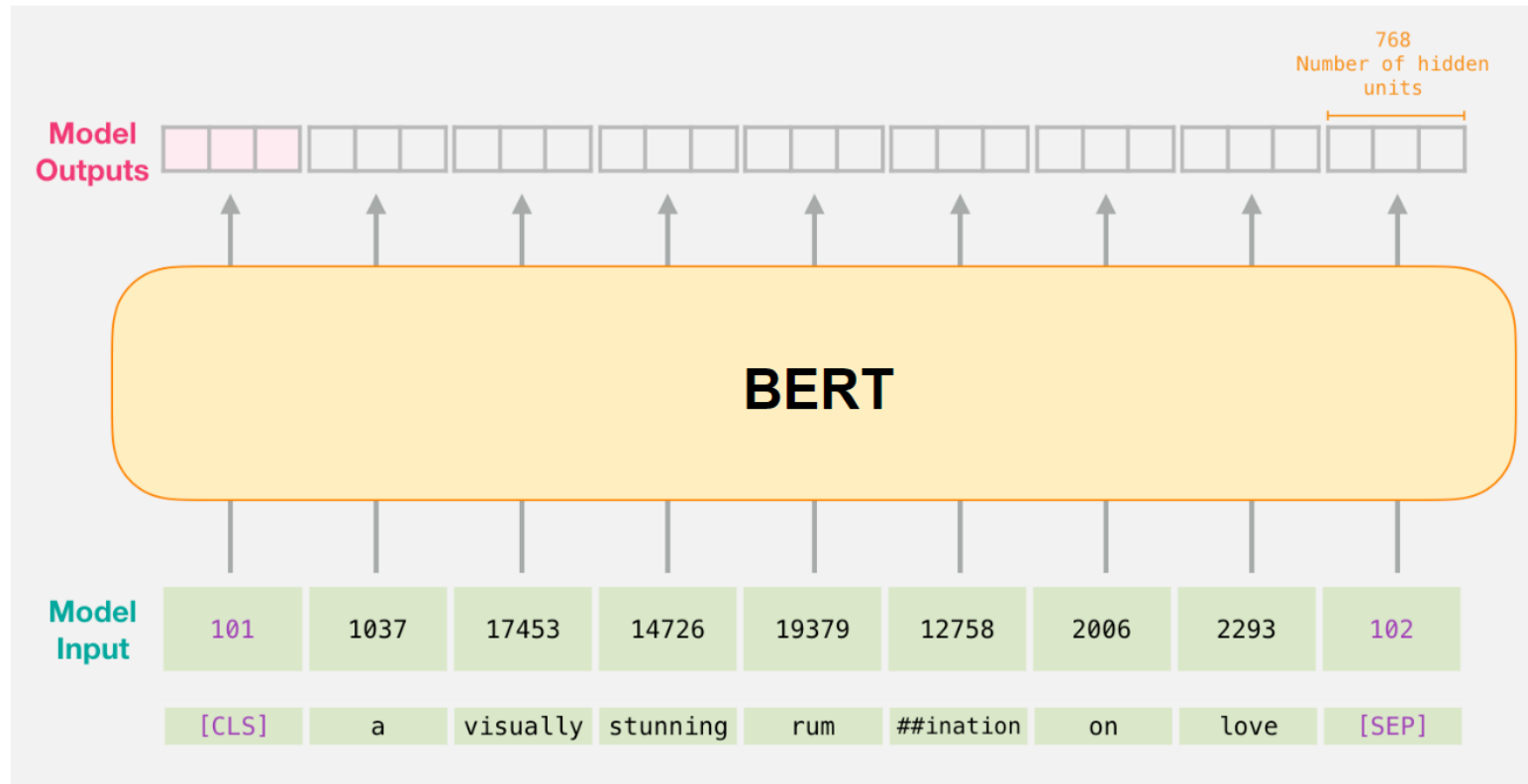
Attention Only

Attention Mechanism. The model calculates the contribution each hidden state should make to the final prediction.



u, w, v (weights matrices)

BERT



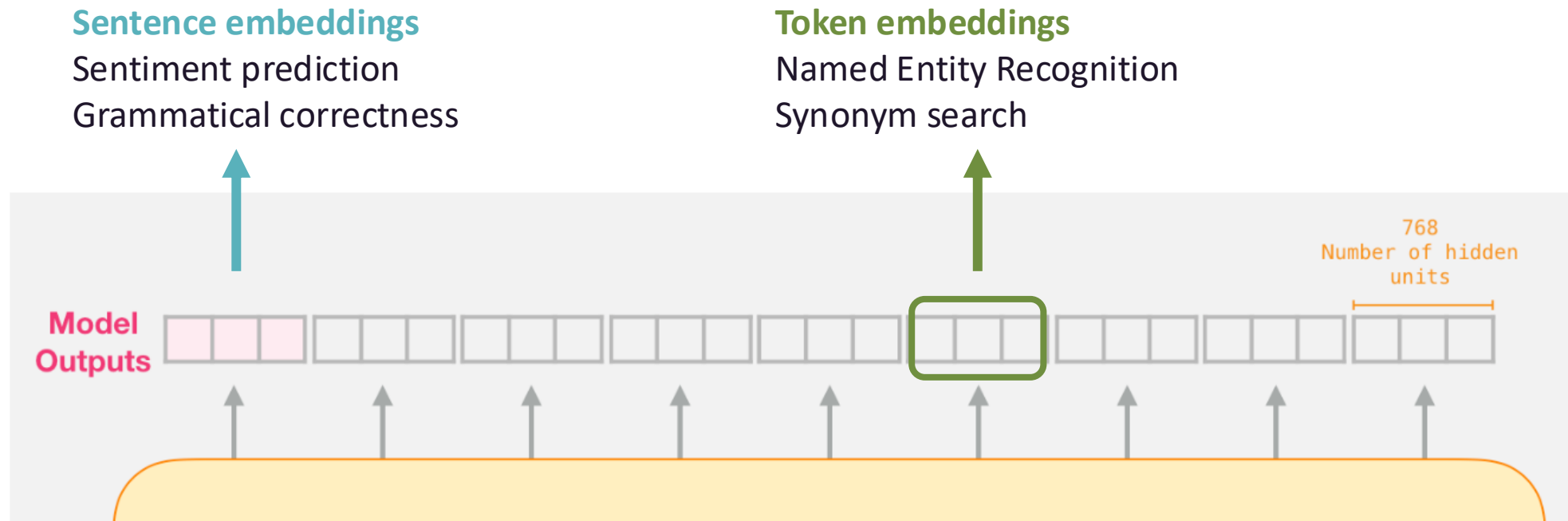
BERT computes **contextual embeddings** for each token of the input:

1. It returns embeddings for each token
2. Encode context

Turned out quite powerful on many NLP tasks.

What you can do with the outputs

- Make predictions with your favorite ML/stat model
 1. Sentence embeddings (on the first token, or averaged representation)
 2. Token embeddings
- Feed the sequence further to generate outcome



Hands-on Exercise: (A1) BERT Overview

Location: notebooks/a1-bert-simple

Content:

1. Introduction to PyTorch and Hugging Face
2. Tokenization of text
3. Masked Language Modeling with ModernBERT

Look for: Exercises 1,2 and 3

Time: 20 min

Setup: Instructor-led



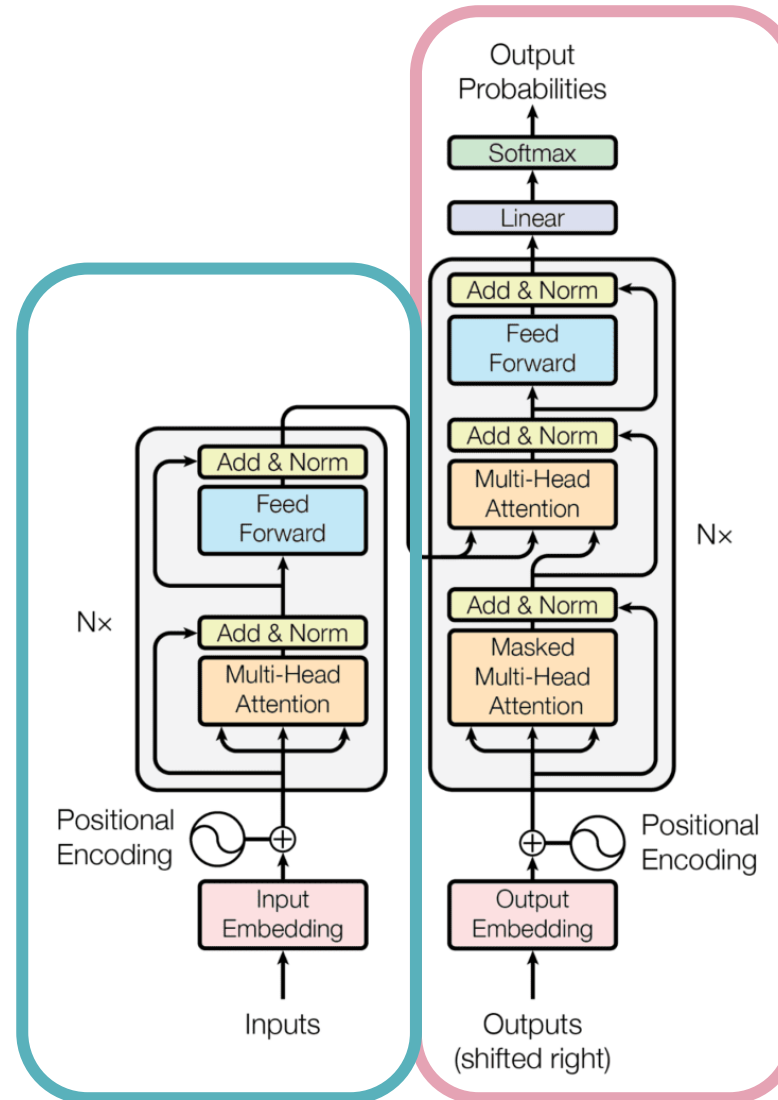
Workshop Page

Warner, Benjamin, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher et al. "Smarter, better, faster, longer: A modern bidirectional encoder for fast, memory efficient, and long context finetuning and inference." In *Proceedings of the 63rd annual meeting of the association for computational linguistics (volume 1: Long papers)*, pp. 2526-2547. 2025.

Transformers

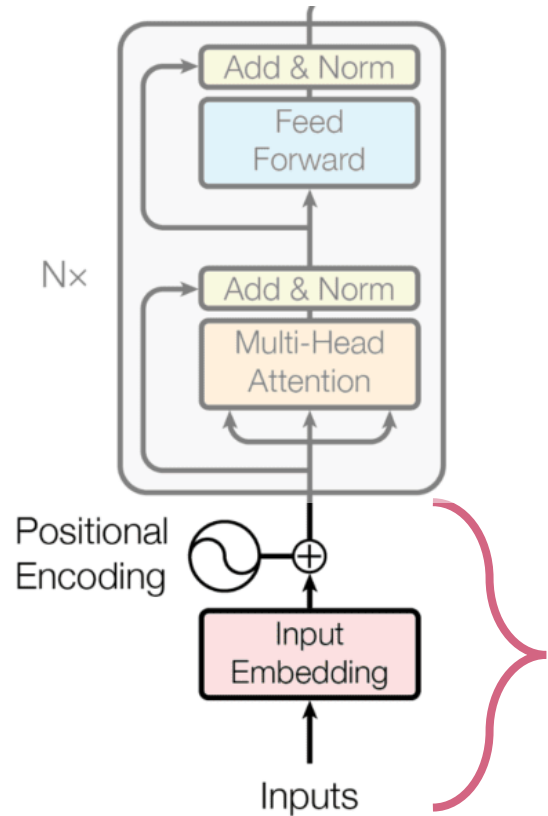
Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 2019.

BERT:
Bidirectional encoder
representations from transformers



GPT:
Generative Pretrained Transformer

BERT : Tokenization



Positions

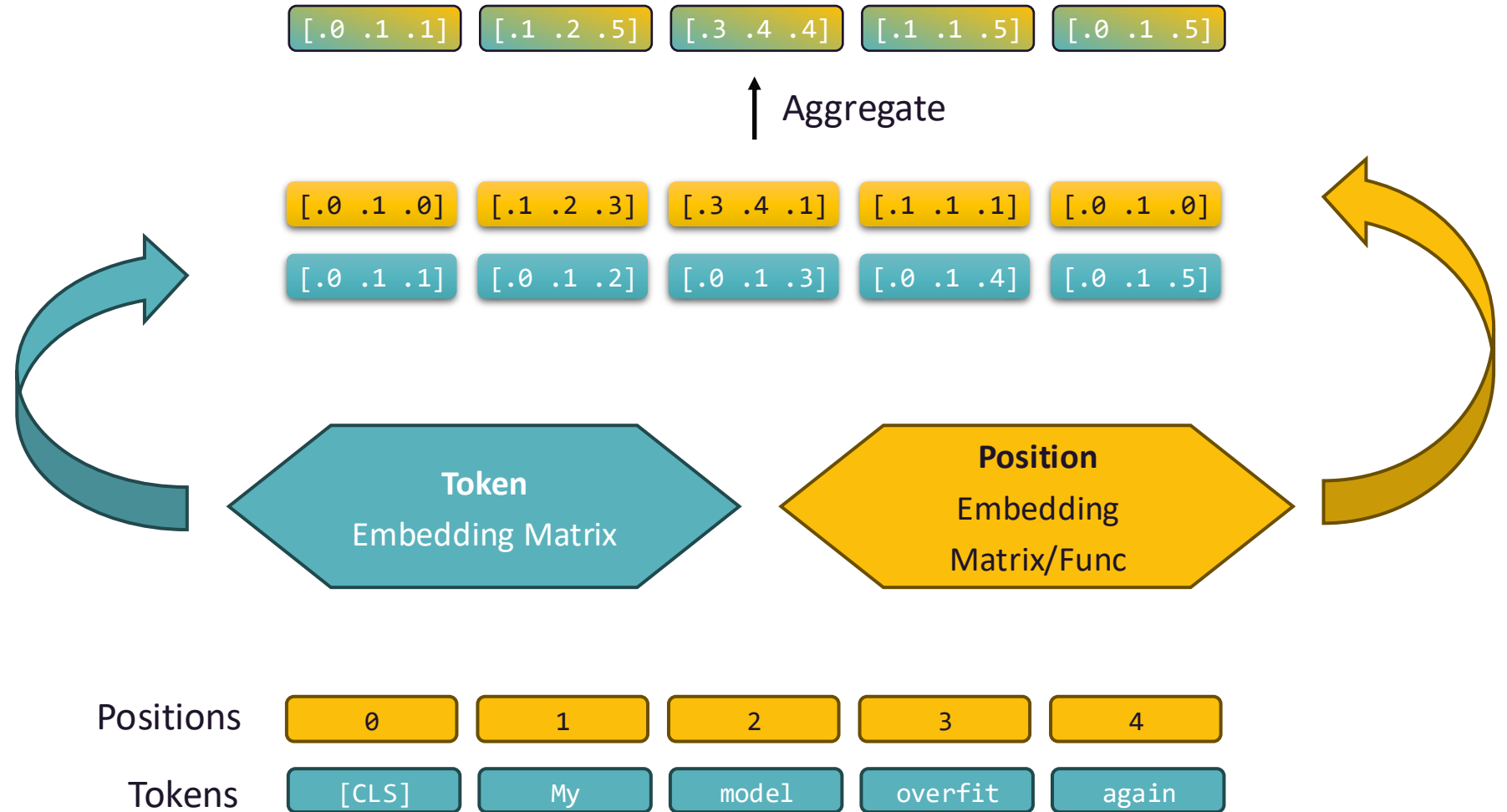
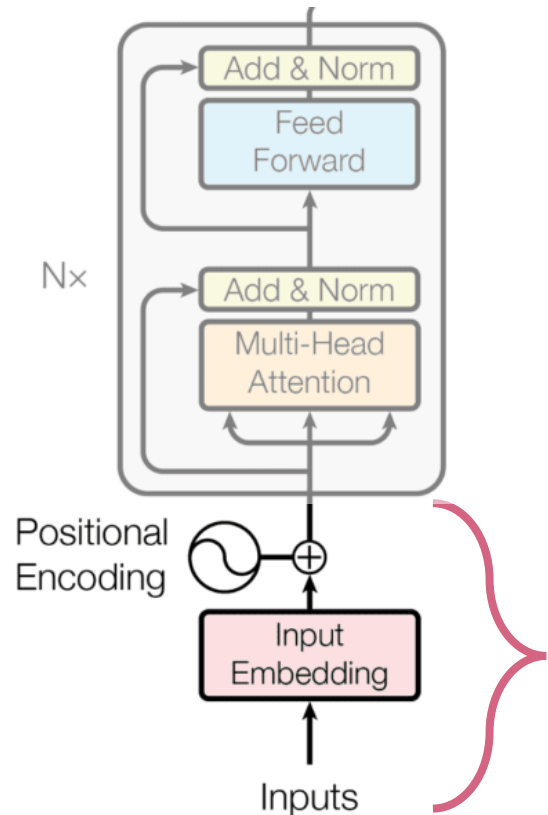


Tokens

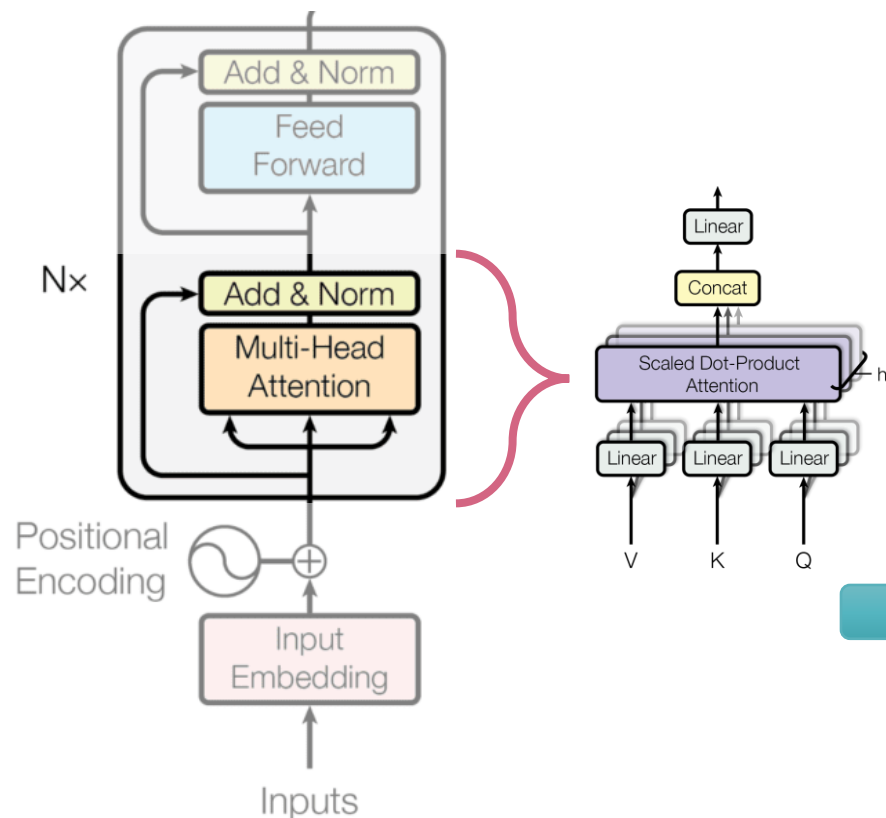


Input: **"My model overfit again"**

BERT : Embeddings



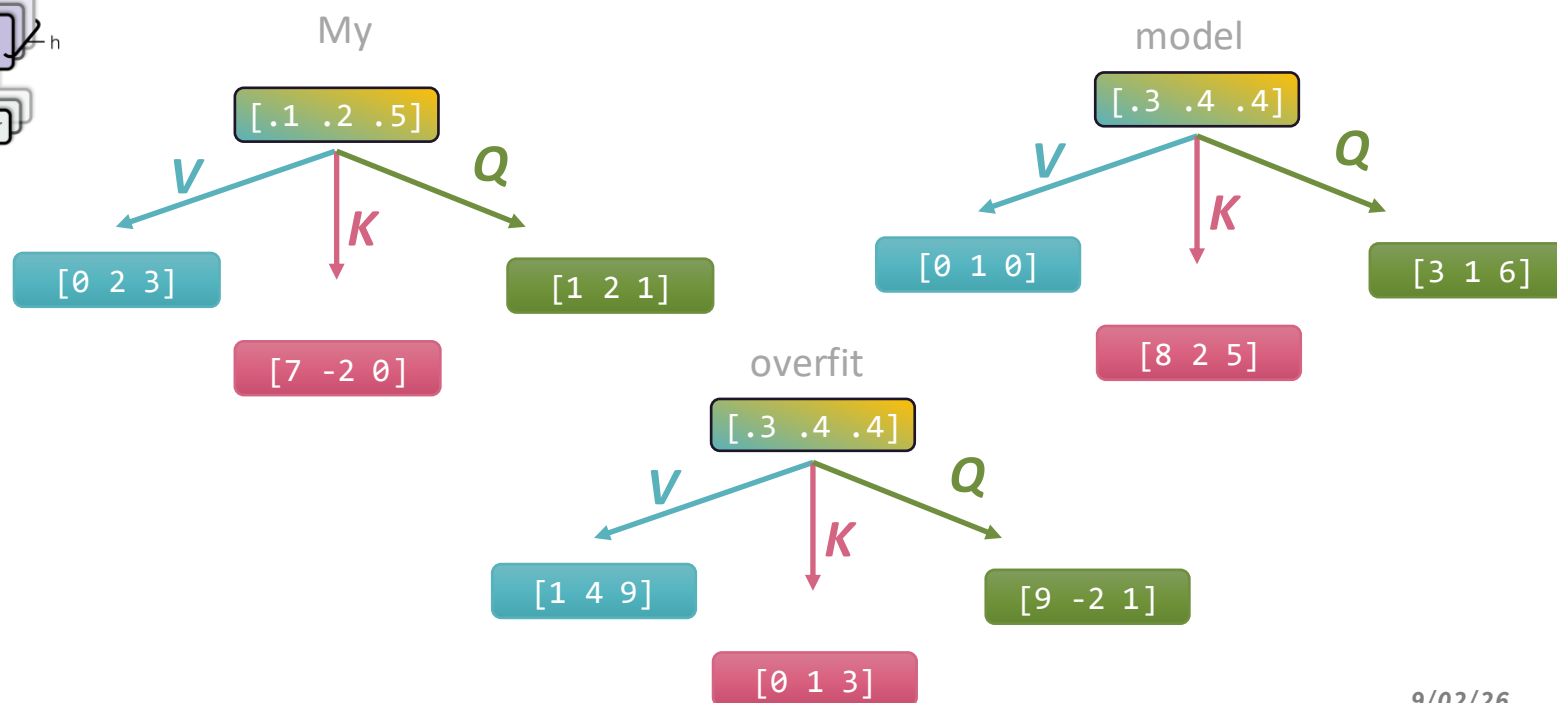
BERT: Self-Attention



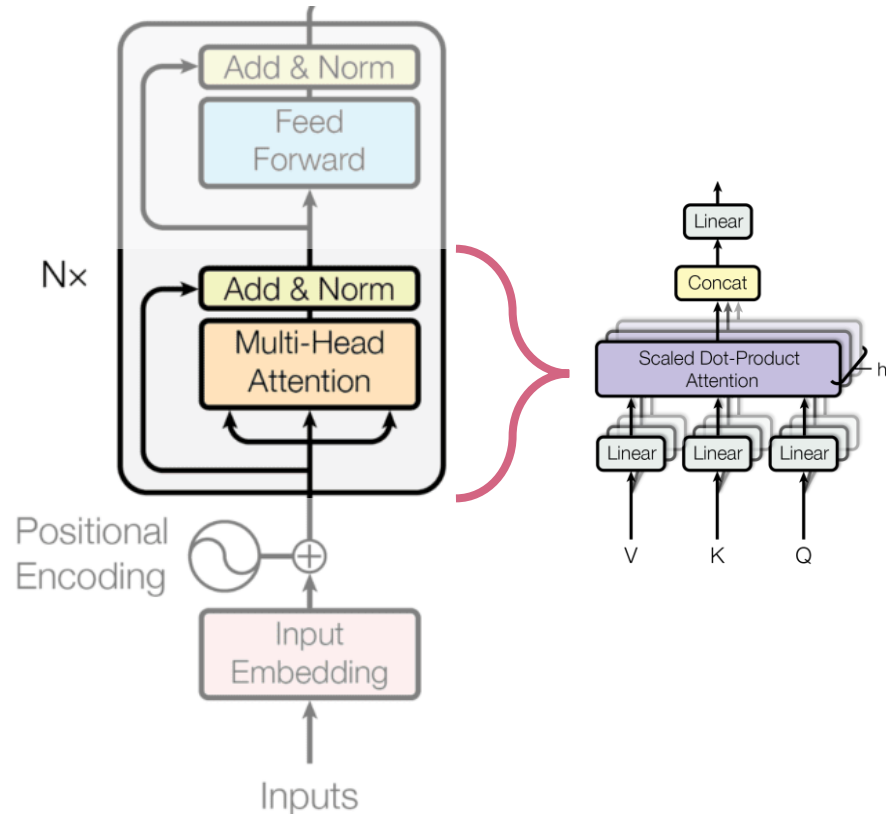
1. Get embeddings from embedding layer (or previous encoder)

	My	model	overfit
Aggregated Embeddings	[.1 .2 .5]	[.3 .4 .4]	[.1 .1 .5]

2. Transform each embedding into **Key**, **Query** and **Value** (those are just names for transformed versions of embeddings)



BERT: Self-Attention



3. Calculate **attention scores** for each word (dot product):

	query		key		Raw Scores	Softmax Normalized
my	7	-2	0	my	-4	.05
model	7	-2	0	model	-2	.25
overfit	7	-2	0	overfit	-1	.70

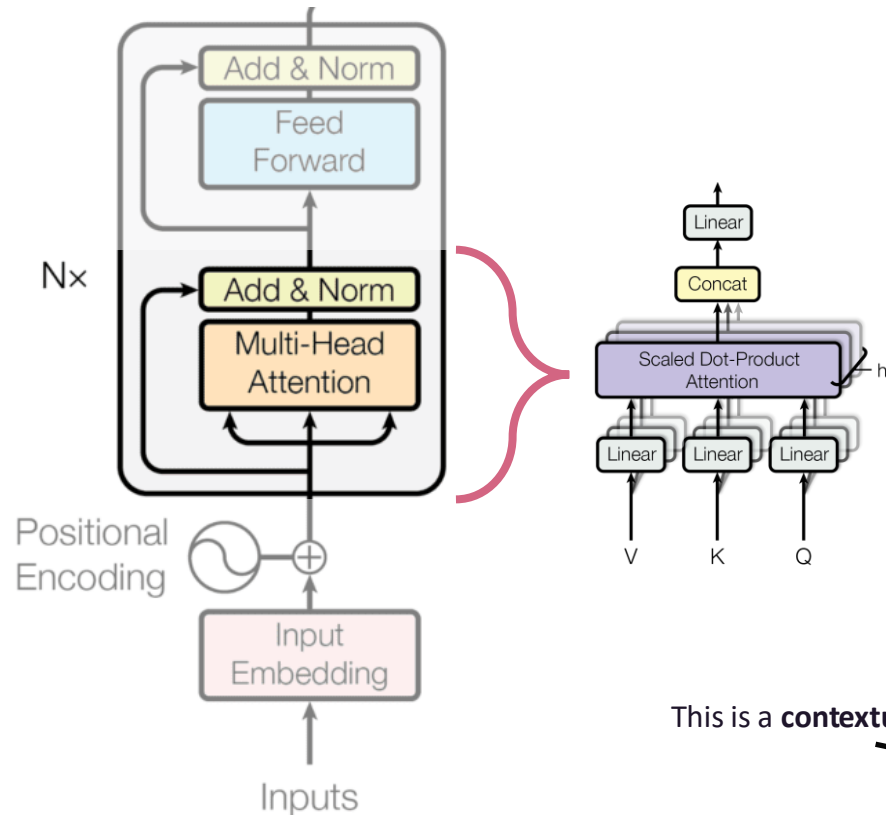
4. Do that for each word and get an **attention matrix**:

$A =$

	my	model	overfit
my	.05	0.25	0.70
model	.10	0.85	0.05
overfit	.50	.10	.40

BERT: Self-Attention

4. Do that for each word and get an **attention matrix**:



$A =$

	my	model	overfit
my	.05	0.25	0.70
model	.10	0.85	0.05
overfit	.50	.10	.40

5. **Compute** the contextualized embedding:

This is a **contextualized** embedding for "my"

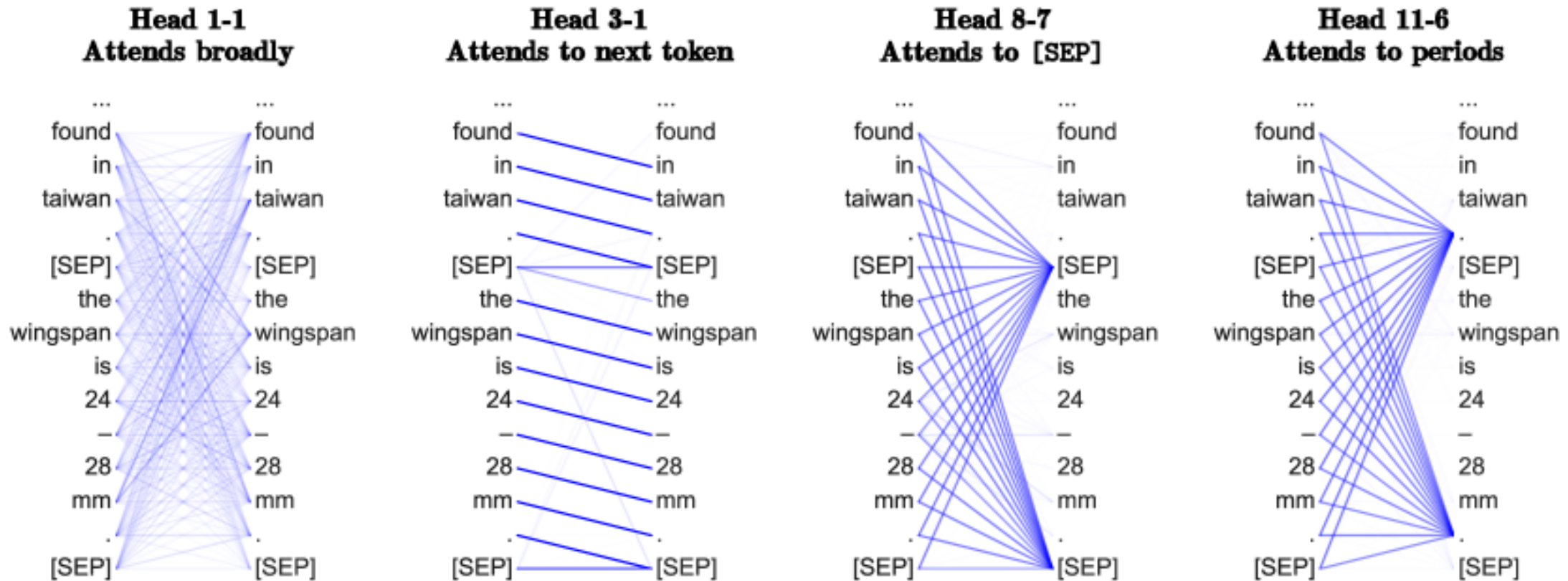
$$\begin{bmatrix} 7 & 1 & 2 \end{bmatrix} = \begin{bmatrix} .05 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 1 \end{bmatrix}_{\text{my}} + \begin{bmatrix} .25 \end{bmatrix} \cdot \begin{bmatrix} 3 & 1 & 6 \end{bmatrix}_{\text{model}} + \begin{bmatrix} .70 \end{bmatrix} \cdot \begin{bmatrix} 9 & -2 & 1 \end{bmatrix}_{\text{overfit}}$$

The diagram shows the calculation of the contextualized embedding for the word "my". The input vector [7, 1, 2] is equal to the weighted sum of the input vectors for all words in the context. The weights are the attention values from the matrix: 0.05 for "my", 0.25 for "model", and 0.70 for "overfit". Each weight is multiplied by its corresponding input vector, and the results are summed to produce the final contextualized embedding for "my".

Multihead Attention

Clark, K., Khandelwal, U., Levy, O., & Manning, C. D. (2019, August). What does BERT look at? an analysis of BERT's attention. In *Proceedings of the 2019 ACL workshop BlackboxNLP: analyzing and interpreting neural networks for NLP* (pp. 276-286).

Why do we need multiple heads? *They are often specialized in specific aspects.*



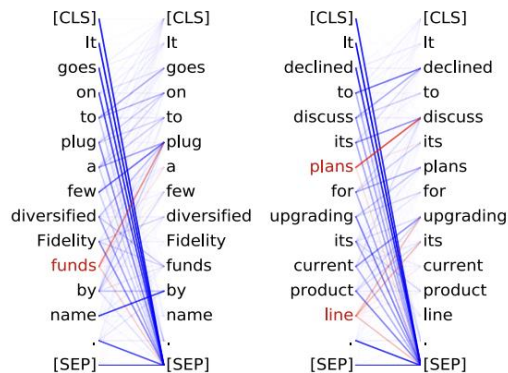
Note: That's **not always** true

Multihead Attention II

Clark, K., Khandelwal, U., Levy, O., & Manning, C. D. (2019, August). What does BERT look at? an analysis of BERT's attention. In *Proceedings of the 2019 ACL workshop BlackboxNLP: analyzing and interpreting neural networks for NLP* (pp. 276-286).

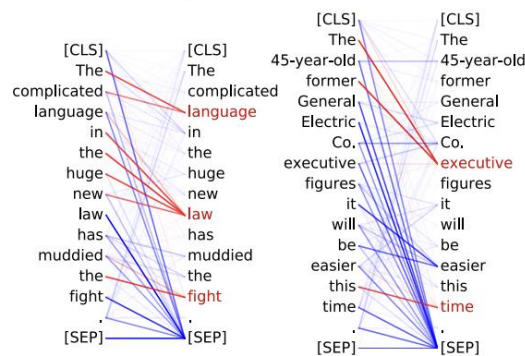
Head 8-10

- **Direct objects** attend to their verbs
- 86.8% accuracy at the dobj relation



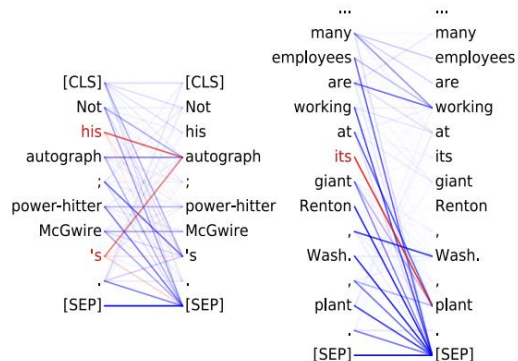
Head 8-11

- **Noun modifiers** (e.g., determiners) attend to their noun
- 94.3% accuracy at the det relation



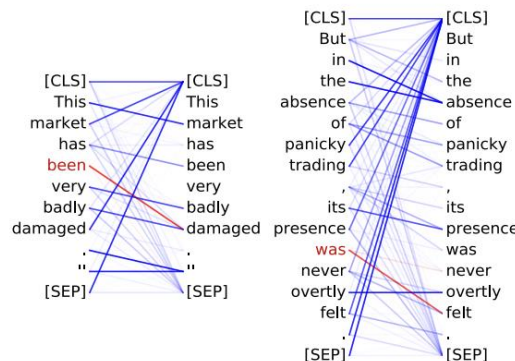
Head 7-6

- **Possessive pronouns** and apostrophes attend to the head of the corresponding NP
- 80.5% accuracy at the poss relation



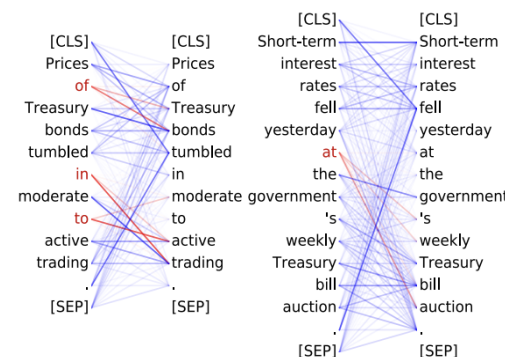
Head 4-10

- **Passive auxiliary verbs** attend to the verb they modify
- 82.5% accuracy at the auxpass relation



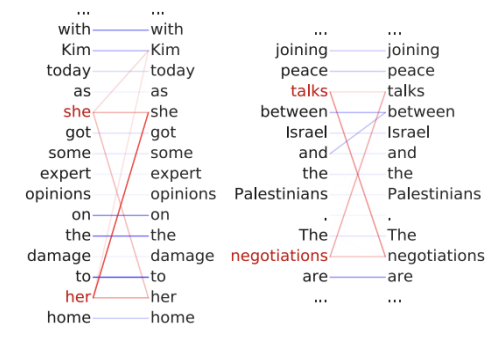
Head 9-6

- **Prepositions** attend to their objects
- 76.3% accuracy at the pobj relation

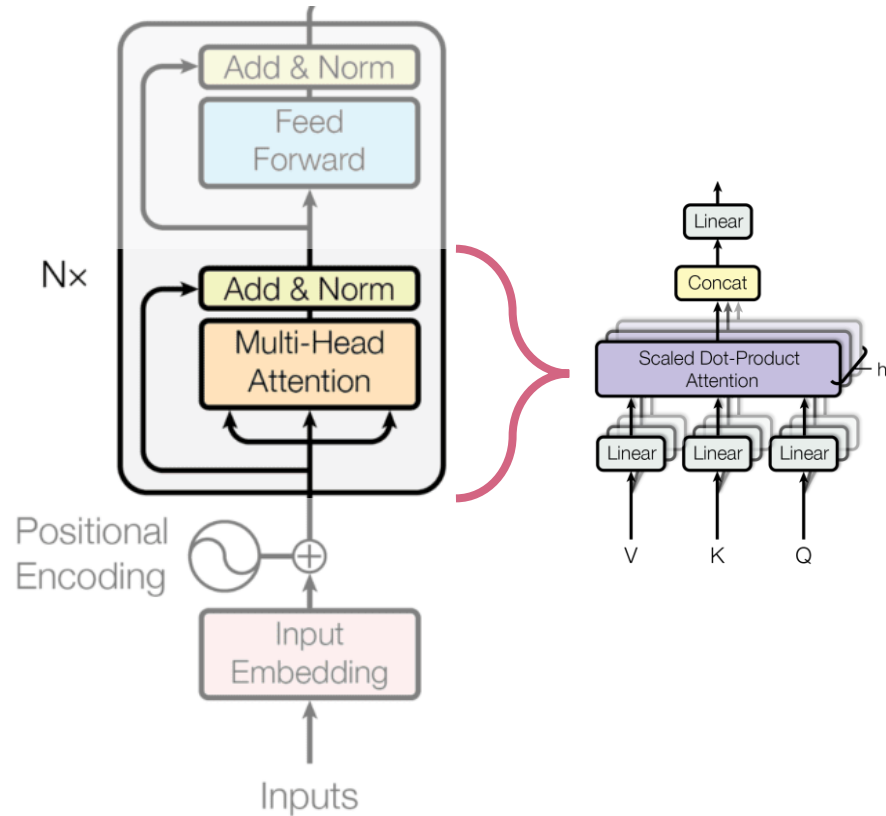


Head 5-4

- **Coreferent mentions** attend to their antecedents
- 65.1% accuracy at linking the head of a coreferent mention to the head of an antecedent



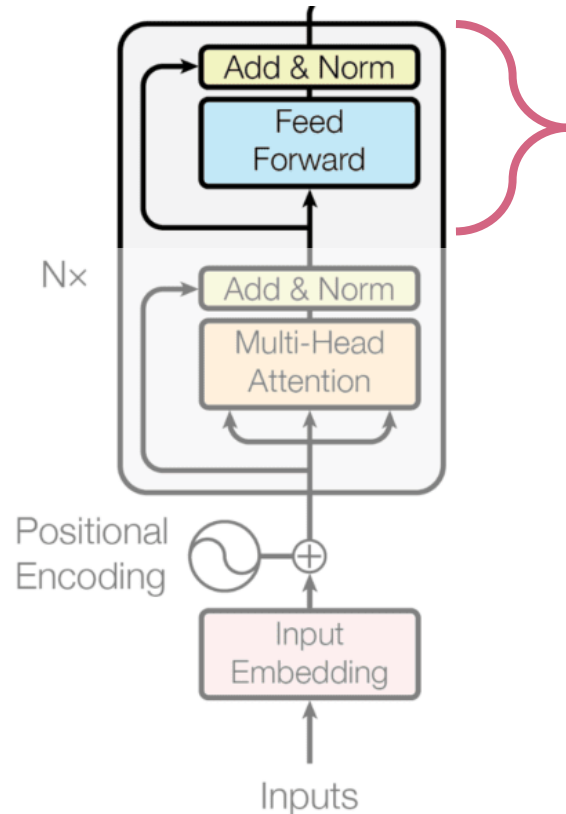
BERT: Residual Connection



5. Add the output of attention block to the initial embeddings

<i>my</i>	<table border="1"><tr><td>7</td><td>1</td><td>2</td></tr></table>	7	1	2	+	<table border="1"><tr><td>.1</td><td>.2</td><td>.5</td></tr></table>	.1	.2	.5	=	<table border="1"><tr><td></td><td></td><td></td></tr></table>			
7	1	2												
.1	.2	.5												
<i>model</i>	<table border="1"><tr><td>9</td><td>2</td><td>2</td></tr></table>	9	2	2	+	<table border="1"><tr><td>.3</td><td>.4</td><td>.4</td></tr></table>	.3	.4	.4	=	<table border="1"><tr><td></td><td></td><td></td></tr></table>			
9	2	2												
.3	.4	.4												
<i>overfit</i>	<table border="1"><tr><td>0</td><td>1</td><td>3</td></tr></table>	0	1	3	+	<table border="1"><tr><td>.1</td><td>.1</td><td>.5</td></tr></table>	.1	.1	.5	=	<table border="1"><tr><td></td><td></td><td></td></tr></table>			
0	1	3												
.1	.1	.5												

BERT: Rest of the Encoder



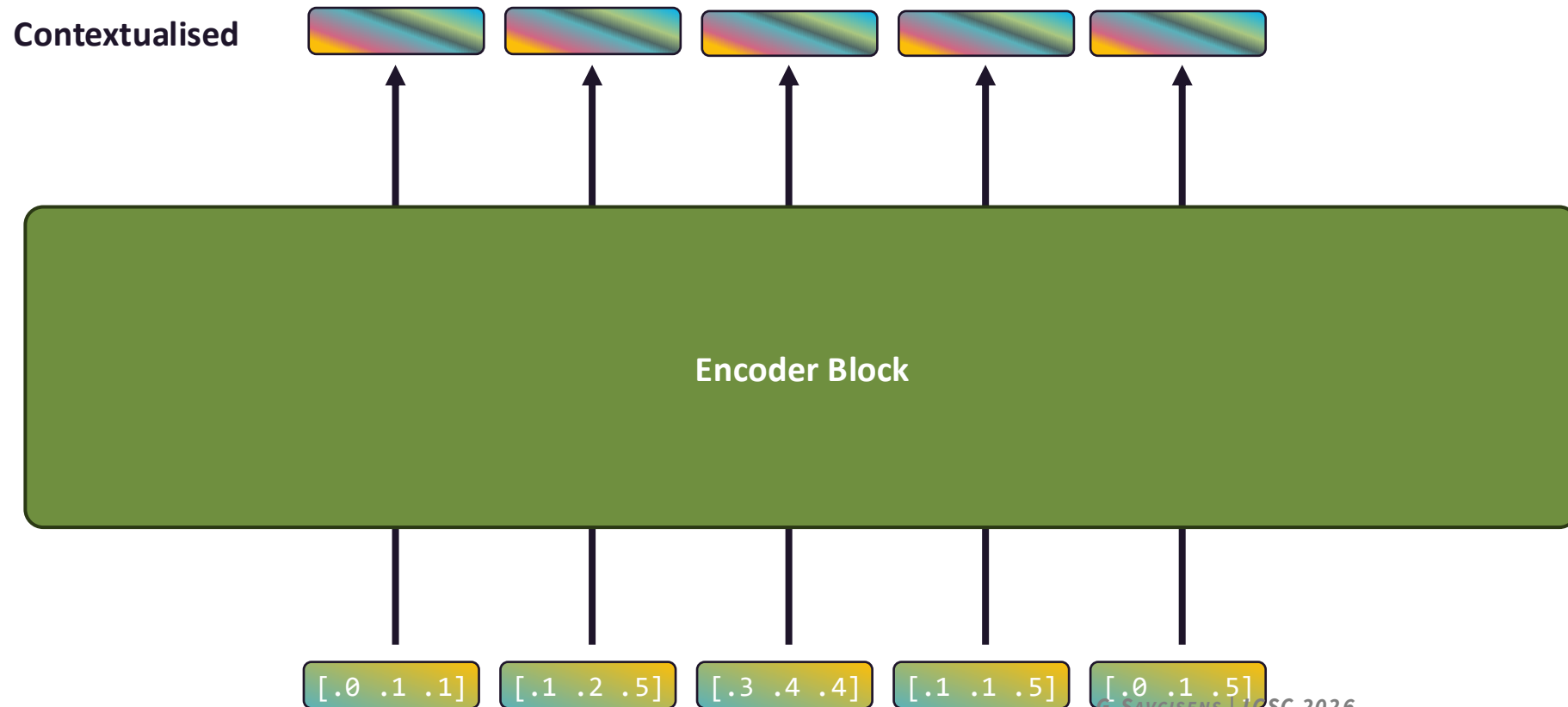
Consists of:

1. Residual connections
2. Position-wise feed forward layer

Here, the information **is not** passed from token to token.

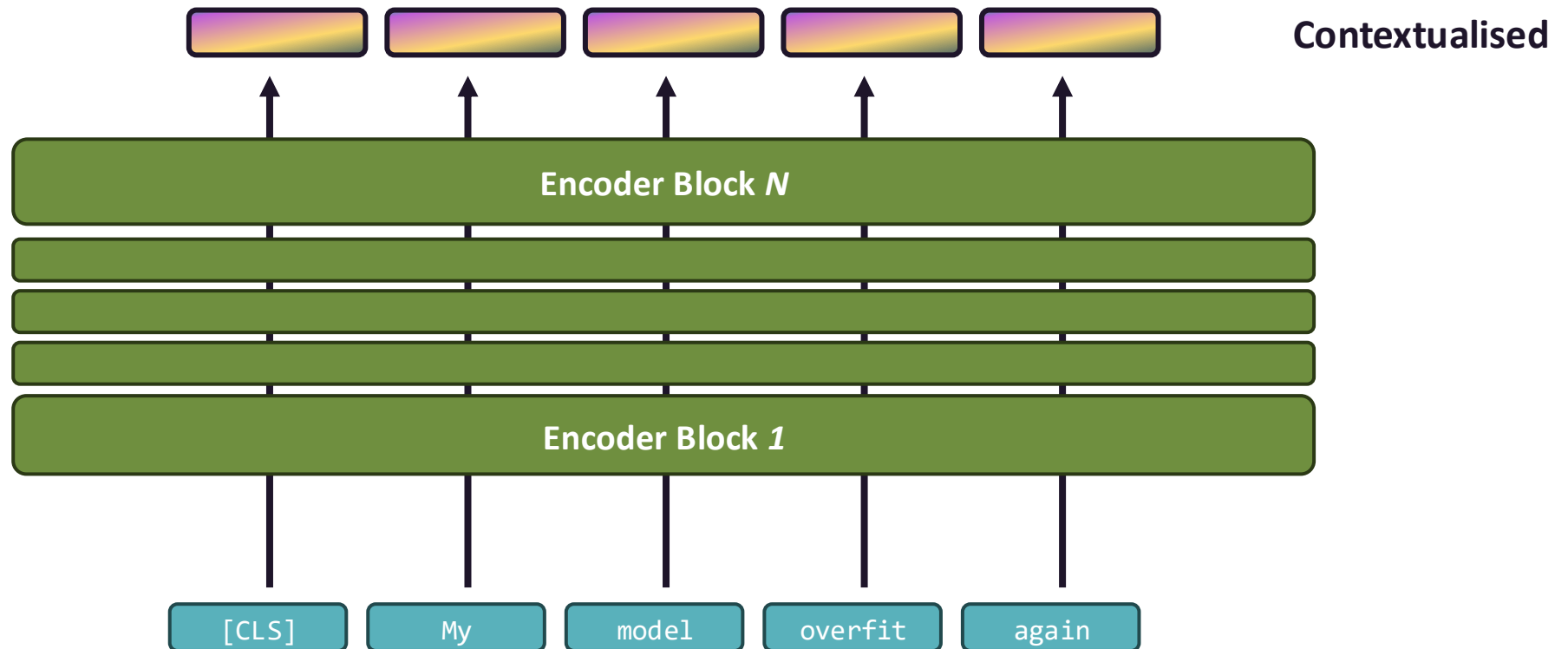
Encoder Output

Representation of the word “My” now contain
Information from other parts of the sequence:
It might contain some aspects of the word “model”



A lot of Encoders

Contextualized token representations **contain rich and nuanced information** about the role of a token in a sequence.



Pen & Paper Exercise

Location: [exercise/pen-paper.md](https://github.com/jackhchenn/exercise/pen-paper.md)

Tasks:

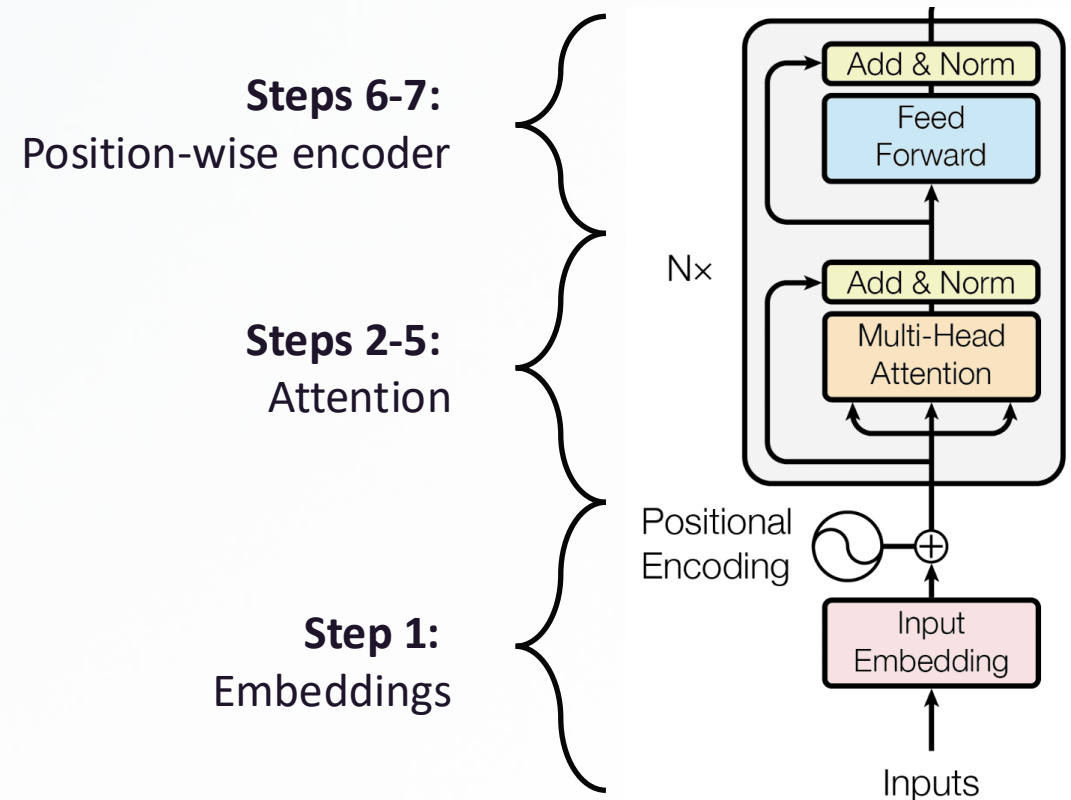
1. Work through the computations of the simplified BERT encoder.
2. Complete steps 1-5
3. If you finish early, go ahead with 6-7.

Time: 15 min

Setup: In Pairs/Groups



Workshop Page



Hands-on Exercise (A2): Embeddings

Location: notebooks/a2-bert-token-embeddings.

Content:

1. Extract and visualize embeddings
2. Use embeddings to estimate word similarities
3. Contextual vs static embeddings

Look for: three "Your turn" cells

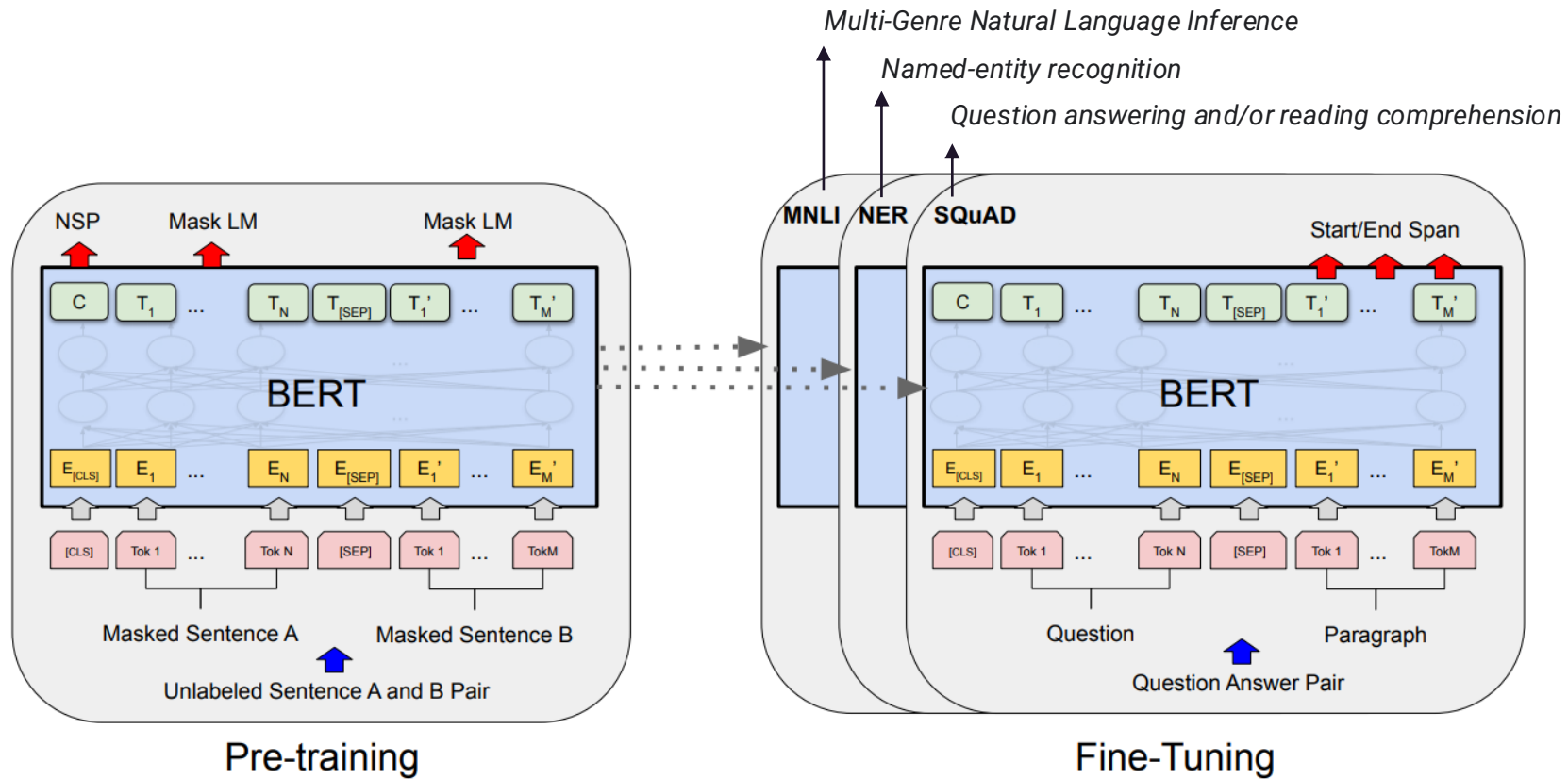
Time: 20 min

Setup: Guided



Workshop Page

Training stages

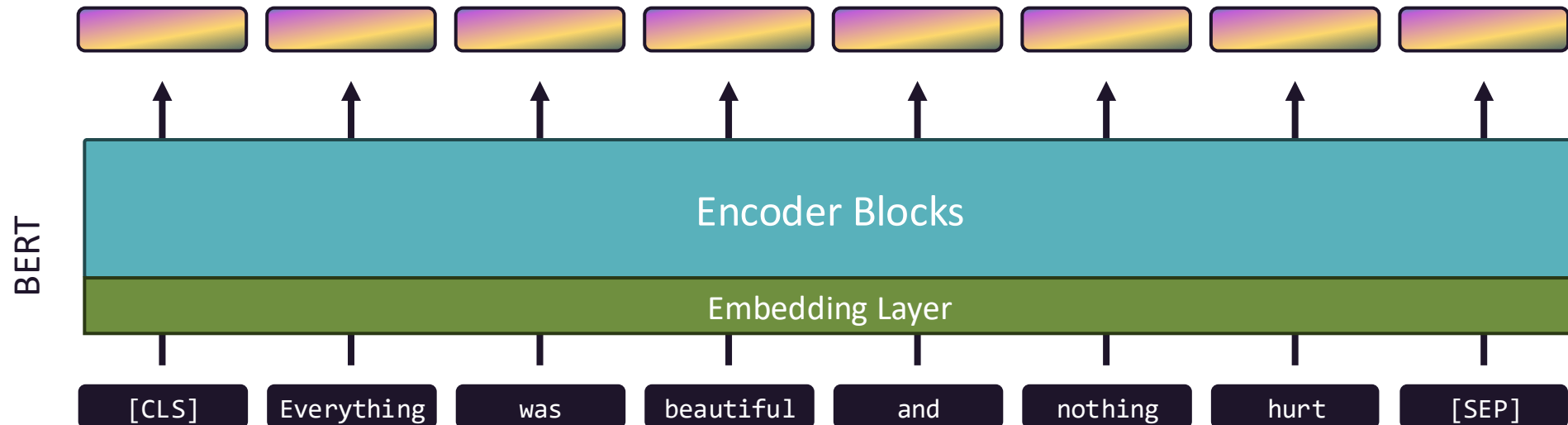


Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding." *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 2019.

Pretraining

Mask 15% of all the tokens (not including **[PAD]**,**[SEP]**,**[CLS]**):

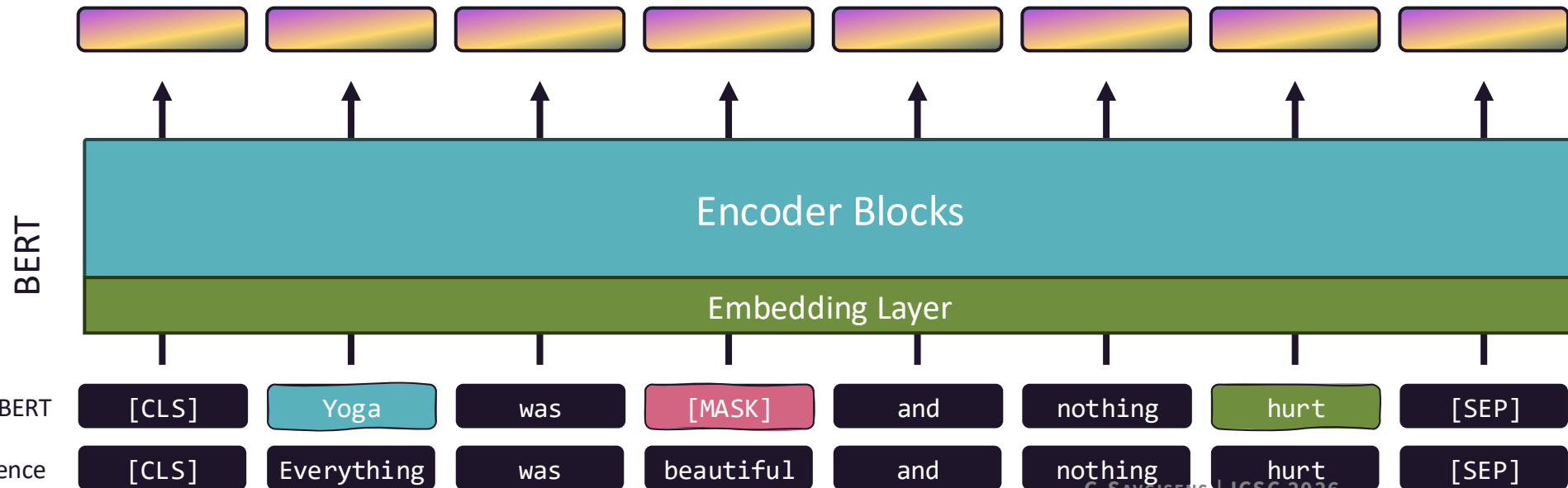
- Keep 10% unchanged
- Substitute 10% with random tokens
- Substitute 80% with **[MASK]** token



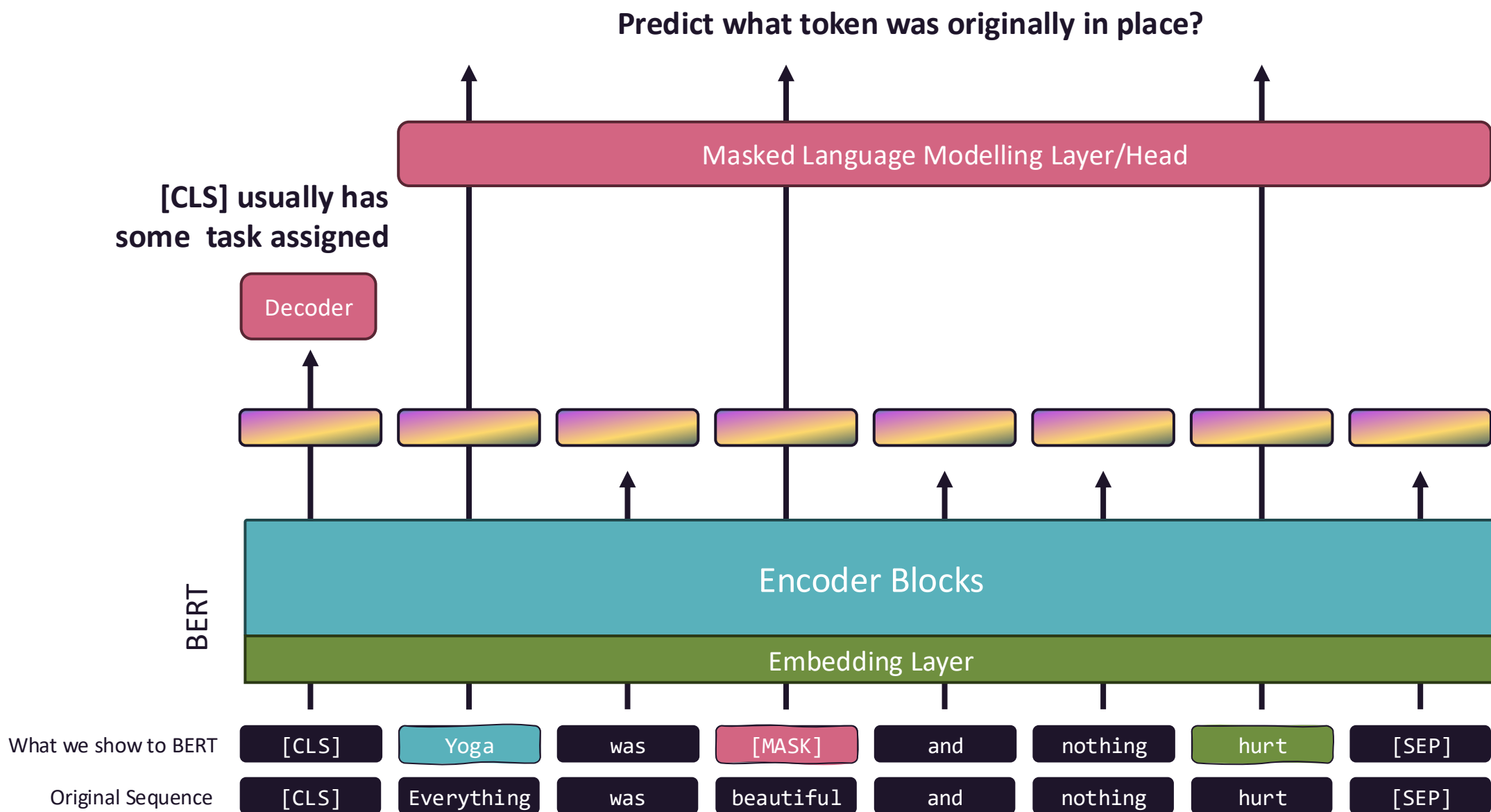
Pretraining

Mask 15% of all the tokens (not including [PAD],[SEP],[CLS]):

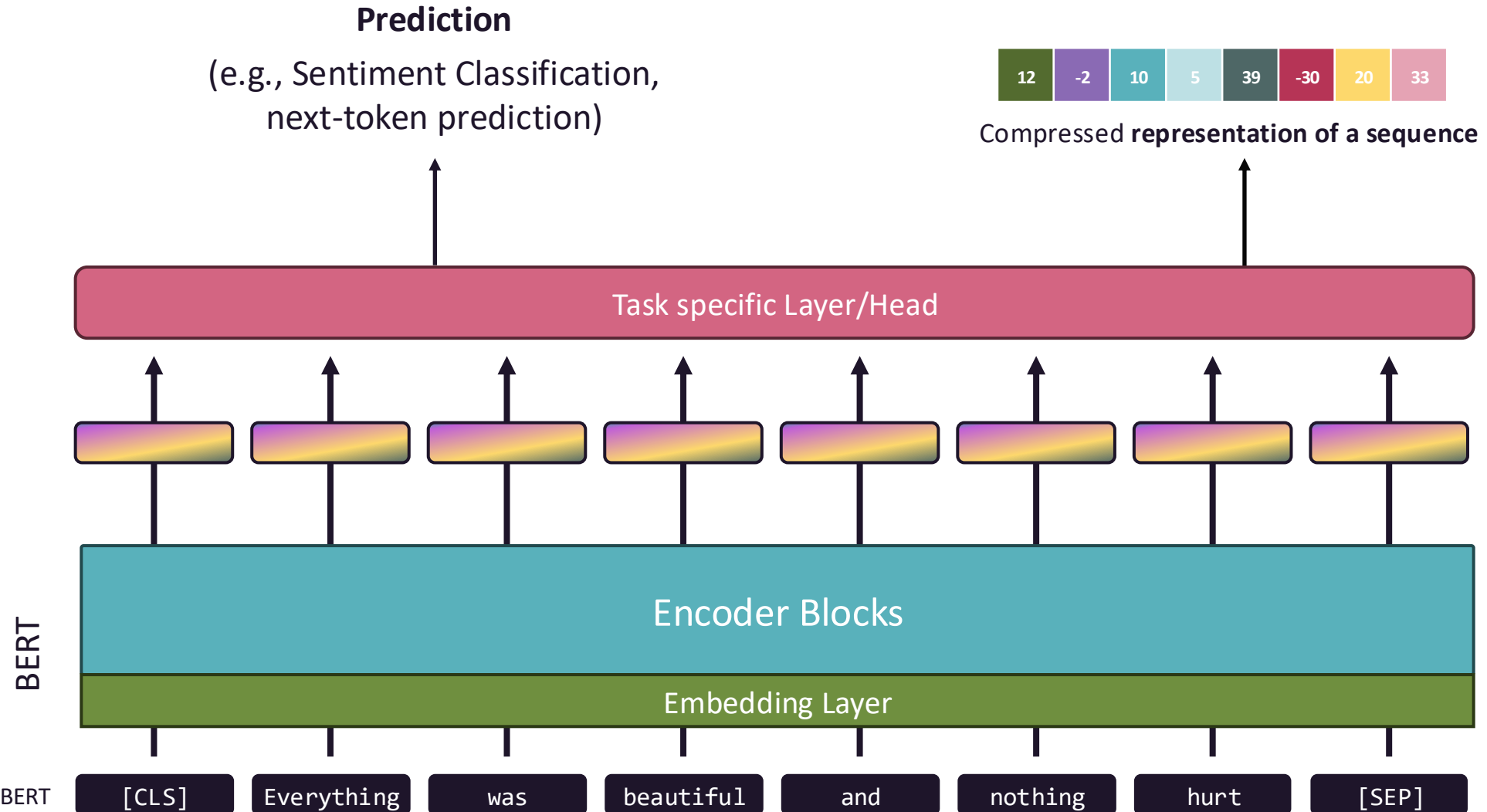
- Keep 10% unchanged
- Substitute 10% with random tokens
- Substitute 80% with [MASK] token



Pretraining



Finetuning



Hands-on Exercise (A3): Probing

Location: notebooks/a3-bert-predictions.

Content:

1. Does BERT embeddings capture truthfulness?
2. Train classification probes on the contextual embeddings.

Time: 15 min

Setup: Guided



Workshop Page

Why borrow from language at all?

A sentence	A life event
a sequence of discrete symbols	a sequence of discrete events
from a fixed vocabulary	from a fixed set of codes
order changes the meaning	order changes the meaning
the same word means different things in different sentences	the same event means different things in different lives
irregular, redundant, incomplete	irregular, redundant, incomplete
no fixed length	no fixed length

Everything that makes text hard is what makes life-course data hard.

So: use the machinery that was built for it.

Block A: Take aways and Questions

- You ran a **150-million-parameter model** and did five things:
 - (A1) tokenization
 - (A1) mask language modeling
 - (A2) visualized embedding space
 - (A2) built a ruler out of anchor words (+ control)
 - (A3) probed every layer for a property nobody trained it on

After the break: we will focus on the transformers for the life-sequences

Questions?

Break

Until 11:00

Transformers for Life-sequences

Block B

Disclaimer

What we cover:

- Encoder-only transformers
- Transformer finetuning
- Analysis of representations
- Predictions with the representations

What we do *not* cover:

1. Autoregressive transformers (that predict next-tokens)
2. Subword tokenization (used by modern LLMs)
3. Variants of positional encodings (used by modern LLMs)

Case study: **Life2vec**

Article | Published: 18 December 2023

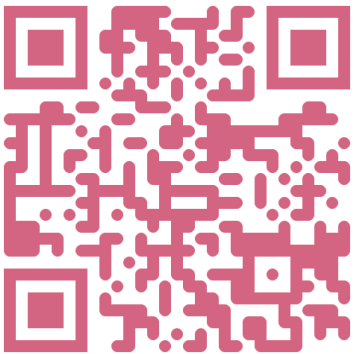
Using sequences of life-events to predict human lives

[Germans Savcicens](#), [Tina Eliassi-Rad](#), [Lars Kai Hansen](#), [Laust Hvas Mortensen](#), [Lau Lilleholt](#), [Anna Rogers](#), [Ingo Zettler](#) & [Sune Lehmann](#) 

[Nature Computational Science](#) **4**, 43–56 (2024) | [Cite this article](#)

33k Accesses | **64** Citations | **2288** Altmetric | [Metrics](#)

Project Page



life2vec provides a novel way to ...

- process **complex, large-scale data** such as **life-sequences**
- predict **life outcomes**
- provide **explainable predictions**

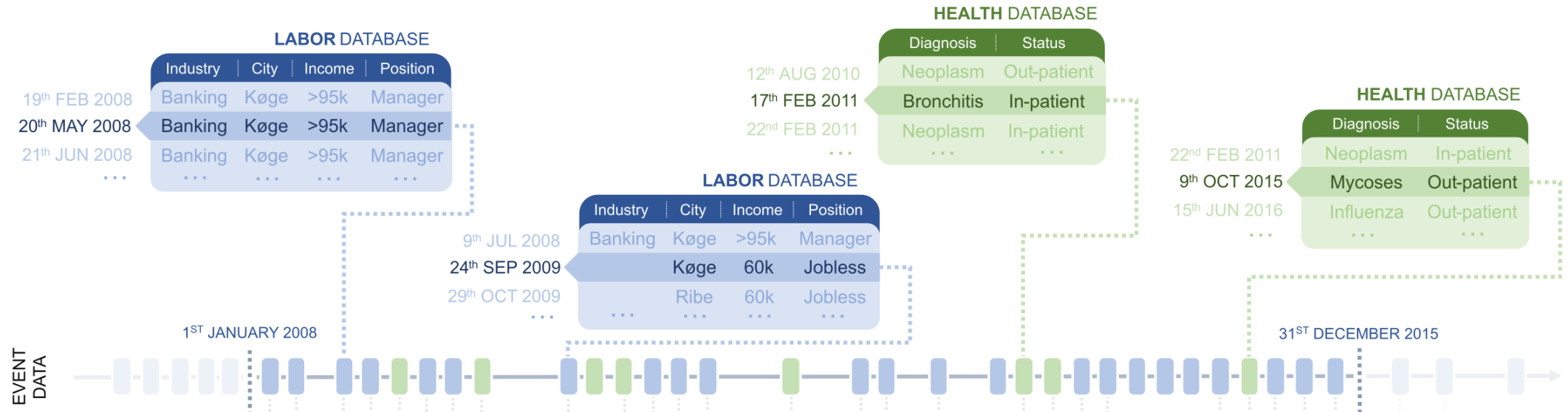
Danish National Register

	People Names, population, health, elections, housing, church, gender equality...
	Social conditions Criminal offences, social benefits for senior citizens, cash benefits, placements...
	Transport Cars, goods transport, passenger transport, infrastructure, traffic accidents...



	<u>Labour and income</u> <u>Employment, unemployment, earnings, income, wealth...</u>
	Education and research Number of students, education programmes, innovation...
	Culture and leisure Film, media, museums, music, digital behaviour, sports...

**Personal raw data is
tied to the Social Security Number (CPR)**

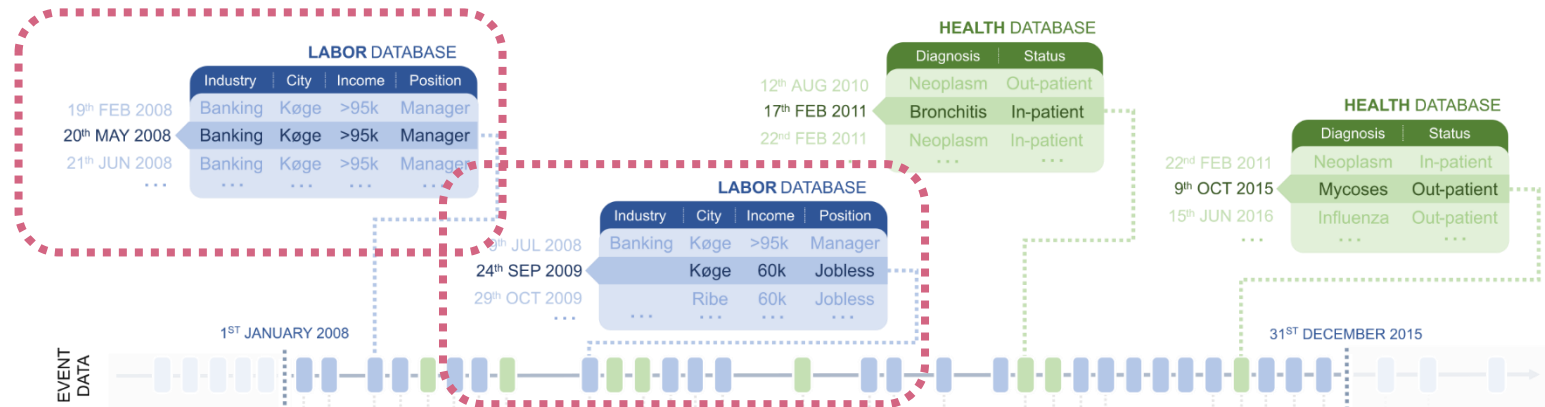


Labour Data

Health Data

Detailed reconstruction of labor
and health trajectories

Labour Data



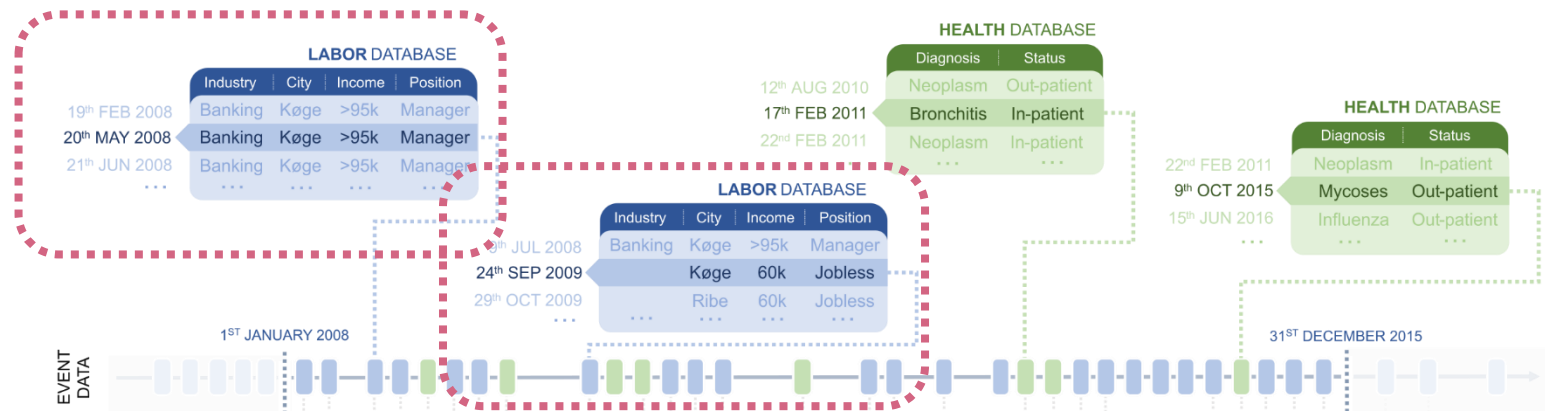
Records of any reported and taxable income:

1. Each record has around 70 features
2. Hourly precision
3. Features based on hierarchical categories
4. *Timespan*: 2008-2020

We focus on:

- *Income*
- *Residence*: Country of Origin / Citizenship / Address
- *Socio-economic status*: Age / sex / Employment status, etc.

Labour Data



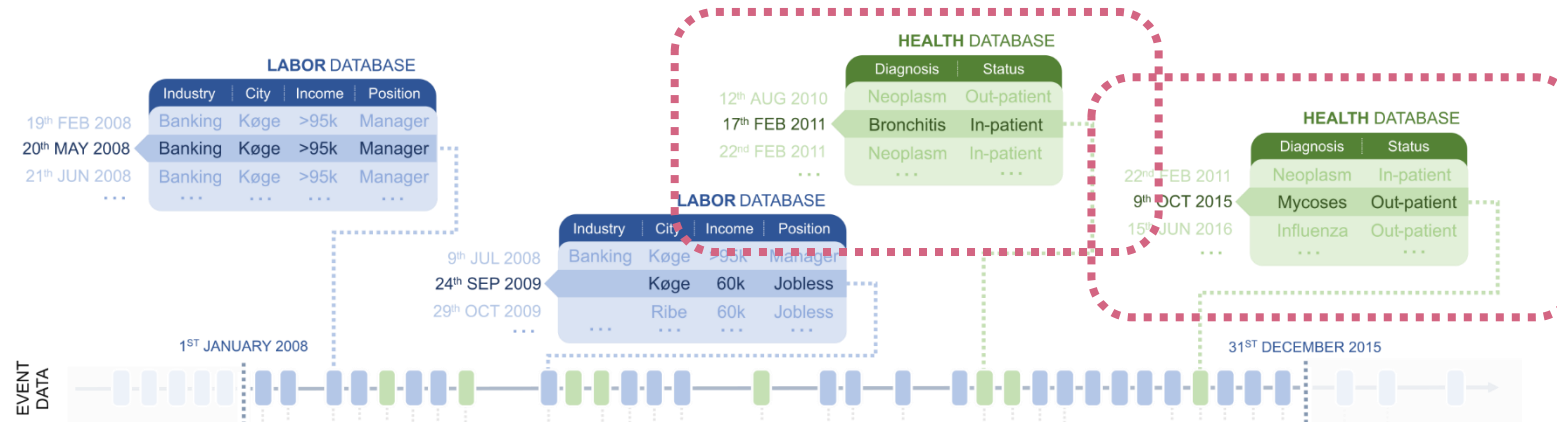
Example of codes describing the **Industry**

DB07 Code	Interpretation
C	Manufacturing
18	Printing and Reproduction of Recorded Media
18.1	Printing and Related Services
18.14	Bookbinding and Similar Services

Example of codes describing the **Occupation**

ISCO-08 Code	Interpretation
2	Professionals
26	Legal, Social and Cultural Professional
265	Creative and Performing Artists
2654	Dancers and Choreographers

Health Data



Records of visits to a health practitioner or hospital:

- Focus on 3 features
- Diagnoses encoded in the ICD10 System

Features we use:

- Diagnosis (Initial, no follow-ups)
- Patient type: inpatient, outpatient, and emergency
- Urgency: Urgent, Non-urgent

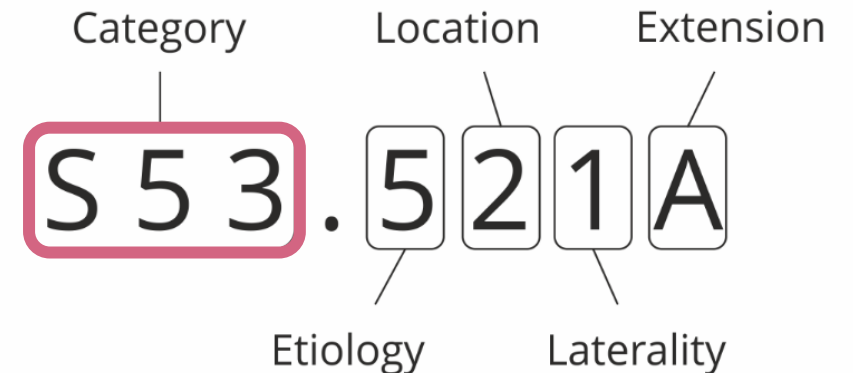
Health Data

ICD-10 Code	Interpretation
S01	Open wound of head
S01.3	Open wound of ear
S01.35	Open bite of ear
S01.352	Open bite of left ear
S01.352D	Open bite of left ear (subsequent encounter)

Examples of ICD10 codes:

- **Y93.D**: Activities involved arts and handcrafts
- **W61.62XD**: Struck by a duck, subsequent encounter
- **H47.51**: Disorders of visual pathways in (due to)
inflammatory disorder

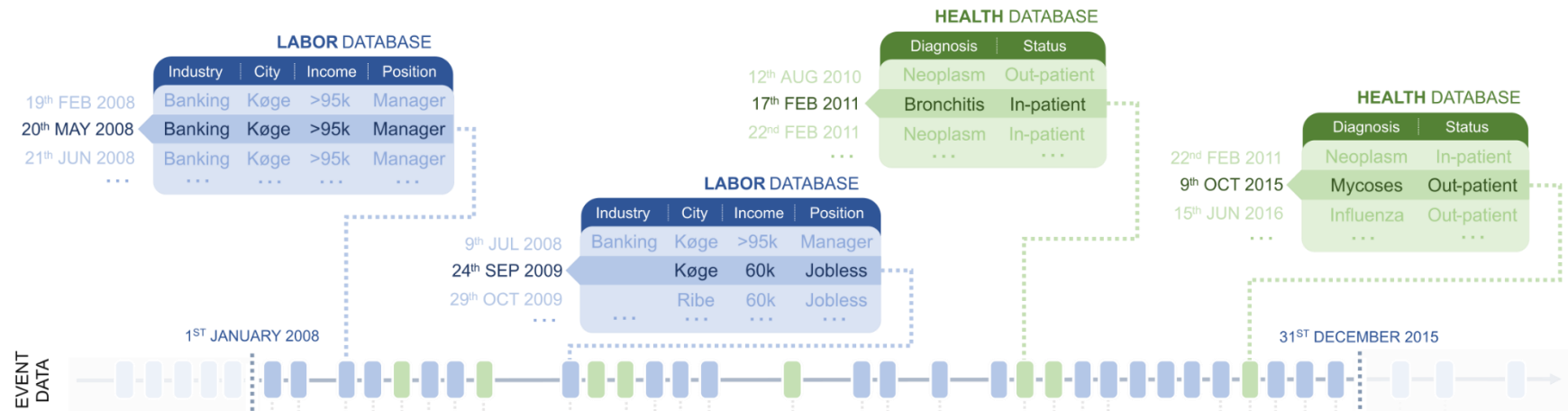
ANATOMY OF AN ICD-10 CODE



ICD-10 code for torus fracture of lower right end of right radius, initial encounter for closed fracture

Power of National Registries

The National Registry is a source of **fine-grained** information about the **progression of one's life**.

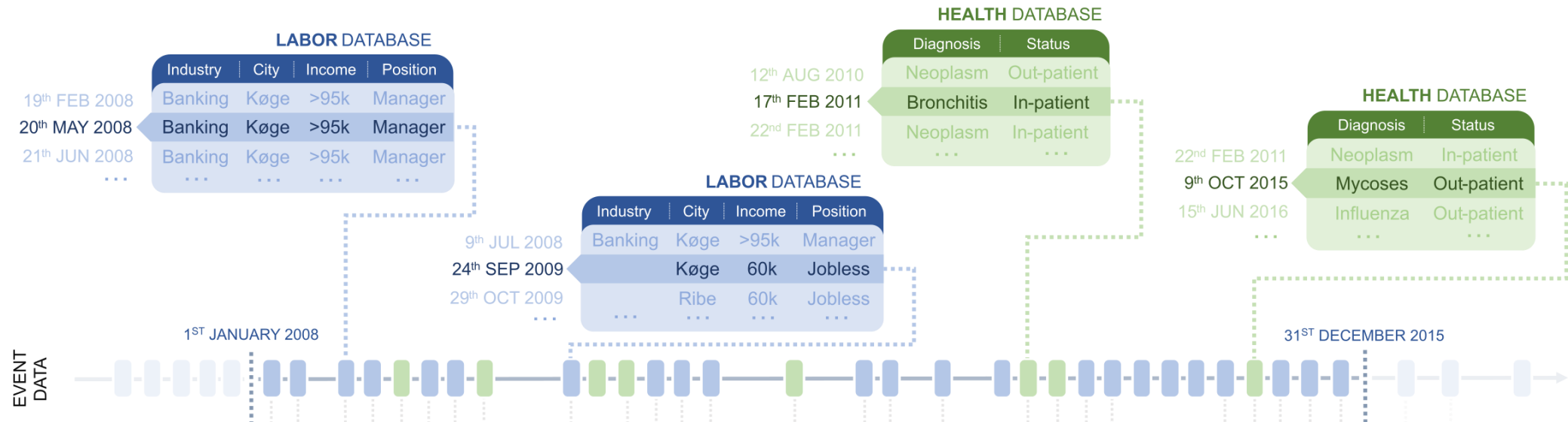


How can we allow for a more **nuanced exploratory analysis** without trading of on the details?

Socio-economic and Health Language

How do we transform long tabular data to sequences?

Forming Language



Tabular to Textual Representation?

Forming Language

LABOR DATABASE				
	Industry	City	Income	Position
19 th FEB 2008	Banking	Køge	>95k	Manager
20 th MAY 2008	Banking	Køge	>95k	Manager
21 th JUN 2008	Banking	Køge	>95k	Manager
...	...	...	...	...

Convey the content in a
spoken language

*"In May 2008, Riley received >95k
as a manager in a Bank."*

Language allows for *flexible* and *nuanced* communication

Forming Language

LABOR DATABASE				
	Industry	City	Income	Position
19 th FEB 2008	Banking	Køge	>95k	Manager
20 th MAY 2008	Banking	Køge	>95k	Manager
21 th JUN 2008	Banking	Køge	>95k	Manager
...	...	...	...	...

Convey the content in a
spoken language

***"In May 2008, Riley received >95k
as a manager in a Bank."***

Language allows for *flexible* and *nuanced* communication

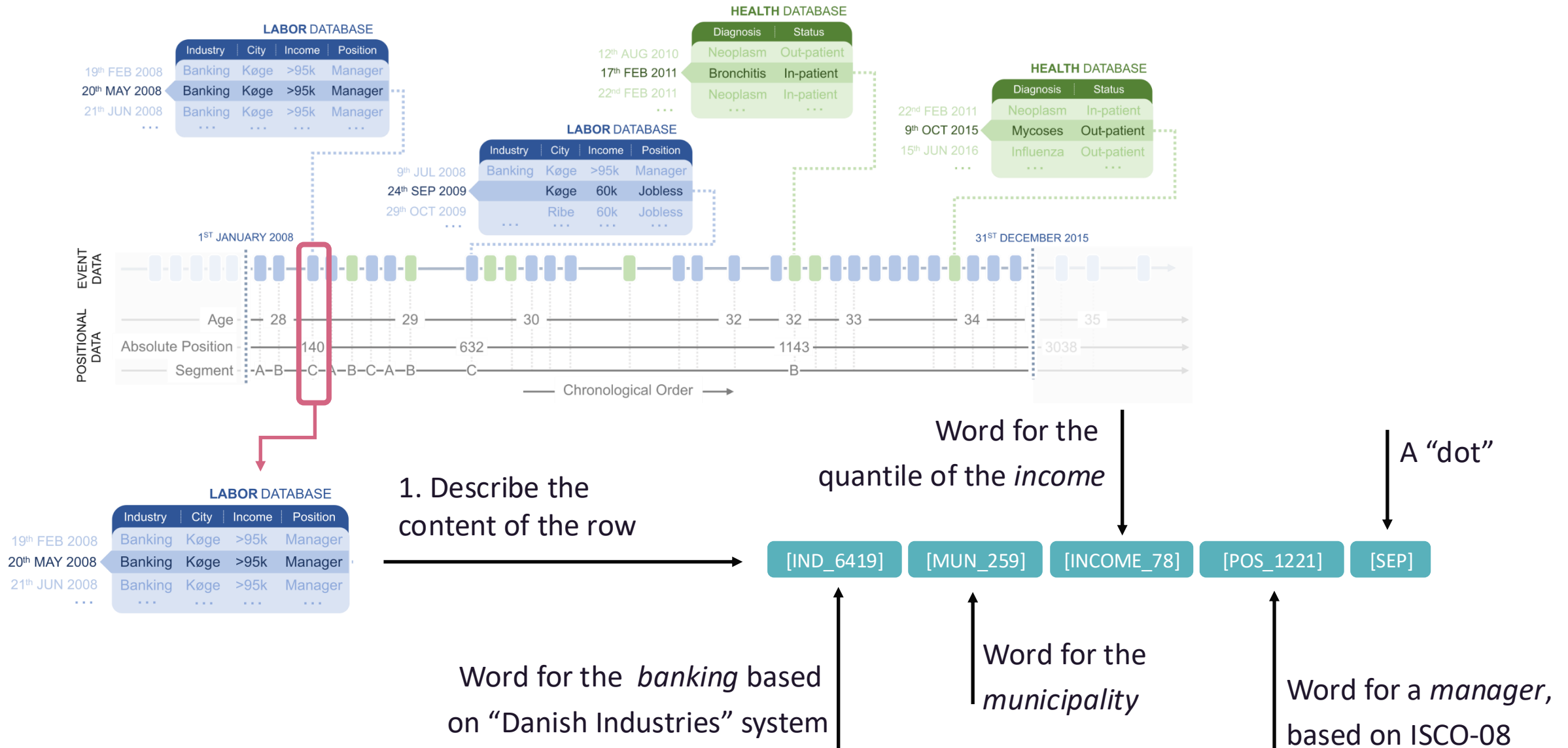
Not all the structures in the English language are of interest to us.

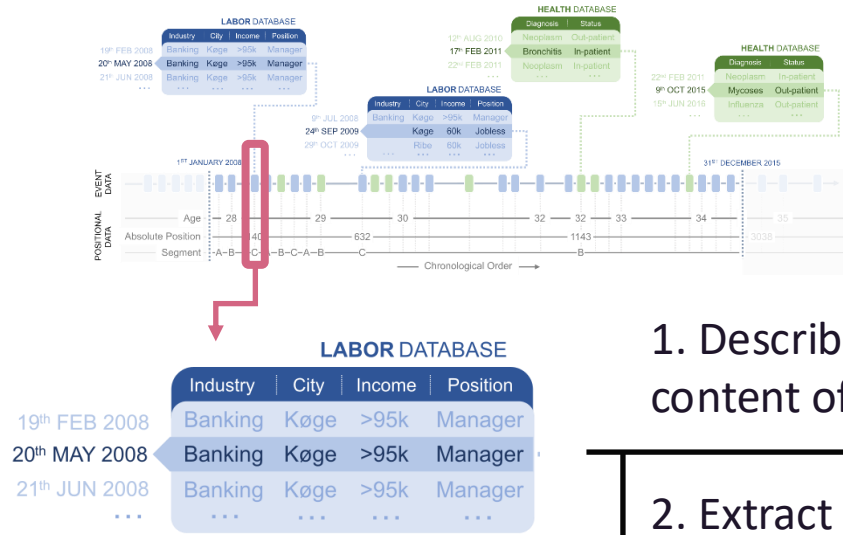
Forming Language

LABOR DATABASE				
	Industry	City	Income	Position
19 th FEB 2008	Banking	Køge	>95k	Manager
20 th MAY 2008	Banking	Køge	>95k	Manager
21 th JUN 2008	Banking	Køge	>95k	Manager
...	...	...	...	...

Convey the content in an **artificial
symbolic language**







1. Describe the content of the row

2. Extract positional information about the event

[IND_6419]

[MUN_259]

[INCOME_78]

[POS_1221]

[SEP]

Age: 28

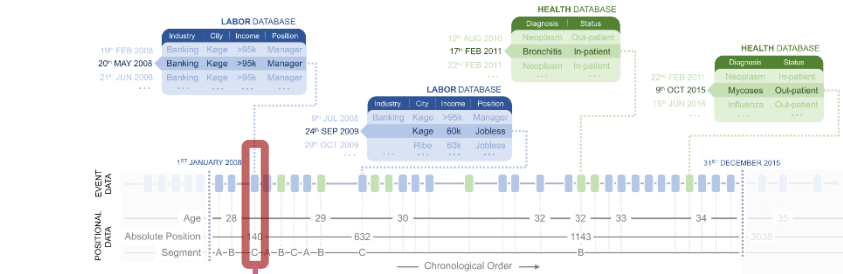
age at the time of the event

Global timestep: 140

number of days since 1st Jan 2008

Segment: C

additional sentence identifier



LABOR DATABASE

Industry	City	Income	Position
Banking	Køge	>95k	Manager
Banking	Køge	>95k	Manager
Banking	Køge	>95k	Manager
...	...	...	...

Age: 28

Global timestep: 140

Segment: C

1. Describe the content of the row

2. Extract positional information about the event

3. Assign this information to tokens

[IND_6419]

[MUN_259]

[INCOME_78]

[POS_1221]

[SEP]

28

28

28

28

28

140

140

140

140

140

C

C

C

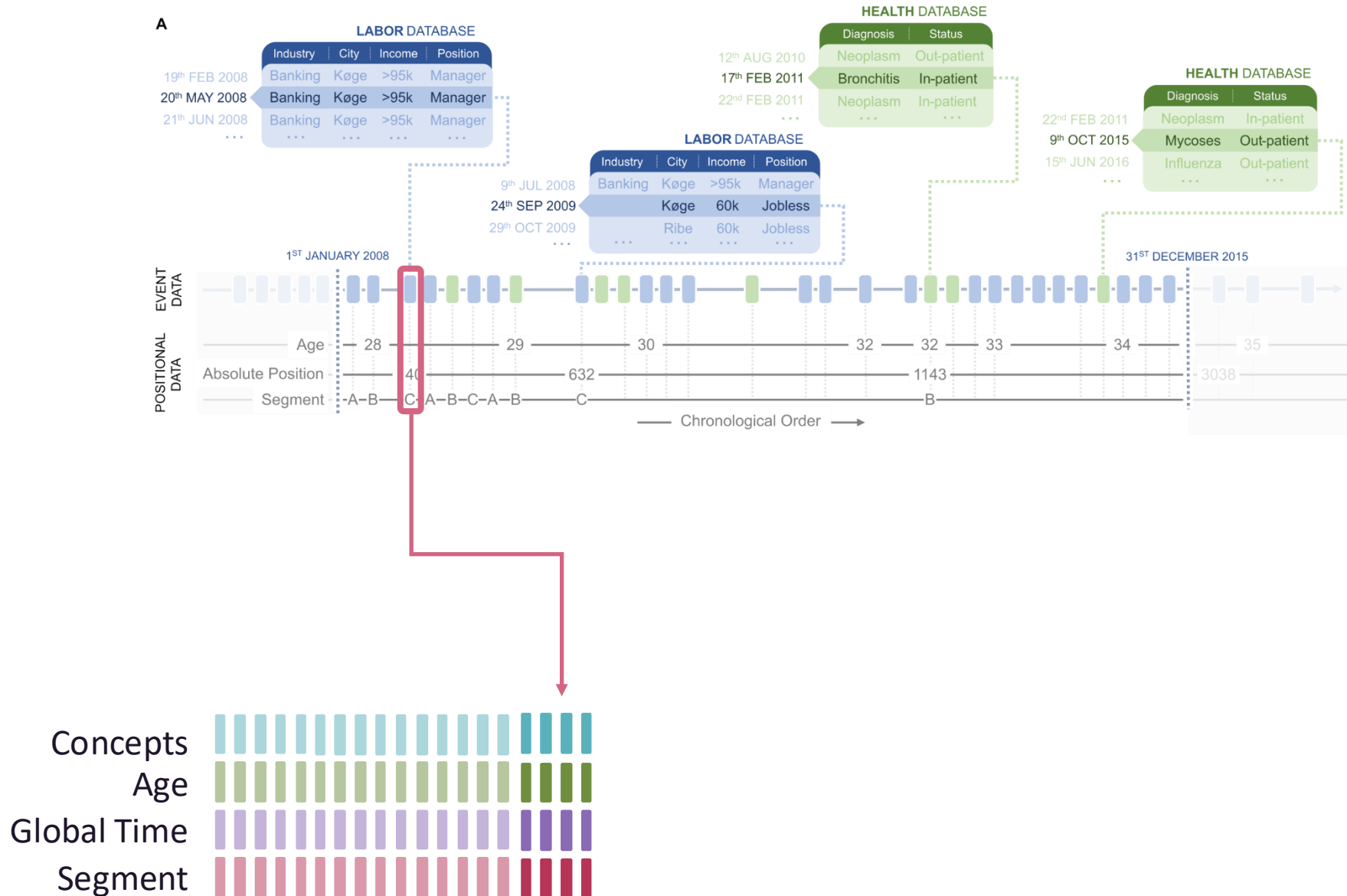
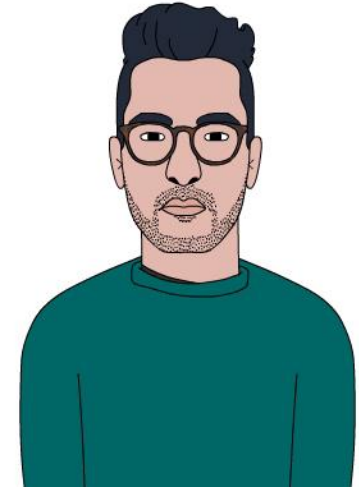
C

C

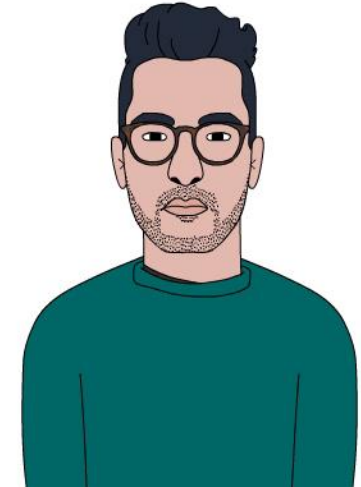
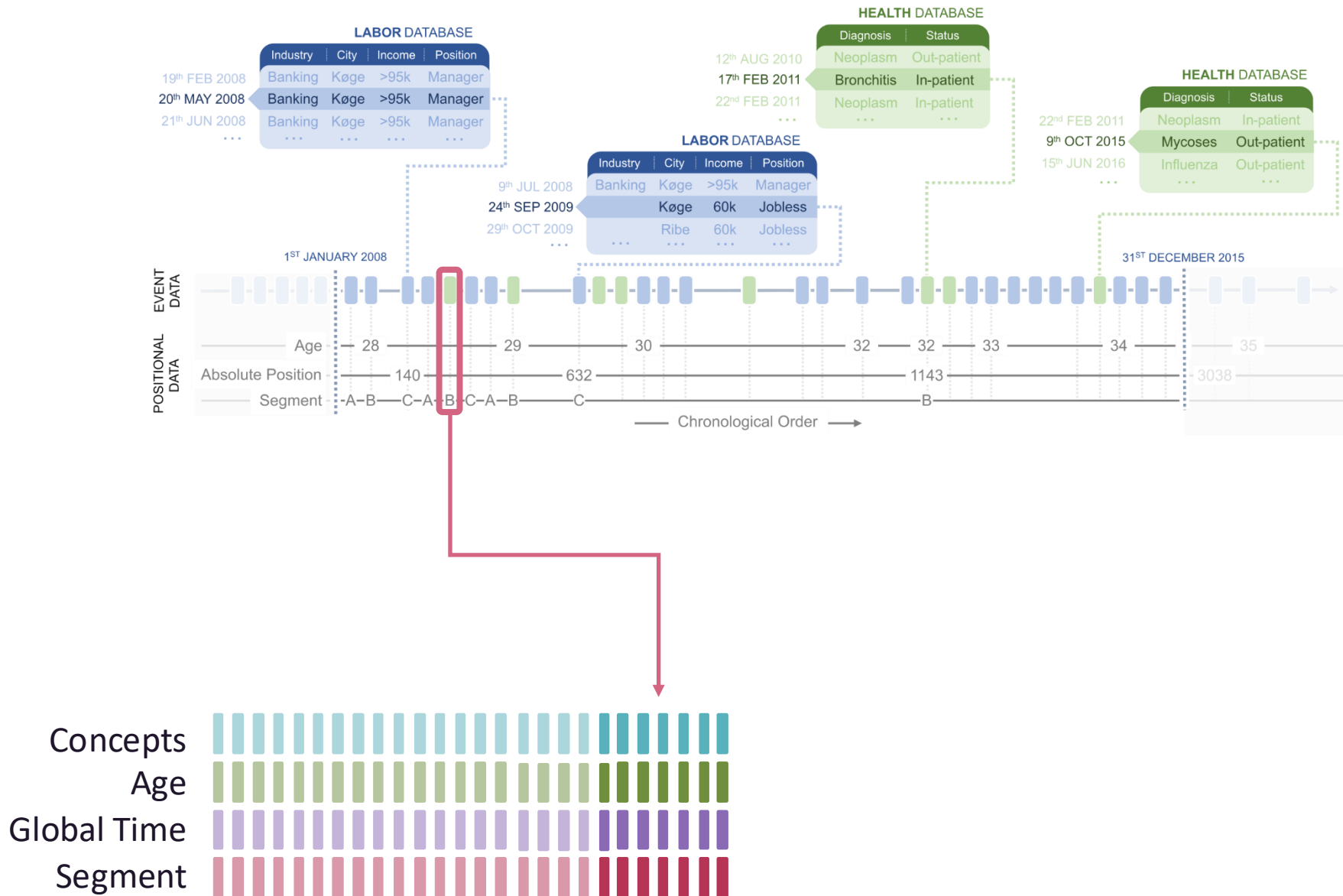
A stylized illustration of a man with dark, wavy hair, thick black-rimmed glasses, and a short, dark beard. He is wearing a teal-colored crewneck sweater. The background is plain white.



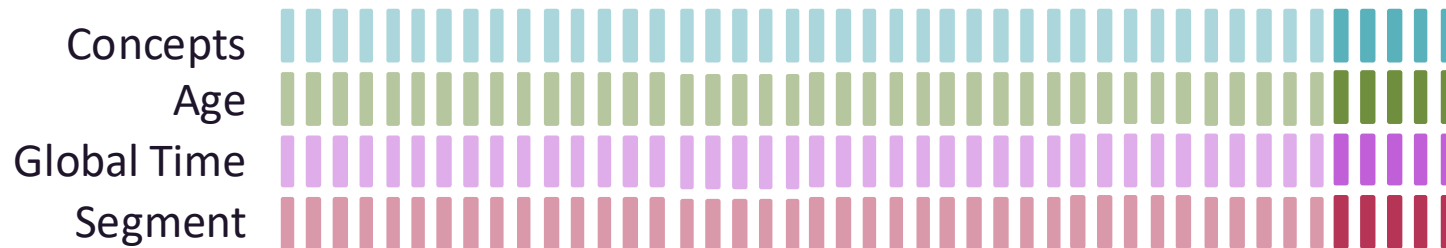
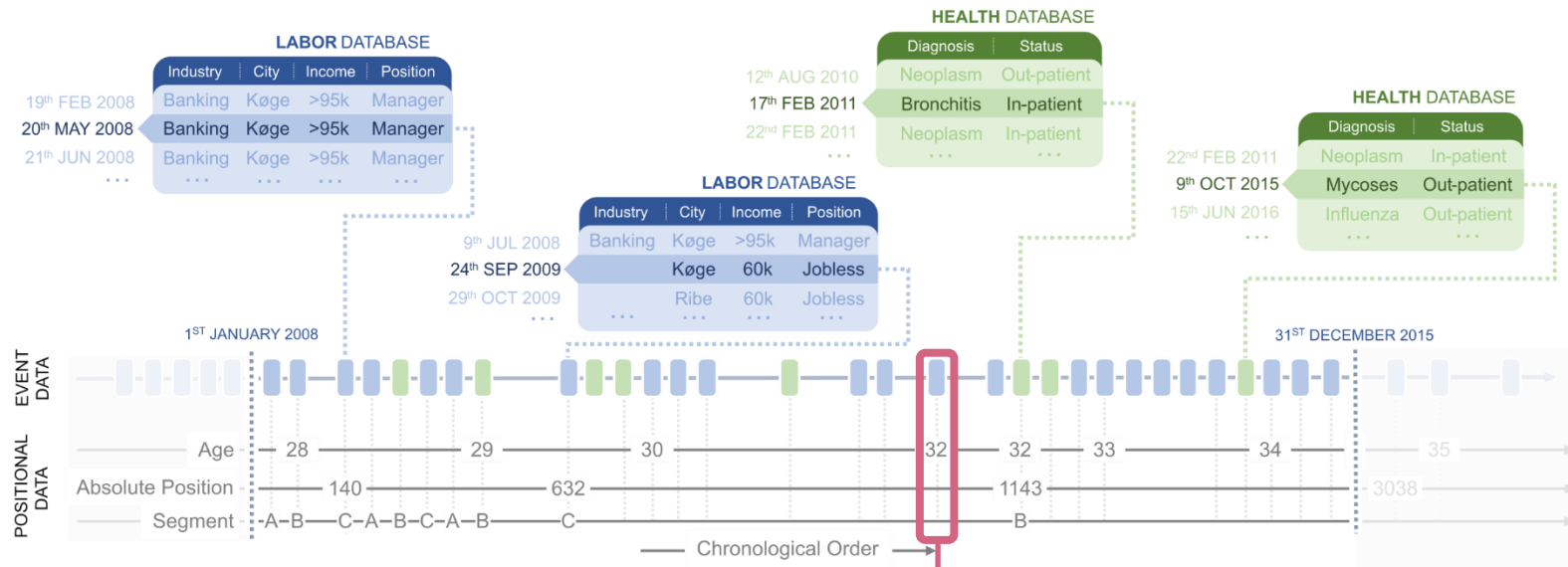
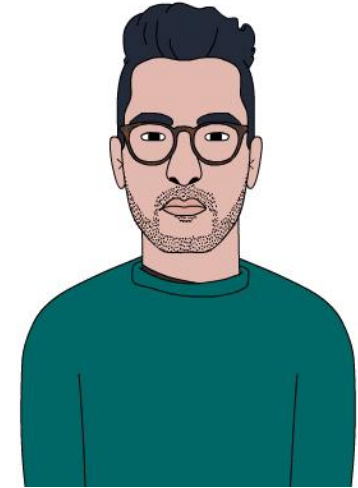
* slightly simplified overview



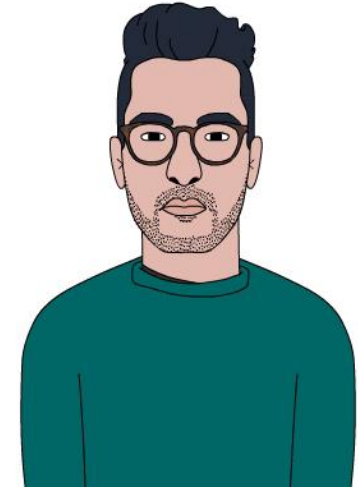
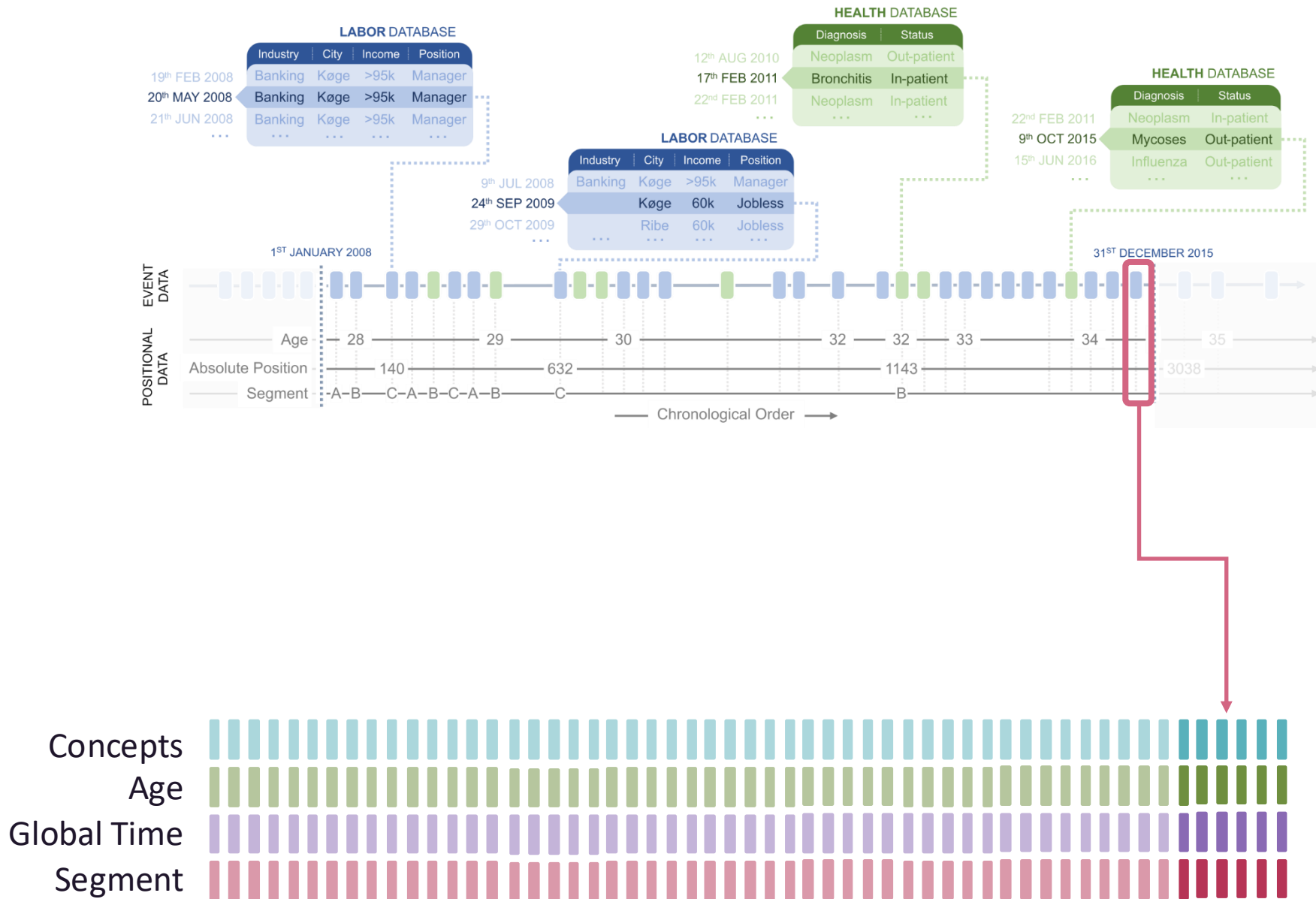
* slightly simplified overview



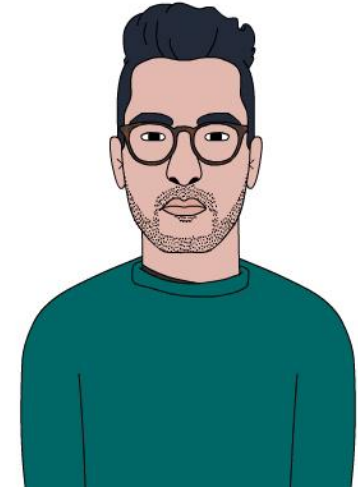
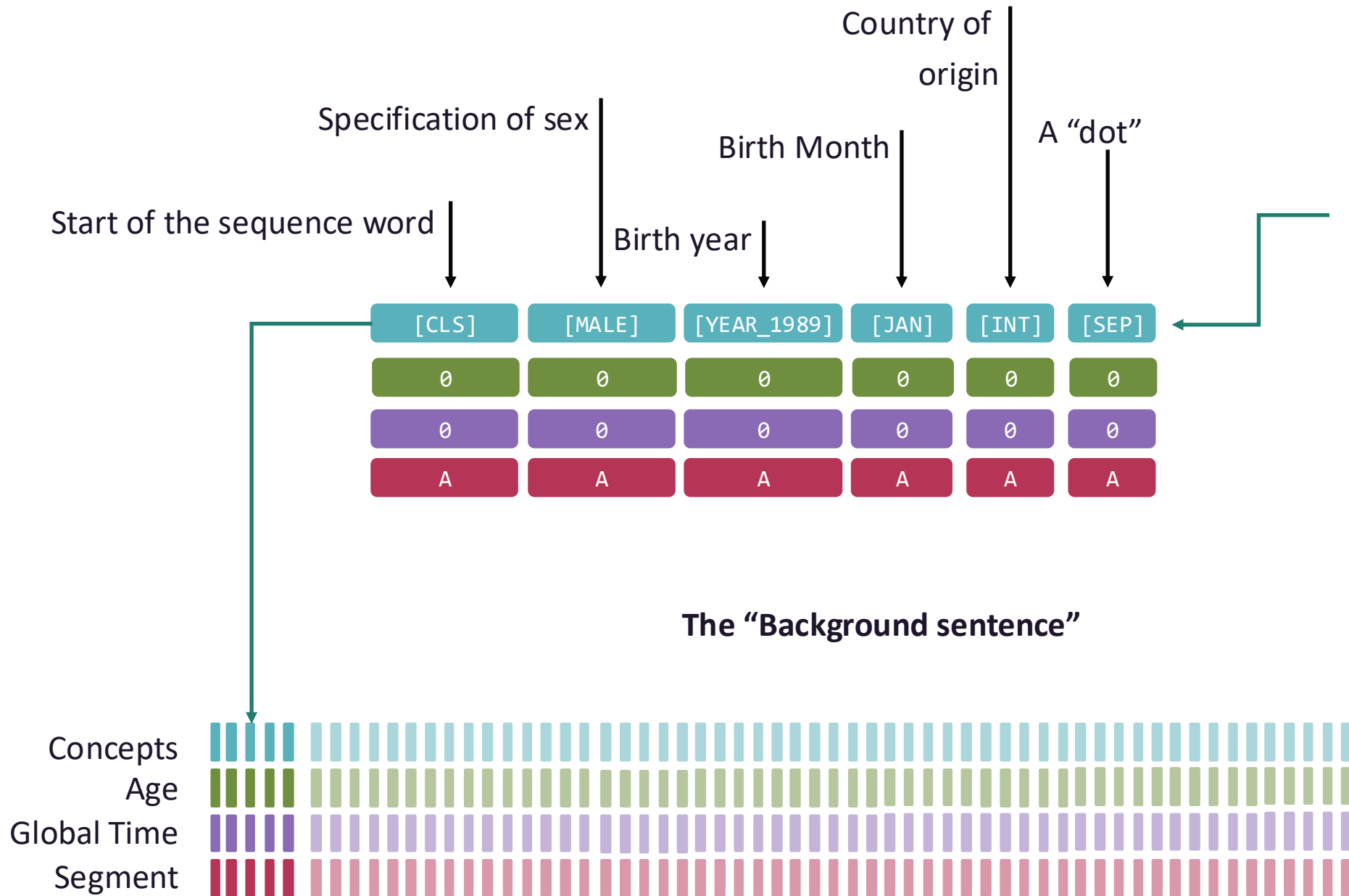
* slightly simplified overview



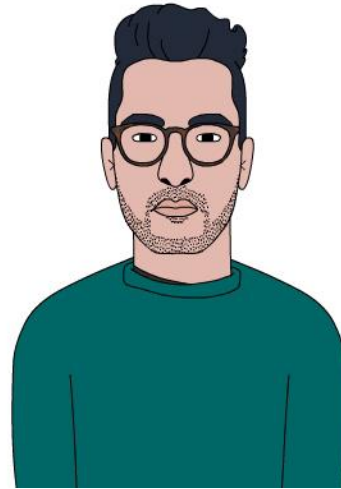
* slightly simplified overview



* slightly simplified overview



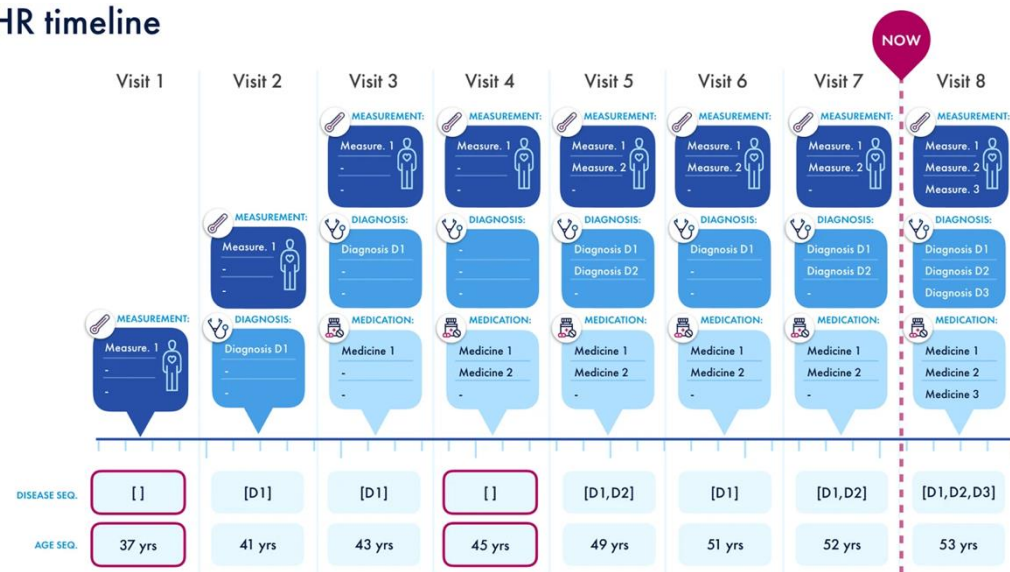
Individual Life Sequence



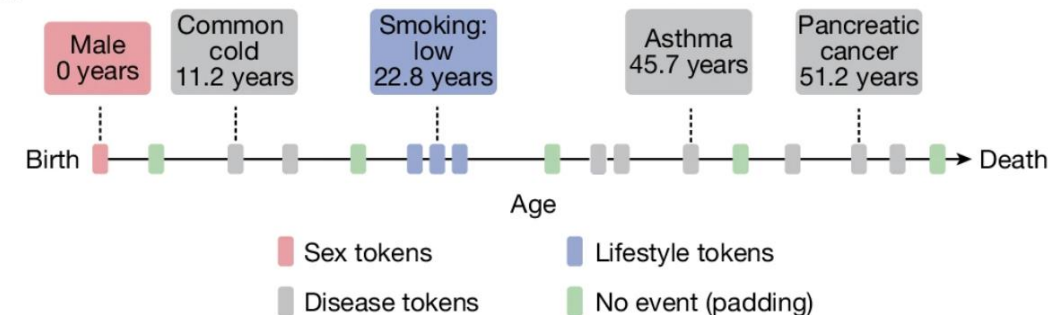
Input to the life2vec model

Other Examples

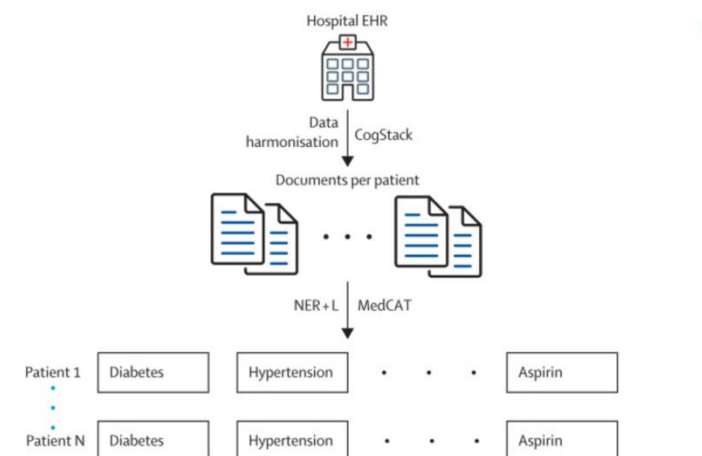
EHR timeline



Li, Y., Rao, S., Solares, J.R.A. *et al.* BEHRT: Transformer for Electronic Health Records. *Sci Rep* **10**, 7155 (2020). <https://doi.org/10.1038/s41598-020-62922-y>

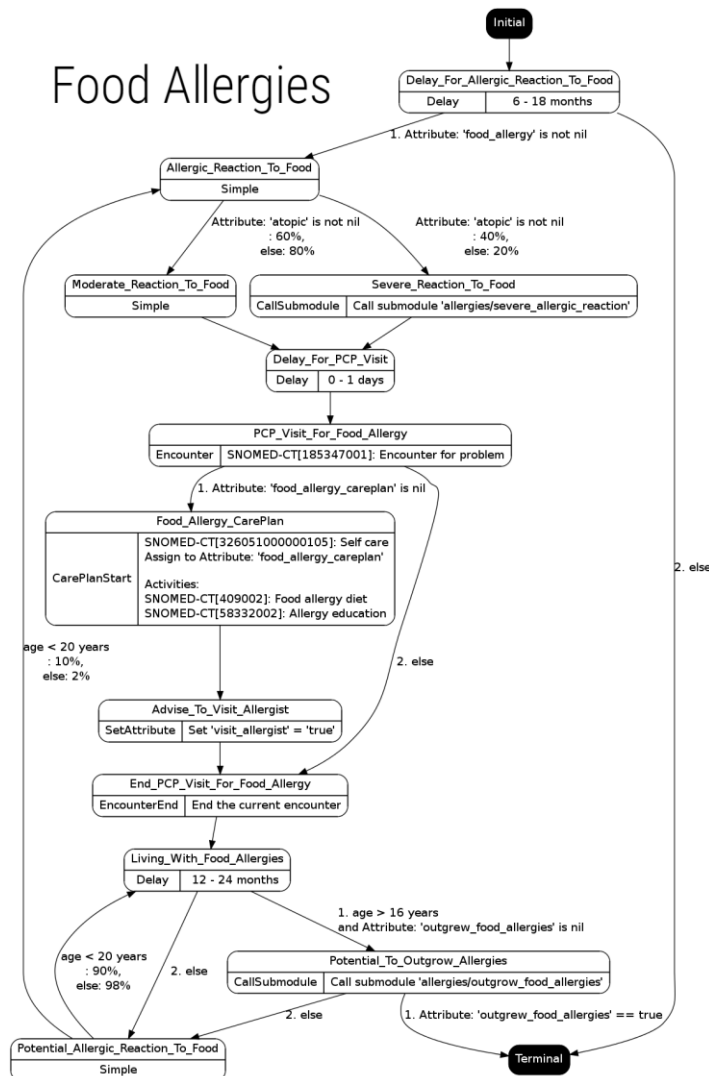


Shmatko, A., Jung, A.W., Gaurav, K. *et al.* Learning the natural history of human disease with generative transformers. *Nature* **647**, 248–256 (2025). <https://doi.org/10.1038/s41586-025-09529-3>



Kraljevic Z, Bean D, Shek A *et al.* **Foresight—a generative pretrained transformer for modelling of patient timelines using electronic health records: a retrospective modelling study** *The Lancet Digital Health*, 6, e281-e290

Our Data: Synthea



We work with the **Synthea** data:

1. Synthetic Patient Records
2. Generates EHRs including *medications, immunization, allergies, conditions, visits*
3. Mimics specific cohorts of people (via rules).

Our generated data mimics:

1. 100k people from Massachusetts, US
2. 10 years of medical history
3. Age range: 35-65
4. End date: Aug 18th, 2026
5. Cutout/Index date is Aug 18th, 2021
6. You will work with a subset of 50k people

Our **Synthea-BERT** Vocabulary

- **1,036 tokens** = 5 special + 46 background + 985 event types
- **Inclusion criteria:** Events occur ≥ 30 times
 - cut-off 176 rarer types map to [UNK] (0.008% of all events)
- **Background:** age band, sex, race, ethnicity, marital status, income in 25 equal-frequency bins.

Hands-on Exercise: (B1) Events to Sequences

Location: `notebooks/b1-events-to-sequences`.

Content:

1. Overview of vocabulary
2. How to convert tables to sequences?
3. How modeling choices affect sequence composition?

Time: 15 min

Setup: Solo run (I am here to help).

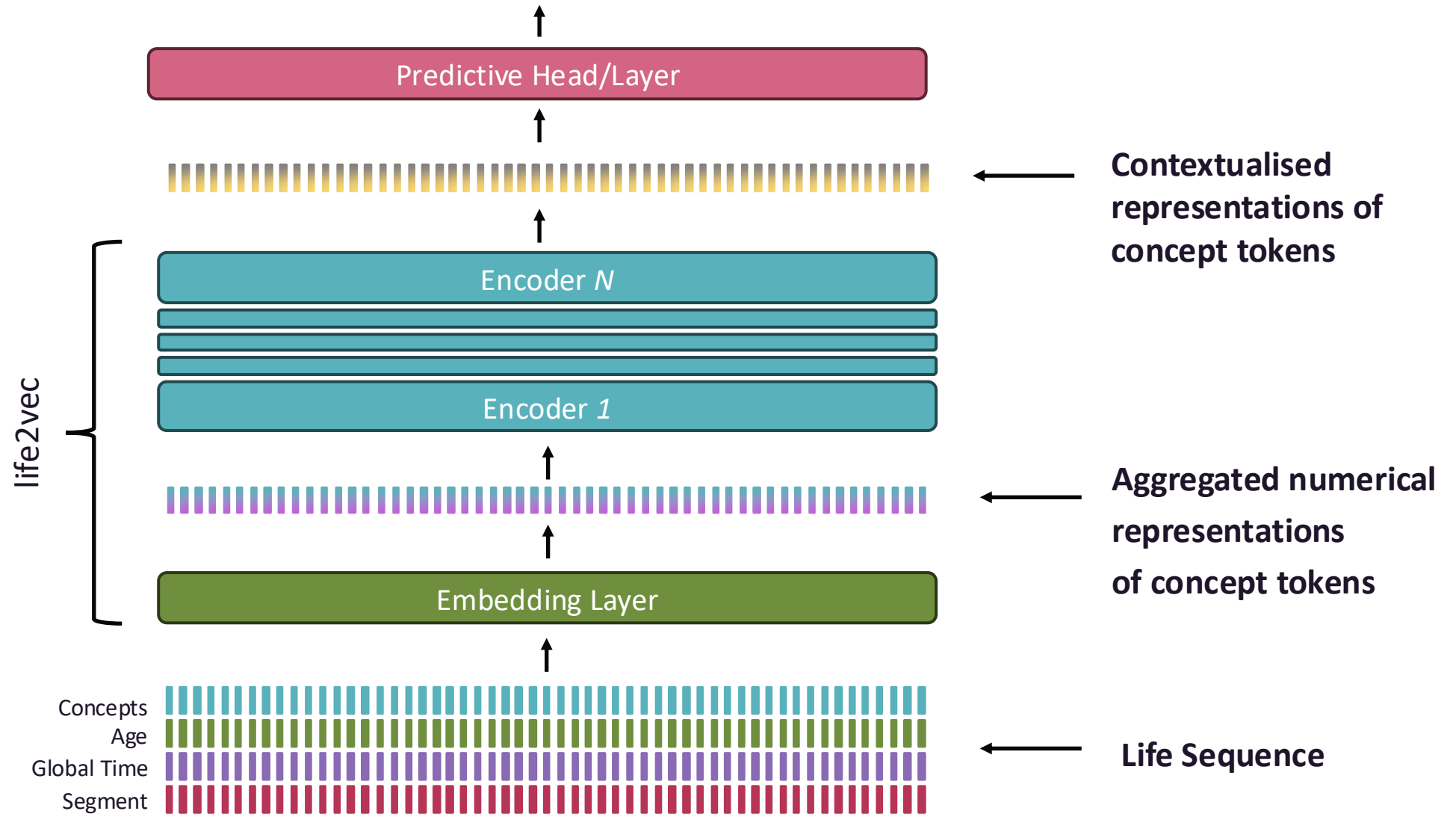
P.S. I precomputed the vocabulary



Workshop Page

Transformers for life-sequences

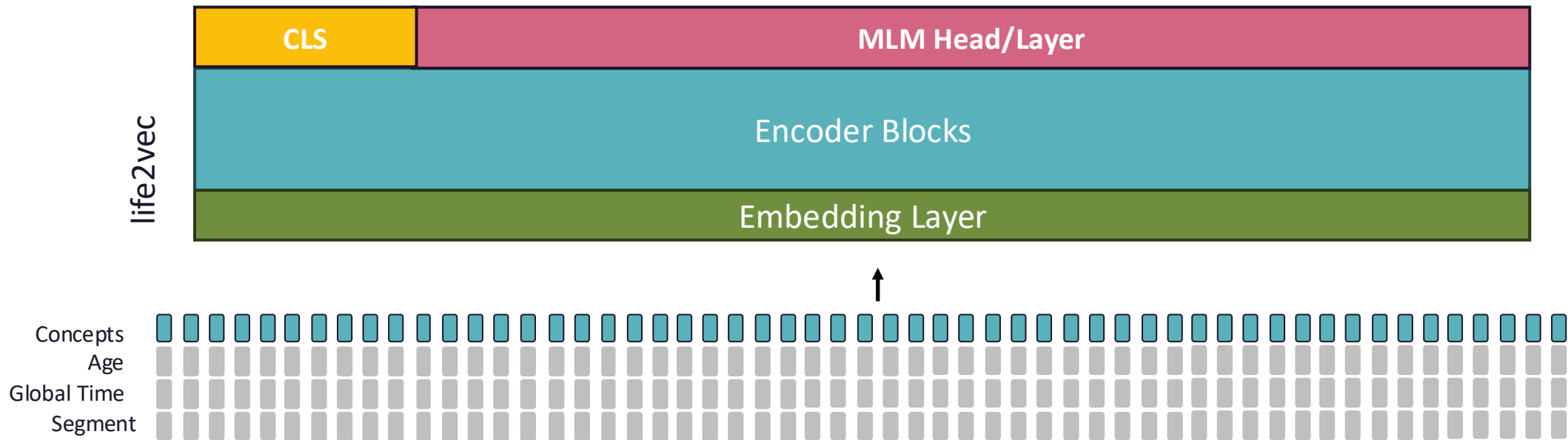
General Pipeline



Life2vec: Pre-training

Mask 15% of all the tokens (not including **[PAD]**,**[SEP]**,**[CLS]**):

- Keep 10% unchanged
- Substitute 10% with random tokens
- Substitute 80% with **[MASK]** token

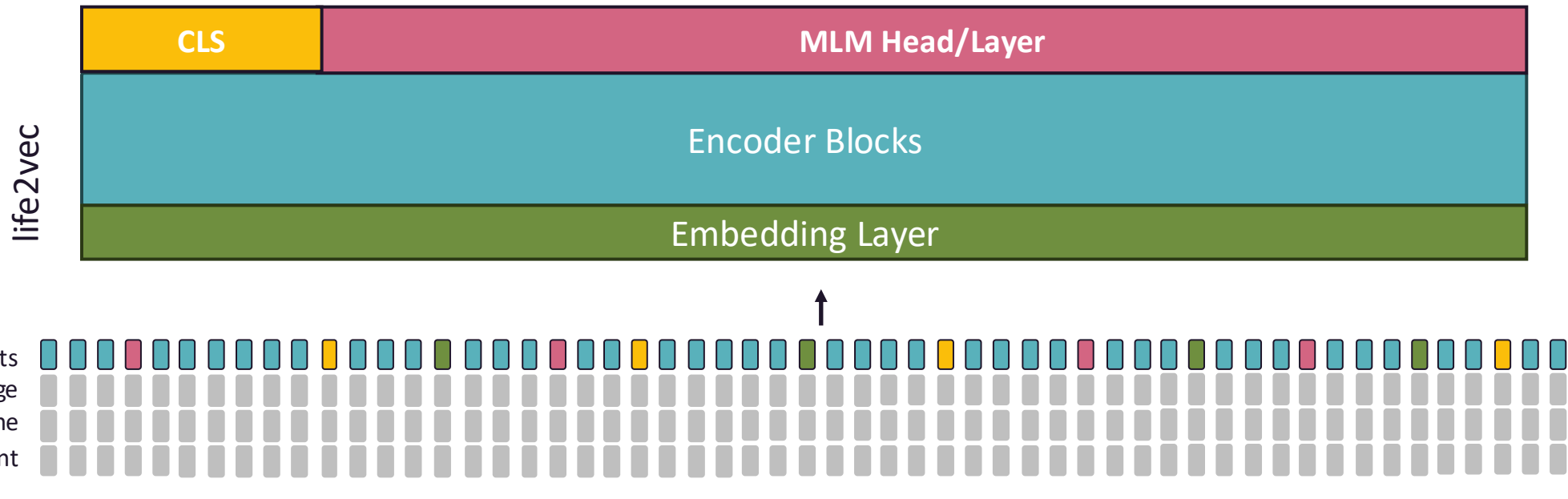


Note: we use our artificial language here, not a natural one (e.g., not English)

Life2vec: Pre-training

Mask 15% of all the tokens (not including **[PAD]**,**[SEP]**,**[CLS]**):

- Keep 10% unchanged
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- Substitute 80% with [MASK] token

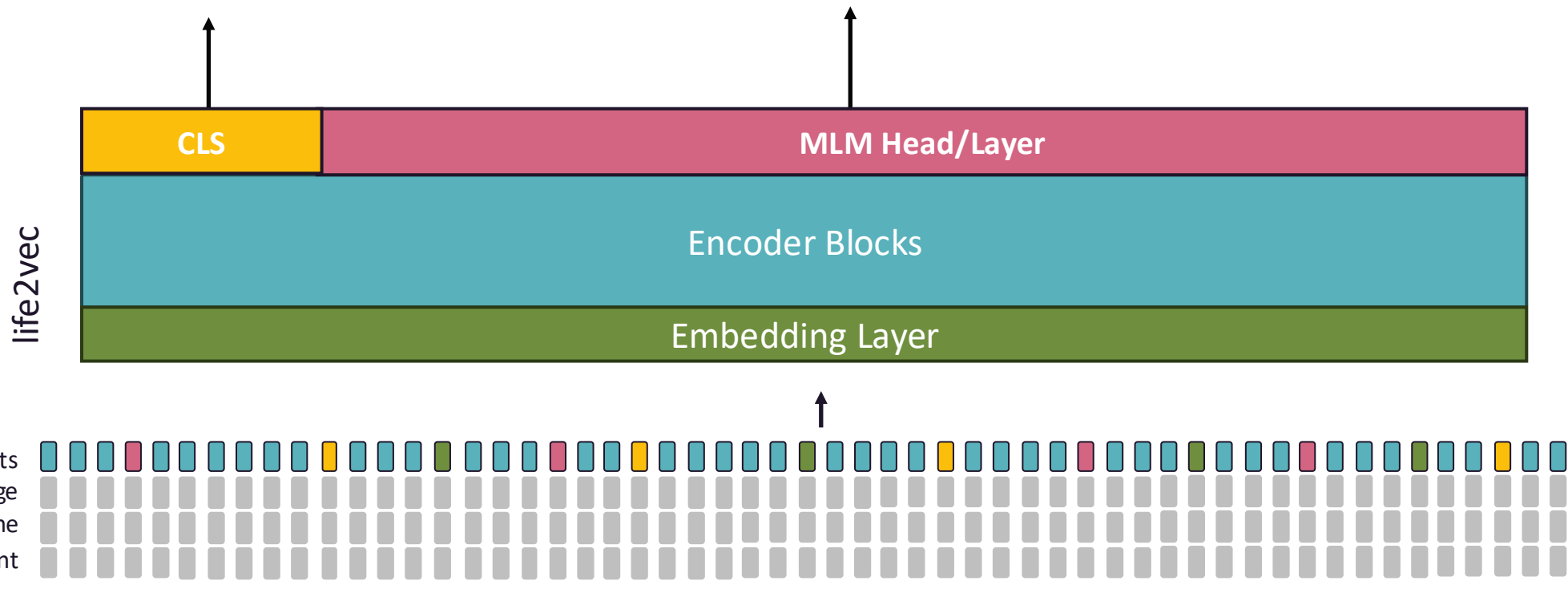


Note: we use our artificial language here, not a natural one (e.g., not English)

Life2vec: Pre-training

Task: Sequence order
(reversed, shuffled, original)

Task: Predict original tokens



Note: we use our artificial language here, not a natural one (e.g., not English)

Synthea-BERT: How do we adapt BERT?

- **Four things encoded** at every position:
 1. token embedding **WHAT** happened learned, 1,036 x 128
 2. absolute position the **ORDER OF EVENTS** (BERT's own)
 3. Time2Vec(age) **WHEN** in the life we added this
 4. Time2Vec(days before) **HOW RECENTLY** we added this
- **Two clocks**, because "at 40" and "three weeks ago" are different questions, and a position index answers neither.
- Architecture modifications:
 1. ReZero gates (to speed up the training)
 2. Tied embeddings (to speed up the training)

Synthea-BERT: Pretraining

I have pretrained it before the workshop (code available).

1. Dataset:

1. Pretraining used 106,252 patients all usable patients minus the 10,000 workshop test patients (**the GitHub repo data contains only 50k**).
2. Sequence length: 128 tokens max

2. Training:

1. Masking fraction: 30%, no task on the CLS token
2. 40 epochs (takes approximately one hour)
3. No hyperparameter tuning.

Transformer for Health records

	ModernBERT-base (block A)	Synthea-BERT (block B)
Trained by	Answer.AI	Us (I pretrained it already)
Layers	22	4
Heads per layer	12	4
Hidden size	768	128
Vocabulary	50,368 subword pieces	1,036 concept tokens
Max input size	8,192 tokens	128 tokens
Parameters	~150 M	959,124 (~3.8 MB on disk)

Hands-on Exercise: (B2) Synthea-BERT

Location: notebooks/b2-pretraining-and-masked-events.

Content:

1. Architecture of Synthea-BERT
2. How do we encode time?
3. Masked Language Modeling for life-sequences

Look for: Exercises 1, 2 and 3

Time: 15 min

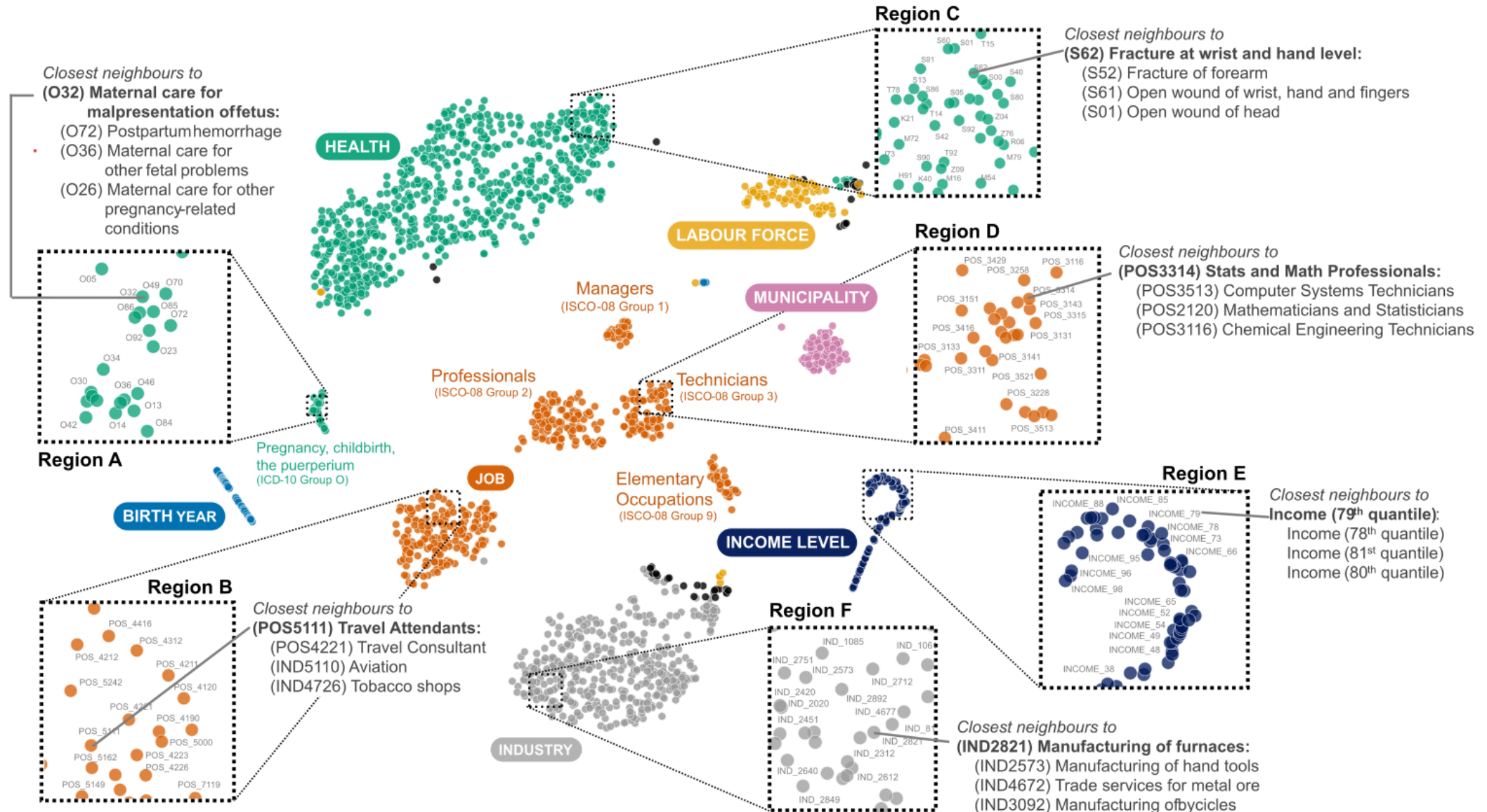
Setup: Guided, with three exercises



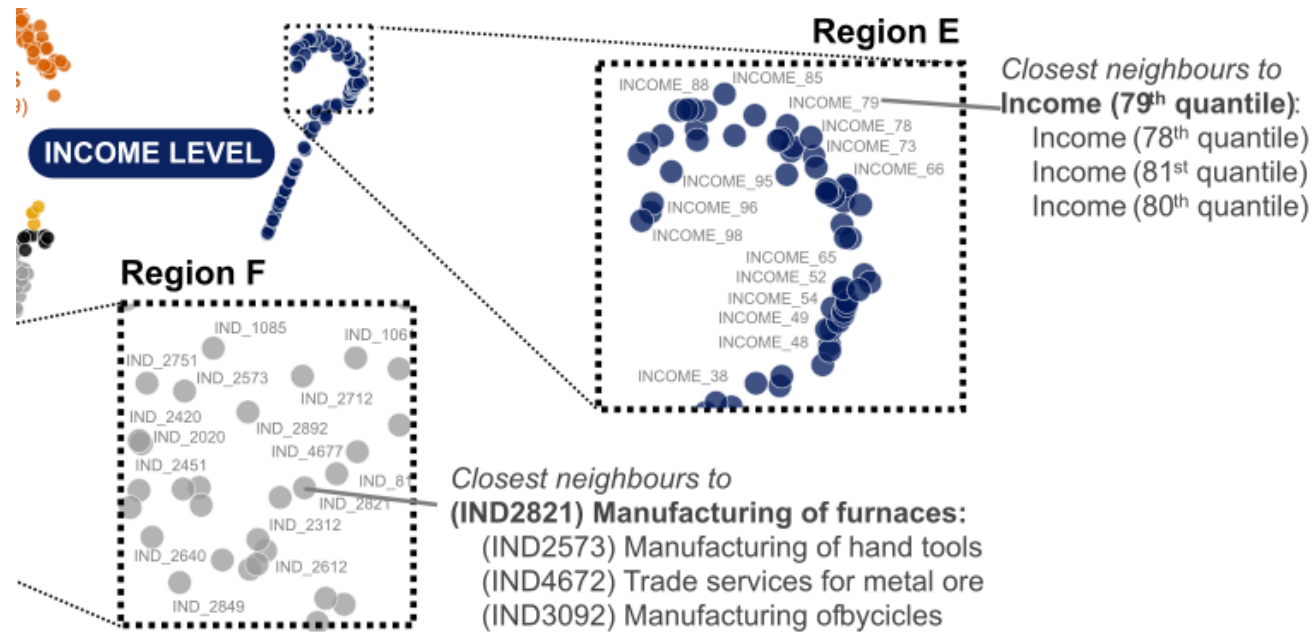
Workshop Page

Life2vec: What did the model **learn**
(about the socio-economic and health language)
on pretraining?

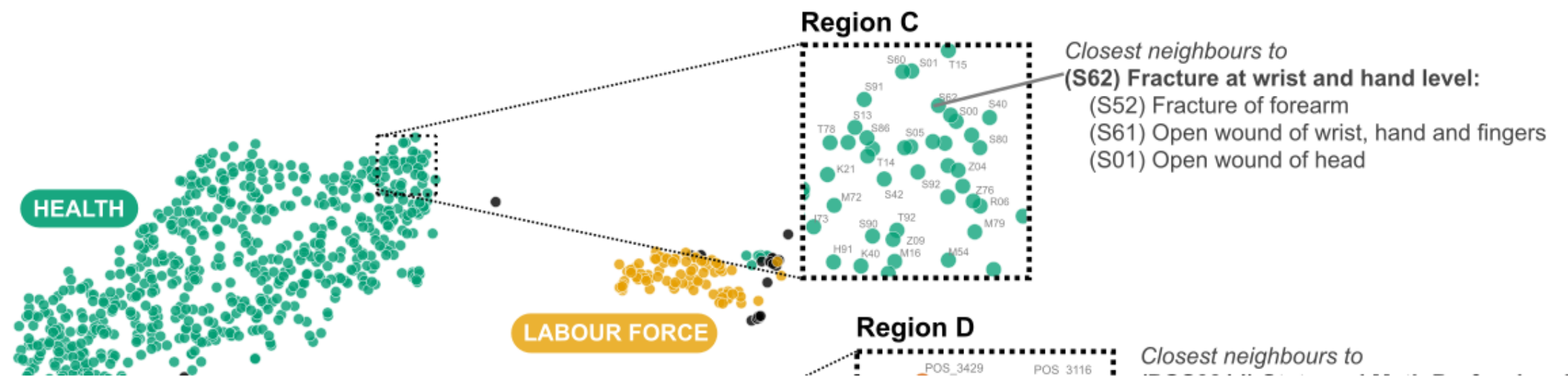
Space of Concept Tokens (with PaCMAP)



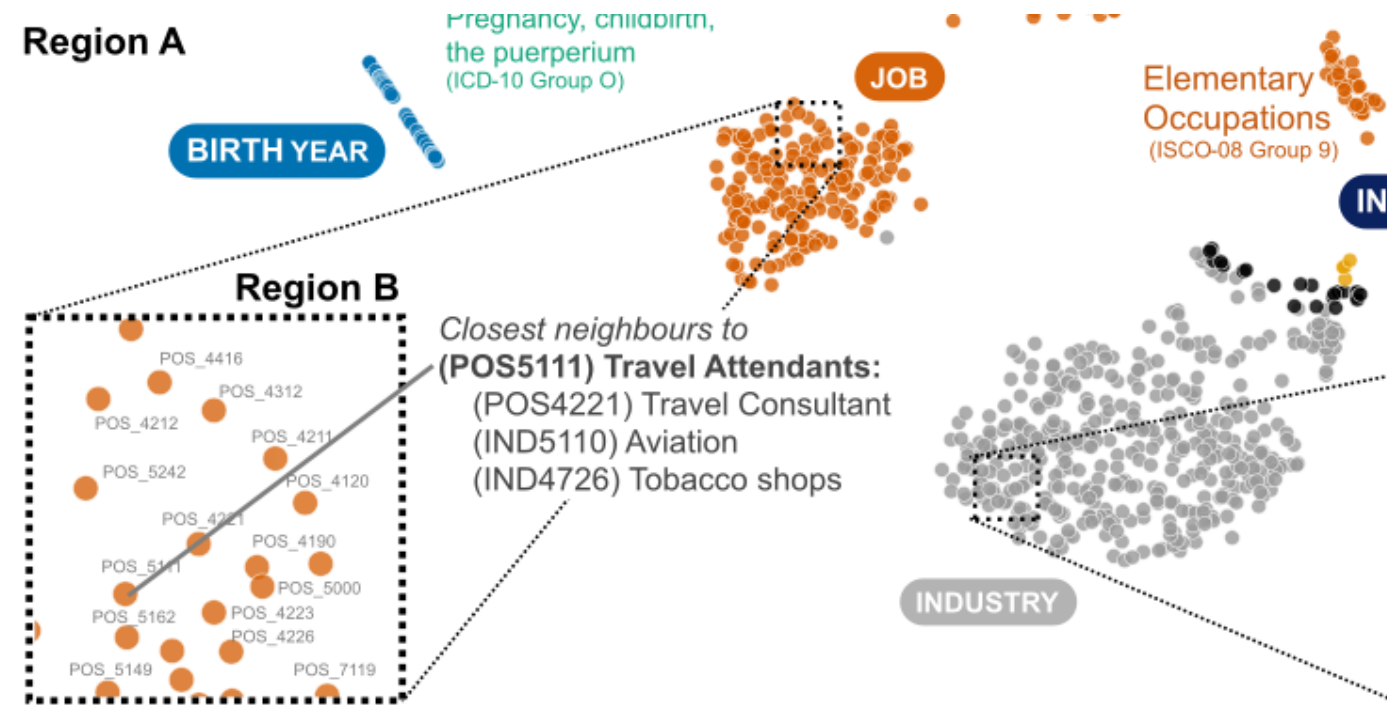
Space of Concept Tokens (with PaCMAP)



Space of Concept Tokens (with PaCMAP)

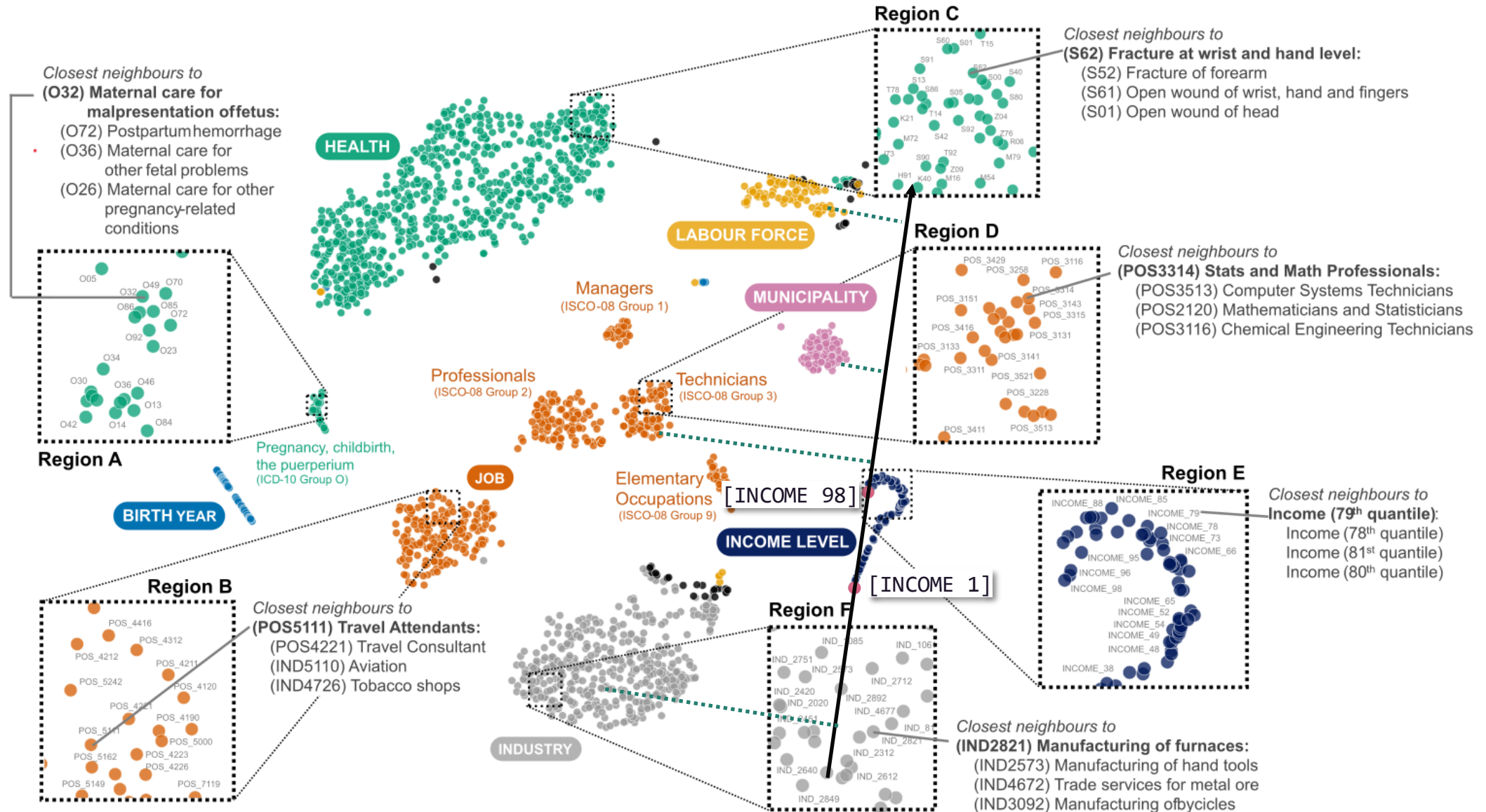


Space of Concept Tokens (with PaCMAP)

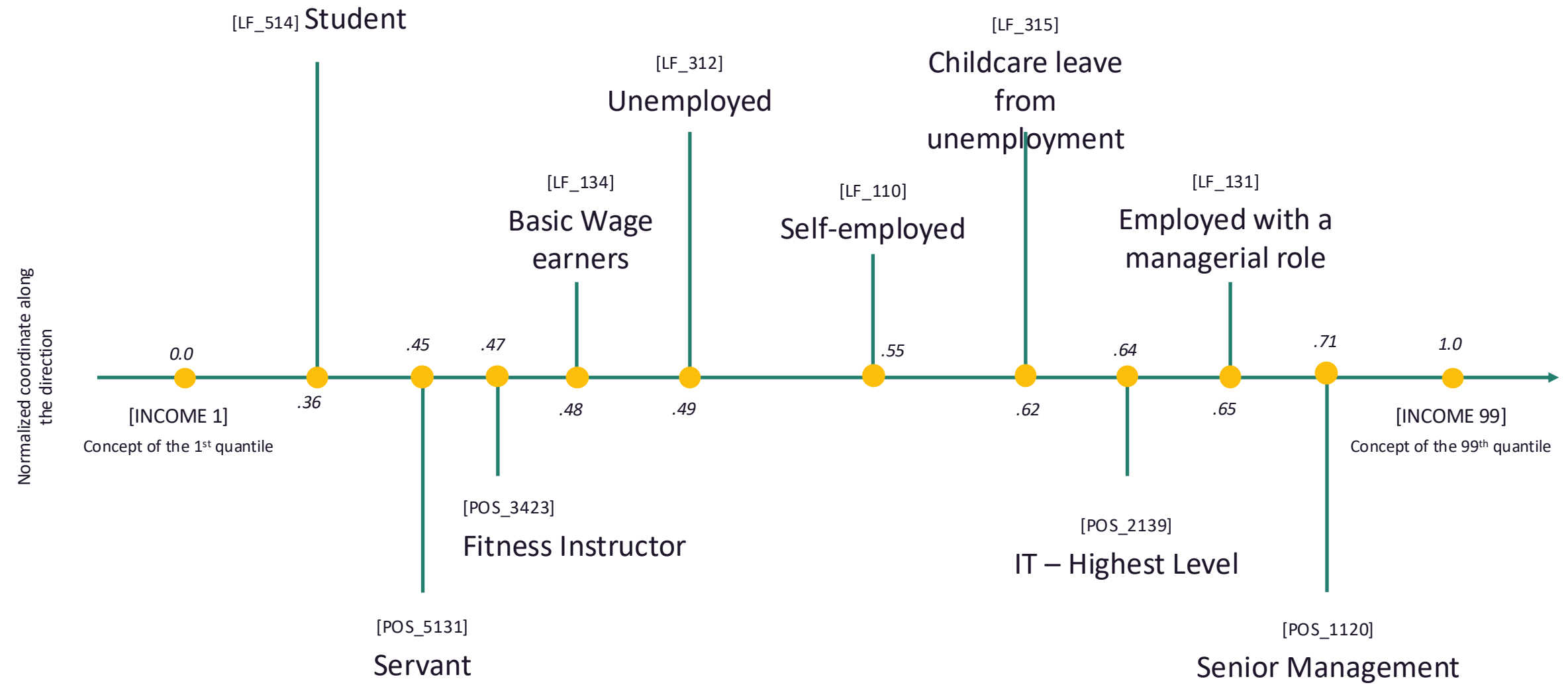


Visually structure corresponds to the structure of the variables

Space of Concept Tokens (with PaCMAP)



Projection to “Income” direction



Projection to “Occupation” direction

The opposite job of a chef and head cook is a physicist .

Chefs and Head Cooks use these skills the most

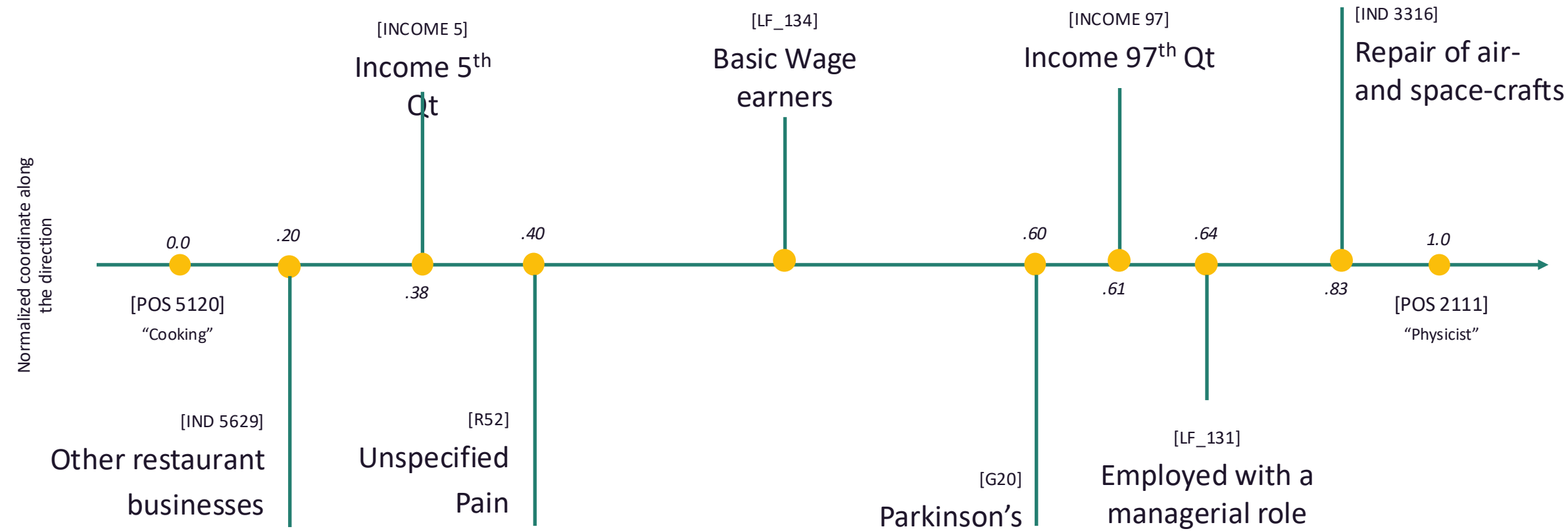
- 1 Management of material resources
- 2 Management of financial resources
- 3 Management of personnel resources
- 4 Coordination
- 5 Negotiation
- 6 Monitoring
- 7 Time management
- 8 Persuasion
- 9 Social perceptiveness
- 10 Learning strategies

Physicists use these skills the most

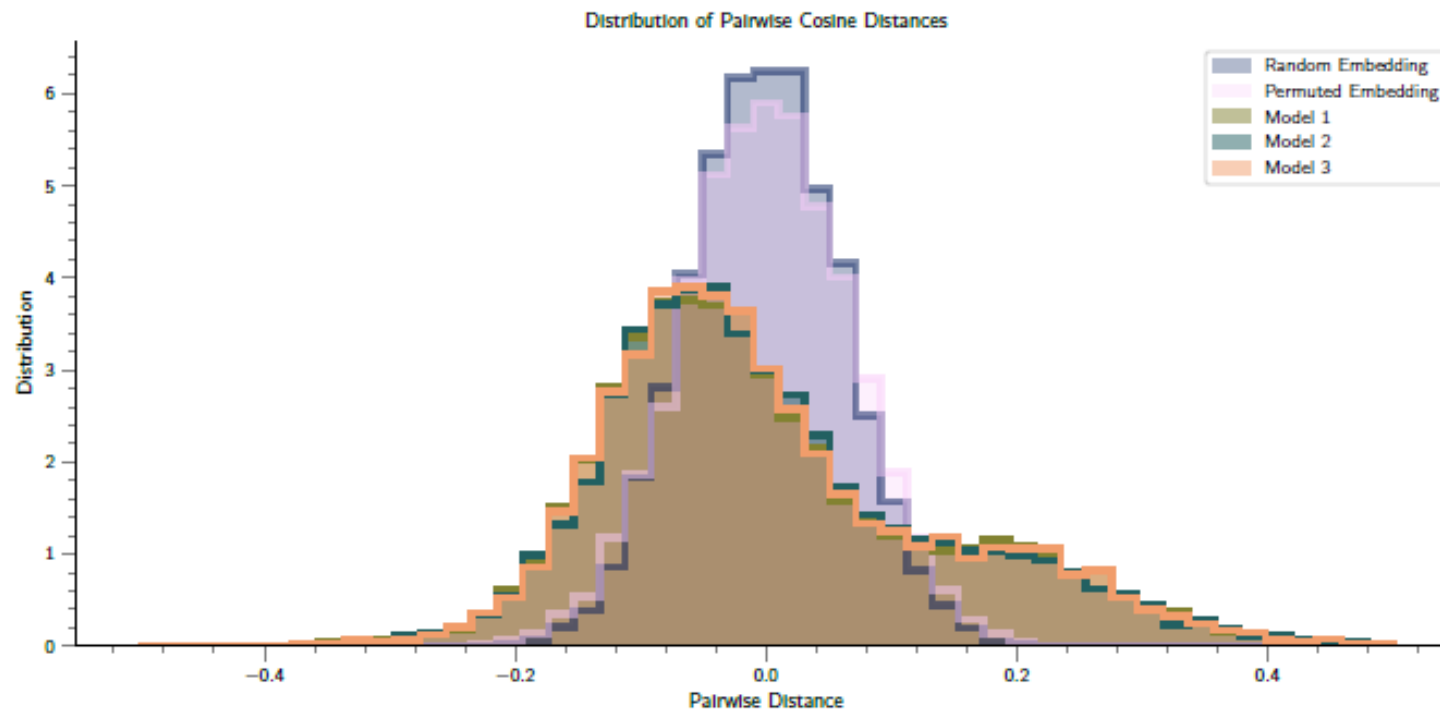
- 1 Physics
- 2 Mathematical reasoning
- 3 Number facility
- 4 Ability to organize groups in different ways
- 5 Information ordering
- 6 Mathematics
- 7 Oral comprehension
- 8 Mathematics
- 9 Originality
- 10 Speech clarity

(n.d.). What Is Your Opposite Job? The New York Times. Retrieved March 11, 2024, from <https://www.nytimes.com/interactive/2017/08/08/upshot/what-is-your-opposite-job.html>

Projection to “Income” direction



Robustness of the space: Permutation test



Models trained on **separate datasets**
and with **different initialization**

Model Comparison	Spearman's ρ
D^1 vs D^2	.668
D^1 vs D^3	.660
D^2 vs D^3	.661
D^1 vs D^R	-.001
D^1 vs D^{1P}	.000

$\rho > .6$ (Strong *monotonic* correlation)¹

Schober, P., Boer, C., & Schwarte, L. A. (2018). Correlation coefficients: appropriate use and interpretation. *Anesthesia & analgesia*, 126(5), 1763-1768.

Hands-on Exercise: (B3) Embeddings

Location: notebooks/b3-event-embedding-space

Content:

1. Embeddings with Synthea-BERT
2. Visualization of the space
3. Concept directions and robustness tests
4. Static vs. Contextual Embeddings

Time: 15 min

Setup: Solo run



Workshop Page

Predictions and Auditing

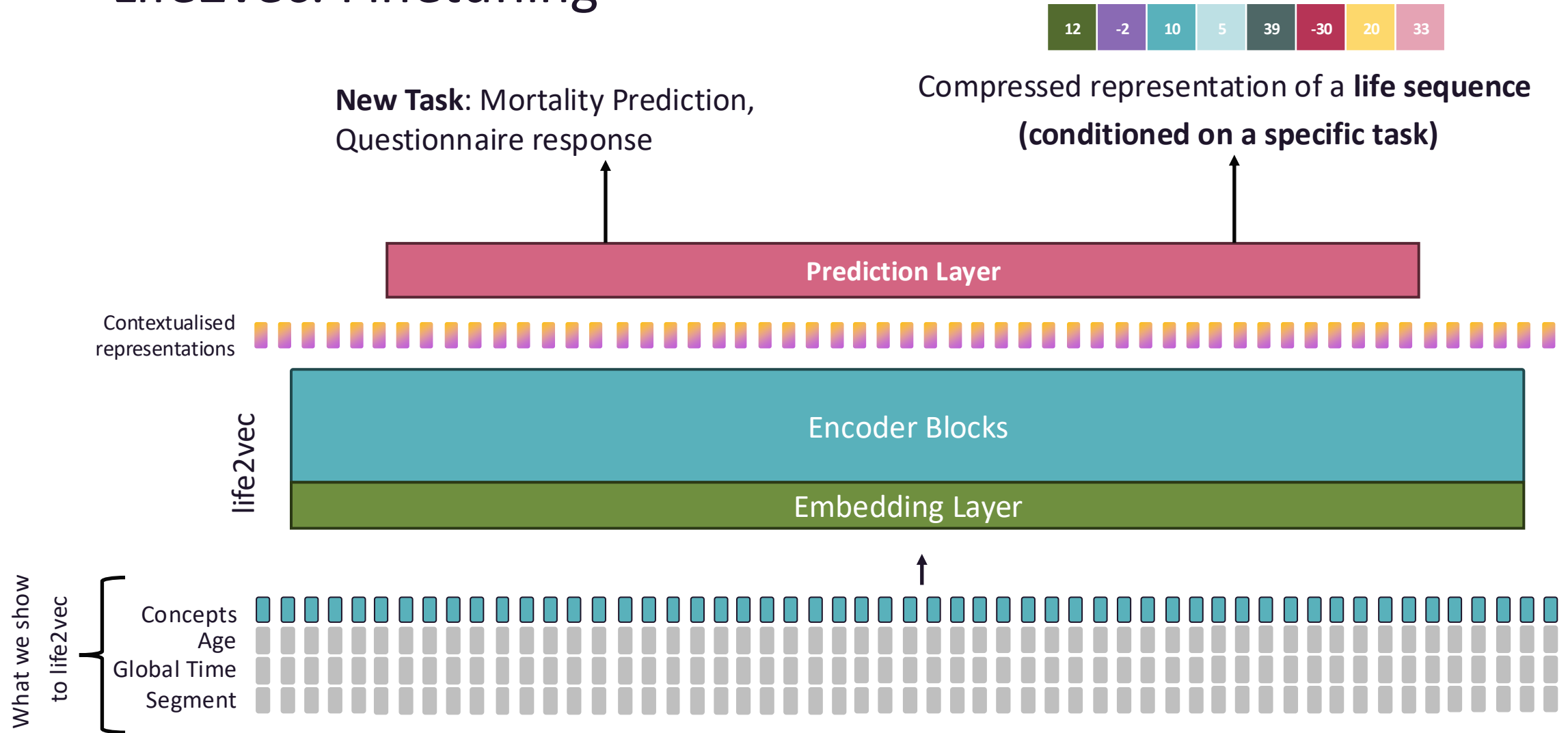
Case of life2vec

On Predictions and Finetuning

- We want (1) **high-predictive power** and (2) **explainability**
- Are learned embeddings "**transferable**" (can be used to solve **many** tasks)?
- Make sure that the predictions are robust and valid.

We will work through couple of life2vec examples and return to Synthea-BERT.

Life2vec: Finetuning



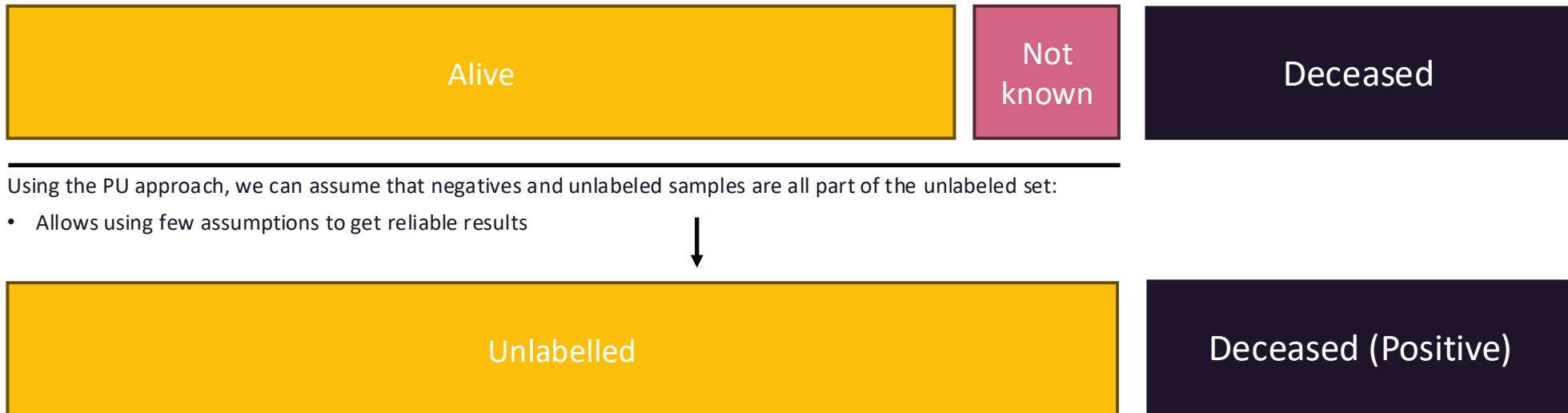
Note: we use our artificial language here, not a natural one (e.g., not English)

Early Mortality

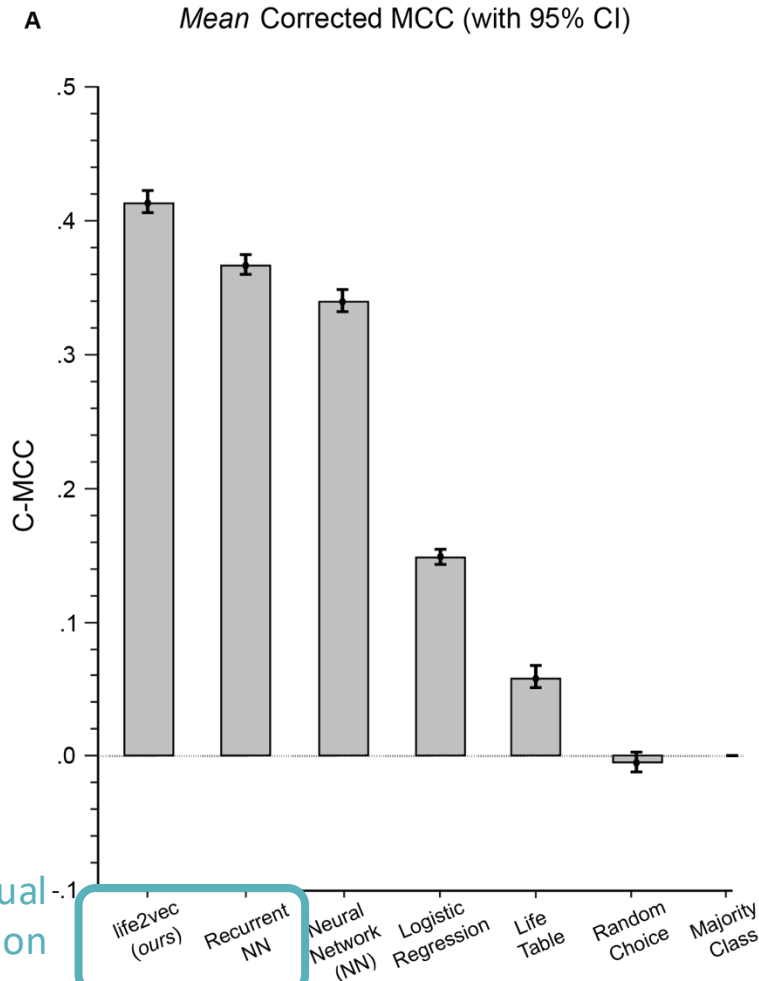
Wang, C., Pu, J., Xu, Z. and Zhang, J., 2021, July. Asymmetric Loss for Positive-Unlabeled Learning. In *2021 IEEE International Conference on Multimedia and Expo (ICME)* (pp. 1-6). IEEE.

- **Task: “Is a person going to be deceased within the next 4 years after 31st December 2015?”**
 - Split people into ones who are marked as dead, and all others
 - Some people do not have “a label”.
 - This is a Positive Unlabelled (PU)-Learning Problem

Why PU Learning? (Mortality Example)



Baselines and Performance Metric



Use textual
representation
of the data

Trained for one specific task

Metrics

1. *Corrected* Matthew's Correlation Coefficient
2. Corrected for the fact that we have missing "labels"
3. MCC measures a level of agreement in unbalanced binary prediction settings.
4. MCC Interpretation:
 - 0: random predictions
 - 1: perfect agreement between predictions and ground-truth
 - -1: total disagreement (predictions are exactly opposite)

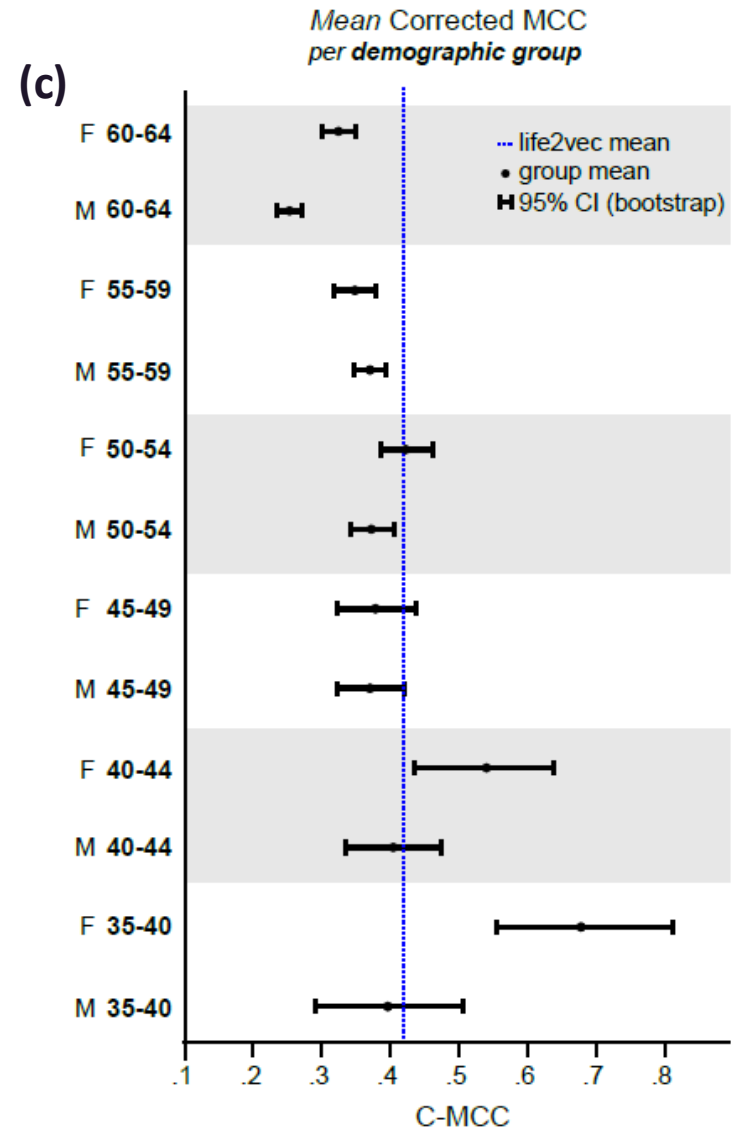
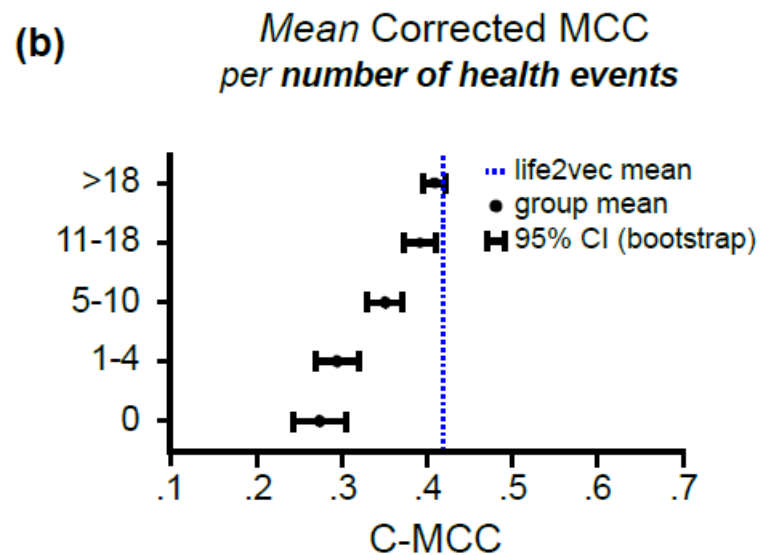
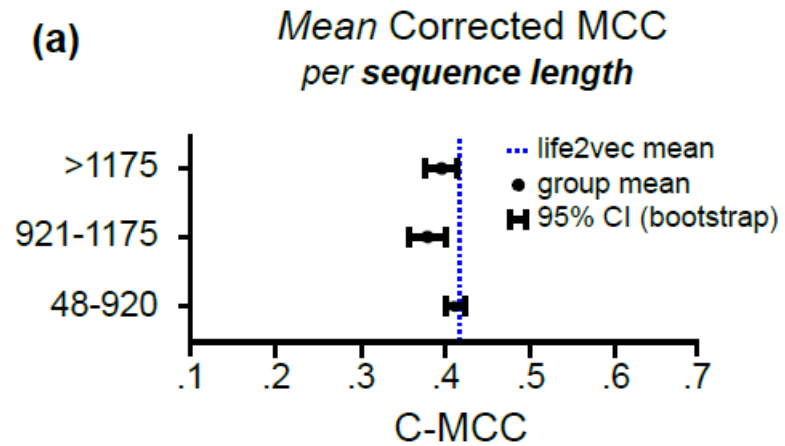
Does not tell us much about how it makes predictions!

Why audit/explain at all?

A score is an outcome, not an explanation.

- **Right for the wrong reason.** Model could be using proxies
- **Right on average.**
- **Right for some, wrong for others.** Subgroup performance is not average performance,
- **Right by unknown means.** The model emits a number. It does not emit the reason, and we usually want to know the reason.
- **Right for a reason that may not be reasonable.** Even a mechanism that generalises can be one you would not defend: a proxy, an artefact of the sampling frame, a shortcut through the encoding.

Performance on subgroups



Interpretation of MCC:
 The higher the value, the more "accurate"/correct the prediction on average (for a specific subgroup)

Performance under different datasets

Retrain the model on different variations of the dataset

Data	C-MCC, 95%-CI	AUL	Vocab Size
Full Labor & Health	0.413 [0.410, 0.422]	0.845	2043
Partial Labor & Health	0.375 [0.367, 0.384]	0.837	1034
Only Full Labor	0.319 [0.312, 0.327]	0.809	1290
Only Partial Labor	0.278 [0.271, 0.285]	0.782	281



Partial Labor: no industry, sector, position and labour force

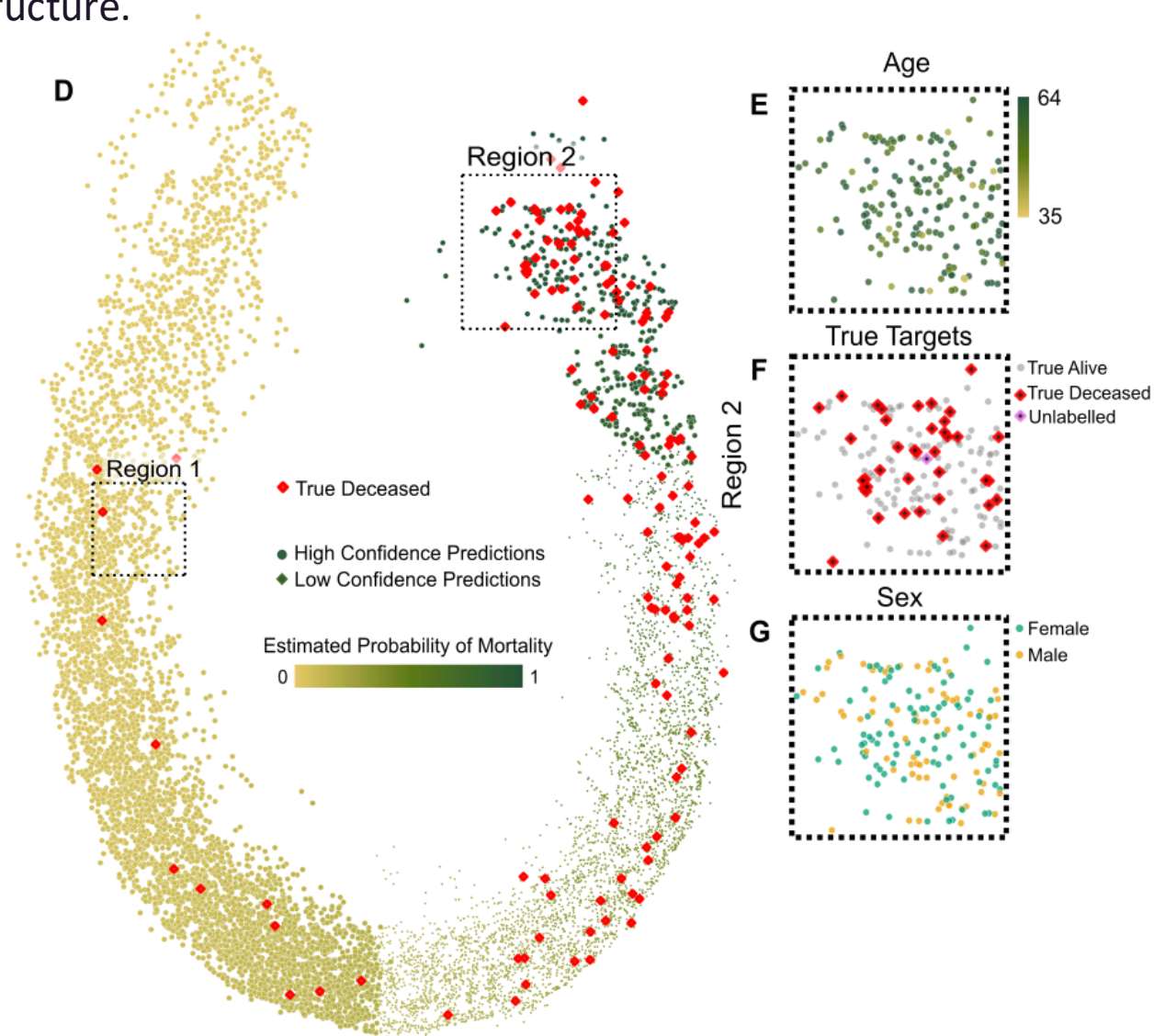
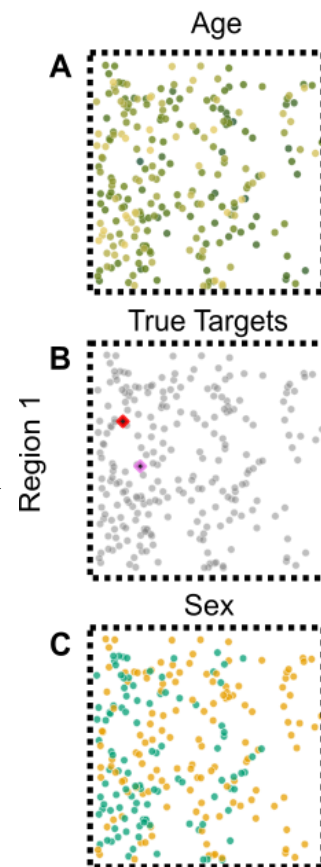
Visualisation of sequence embeddings

Good first step to see if the model captured some structure.

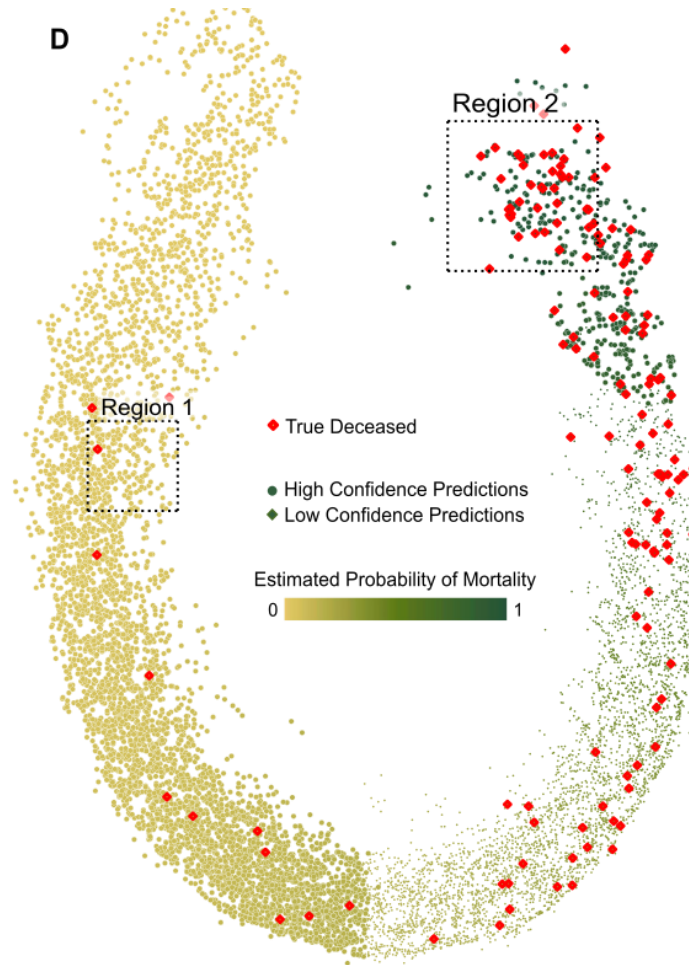
We can look at the low dimensional space of life-summaries.



2D Projection



Interpretation via Attributions



In the Concept space, we can find *somewhat* explainable directions!

- Here, we do not – we need to find them!

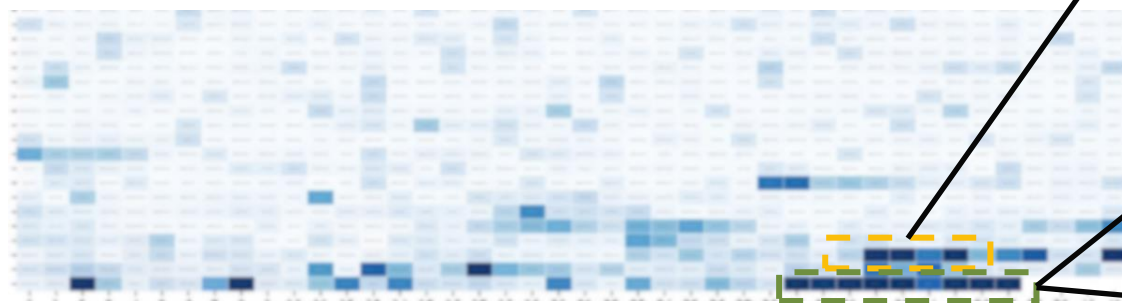
TCAV allows to find these directions

- Interpretation of the **directions of the person-summary** space
- **Sensitivity of the model** towards these directions
- Global Interpretability

Gradients and Attribution

Interpretation of the scores: how large is the change in the output likelihood if we slightly change the embedding of the token.

- Only local explanation (e.g. per sequence), vague interpretation



Sequence of an individual in a textual format
Read: left-to-right, top-to-bottom

If we zoom-in:



Maternity Leave

Flexjob, receives salary

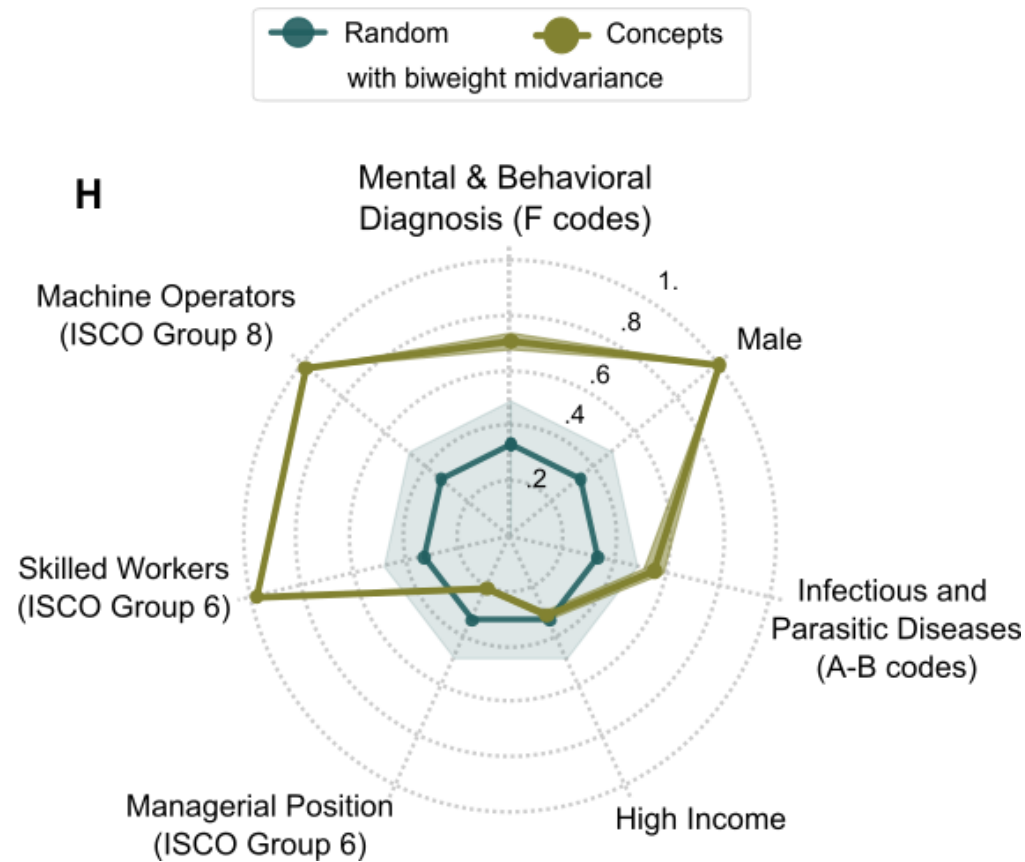


Malignant neoplasm of brain (admitted to hospital)



Pyothorax with fistula (admitted to hospital)

Interpretation via TCAV



- Interpretation of the **directions of the person-summary space**
- **Sensitivity of the model** towards these directions
- Global Interpretability

Self-Reported Personality Assessment

We focused on Extroversion Facets:

- **Sociability** (tendency to enjoy social interactions)
- **Liveliness** (one's typical enthusiasm and energy)
- **Self-esteem** (tendency to have positive self-regard)
- **Boldness** (comfort within a variety of social situations)

Example:

1. In social situations, I'm usually the one who makes the first move

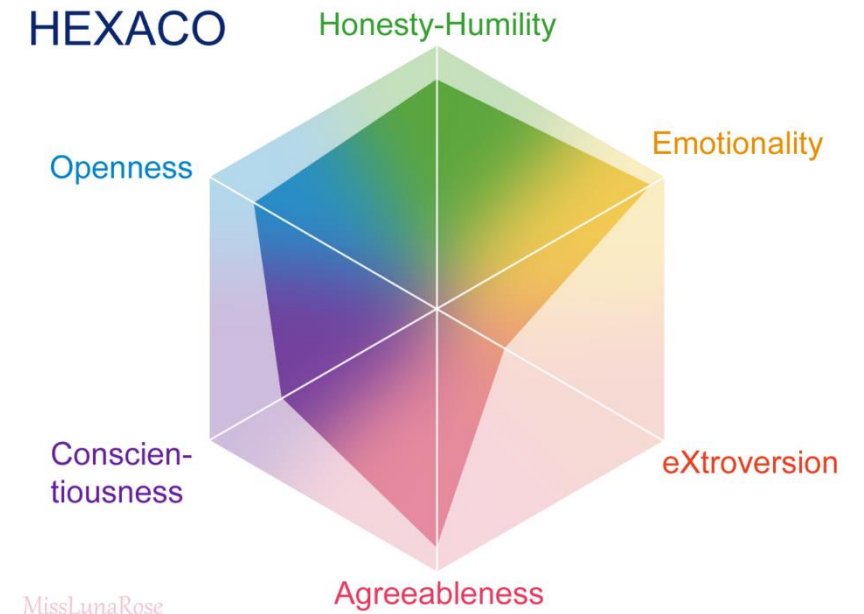


Image source: [Wikipedia](#)

Inventory Descriptions: [The HEXACO Personality Inventory - Revised](#)

Extraversion & Nuance Prediction

Task: “What kind of replies does the person give to the 10 questions evaluating their Extraversion?”

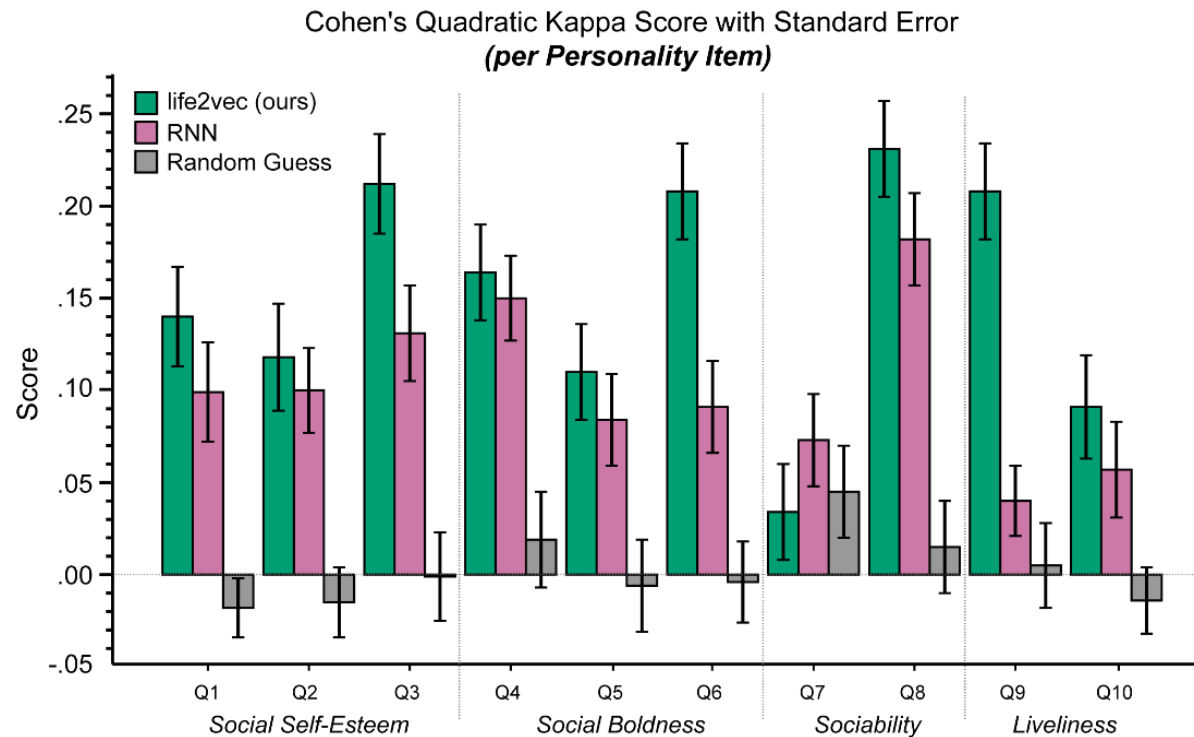
- Multiclass prediction
- Ordinal Classification task (i.e. labels have ordered)
- Highly Imbalanced Data
- *We do not have much data*

Statement:

In social situations, I'm usually the one who makes the first move



Extraversion Prediction



Example of Questions:

6. Most people are more upbeat and dynamic than I generally am (*liveliness*)

7. The first thing that I always do in a new place is to make friends (*boldness*)

Synthea-BERT: Severe Diseases

We will use Synthea-BERT to predict severe diseases.

Task: Given everything recorded before the index date (18 Aug 2021), will this person be diagnosed with a severe condition in the following five years?

Test set: 7,965 eligible test patients, 454 positives (base rate 5.7%)
Always predicting "no" scores 94.3%, so accuracy is not reported again.

What counts as severe:

- An explicit list of 38 SNOMED descriptions (scripts/severe_codes.py).
- Seven codes are excluded on one principle: a code that records the system looking for a disease is not the disease.
- Then we ask why the model assigns the scores it does.

Hands-on Exercise: (B4) Finetuning

Location: notebooks/b4-predicting-severe-disease

Content:

1. Baseline models
2. Finetuning vs raw model outputs
3. Performance evaluation
4. Interpretability
5. (if time allows) demonstration of attribution methods

- **Look for:** Exercises 1 and 2

Time: 20 min

Setup: Solo run



Workshop Page

Block B: Take aways and Questions

You built the whole pipeline on a 959K-parameter model and did five things:

- (B1) turned dated tabular rows into token sequences
- (B2) read masked events,
- (B3) looked at the concept space and built a ruler out of anchor events
- (B4) predicted severe-disease onset
- (B4) applied interpretability tools
- **Questions?**

Conclusion and Questions

Conclusion

What can you do?

1. Design a vocabulary for your own data
2. Read an embedding space, and check yourself
3. Anchor vectors to build a ruler along a direction you actually care about.
4. Static vs contextual, for when the same token means different things in different places.
5. Audit a prediction before you report it

Take aways:

1. The pipeline transfers

- events → tokens → sequences → encoder → representation
- The same five steps appear in life2vec, BEHRT, CAREER, GETNext, MAPLE and Delphi-2M.
- You ran it twice today: with ModernBERT, and on health records with Synthea-BERT.

2. The audit is crucial for these kind of models

Thanks a lot!

Why not Survival Analysis?

Proof-of-Concept

- Test if life2vec embeddings carry any predictive signal.
- PU-learning handles incomplete labels.
- No time-to-event prediction here.

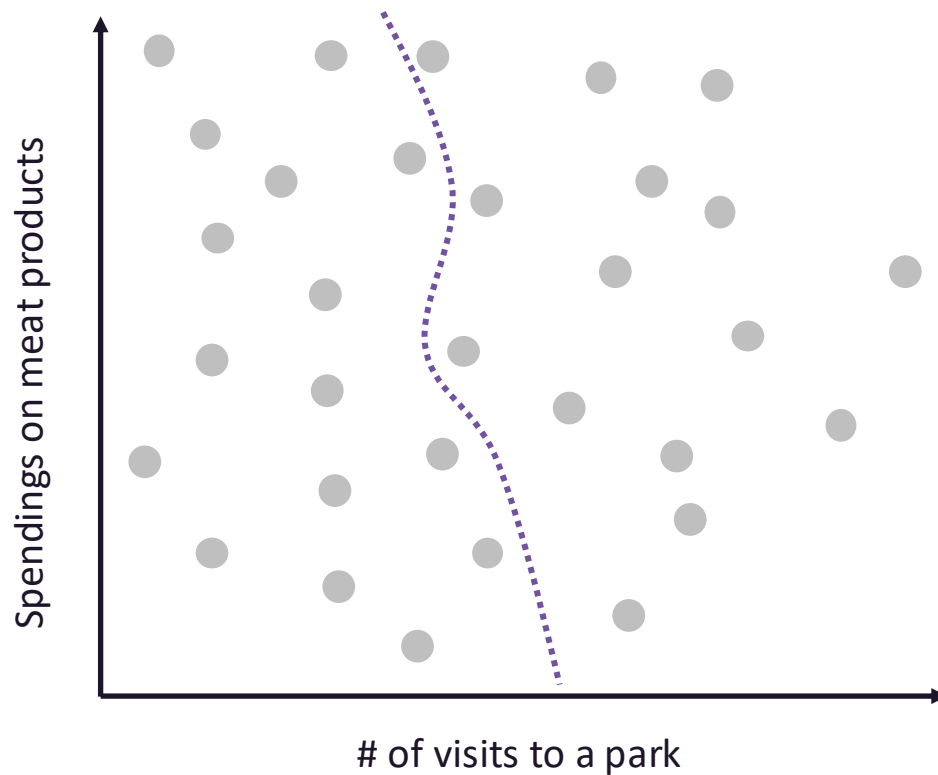
Methodological Fit

- Survival assumes proportional hazards & clean censoring.
- Fixed prediction window $\neq$ natural survival framing.
- Life trajectories are non-linear and complex.

Practical Considerations

- Deep survival models + sequence embeddings were computationally heavy at the time

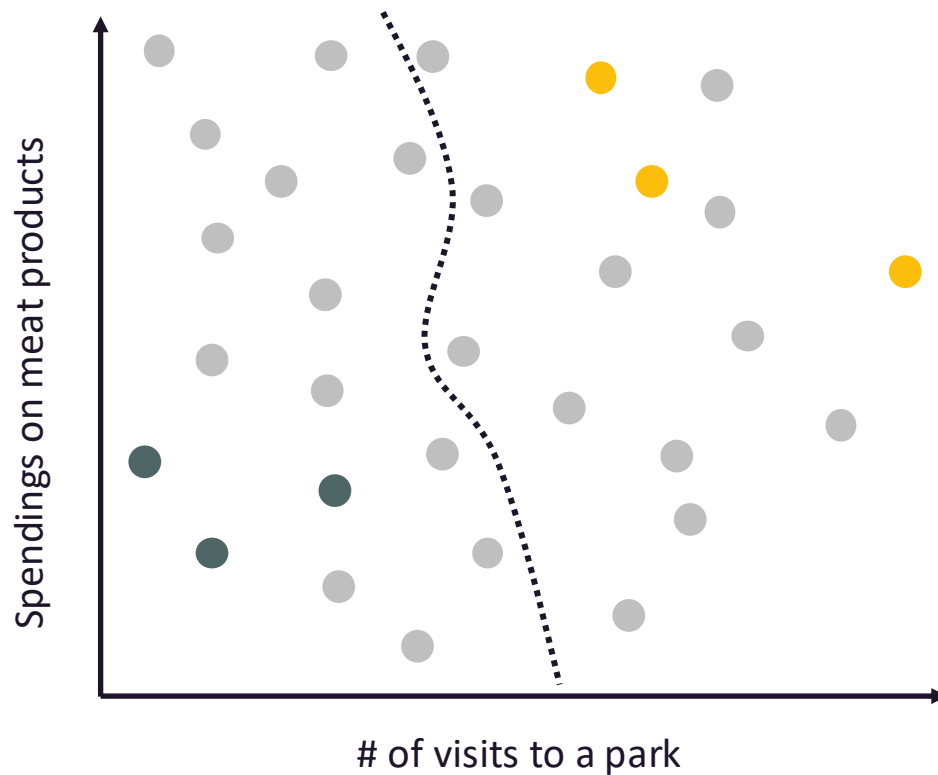
Overview of TCAV



Let's imagine an **algorithm** that predicts **whether a person has a dog**

..... decision boundary of the classifier

Overview of TCAV



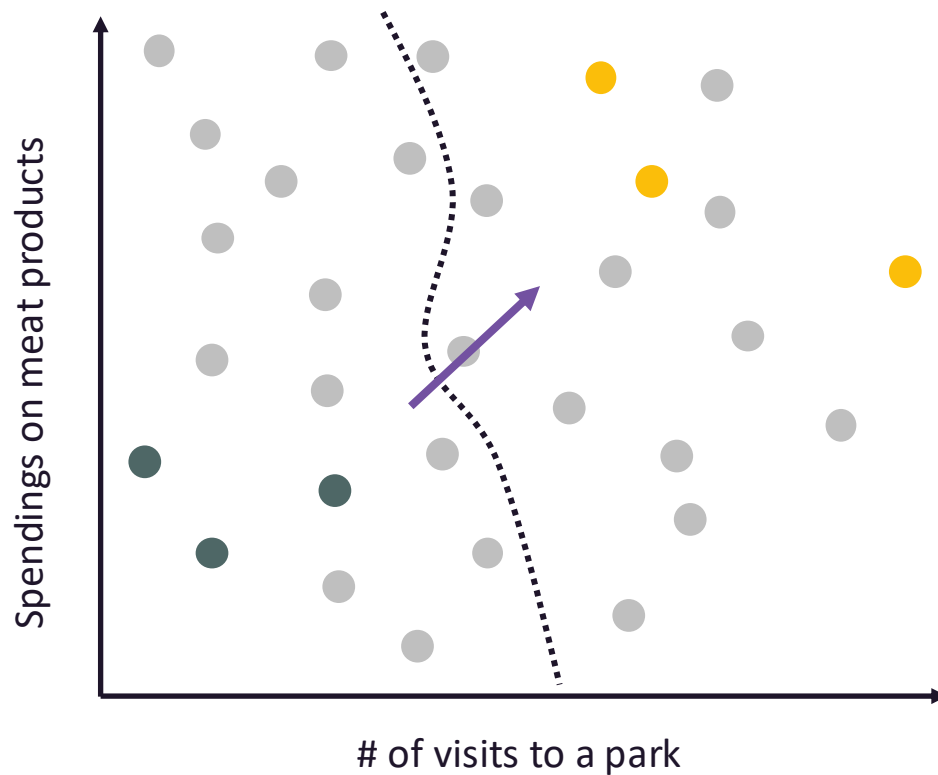
Let's imagine **you have extra information**

..... decision boundary of the classifier

● Lives in a rental

● Owns an apartment

Overview of TCAV



Let's imagine **you have extra information**

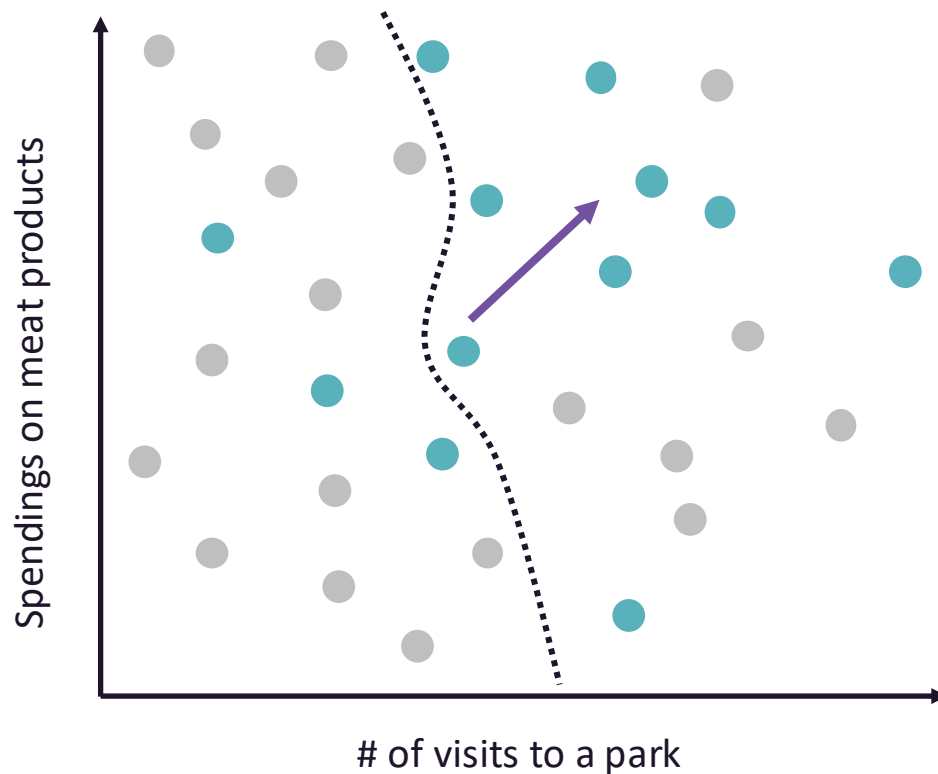
..... decision boundary of the classifier

● Lives in a rental

● Owns an apartment

→ Direction of a concept

Overview of TCAV

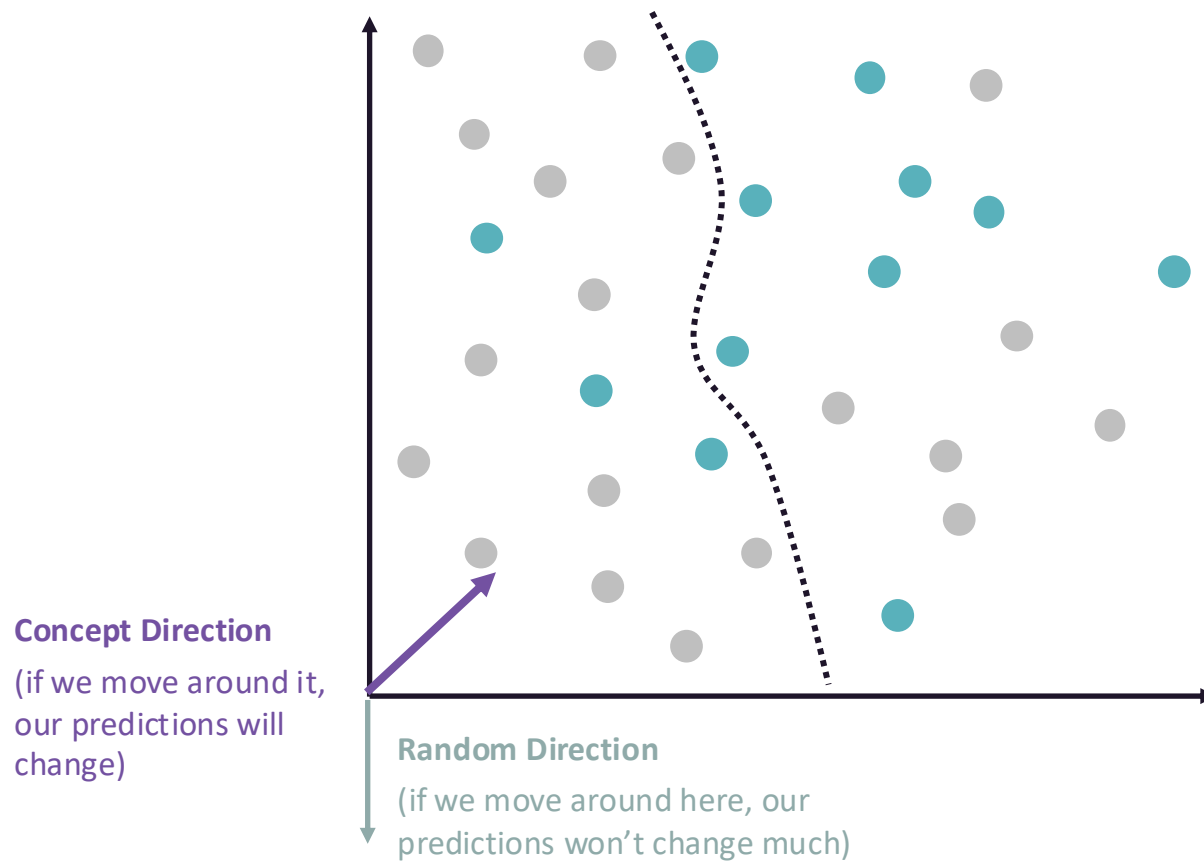


Interpretation:

If we move in a certain direction (the one that is associated with a concept), how strongly would it influence the output of our model (on average)

- decision boundary of the classifier
- Randomly sampled point
- Direction of a concept

Overview of TCAV



Interpretation:

If we move in a certain direction (the one that is associated with a concept), how strongly would it influence the output of our model (on average)